Chapter 1

Questions on Lesson Two

Resistors Connection

Interactive test



The questions signed by ✳ are answered in detail. ● Understand ○ Apply Higher Order Thinking Skills

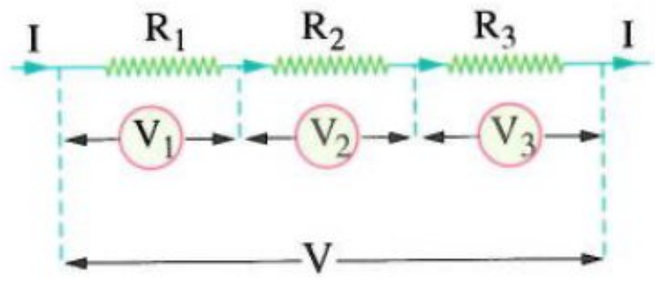
Important guidelines

⇒ Resistors in series:

- To determine the equivalent resistance ($\grave{R}$):

$$\grave{R} = R_1 + R_2 + R_3$$

If N equal resistors each of resistance R are connected in series, then: $\grave{R} = NR$



- To determine the total voltage (V): $V = V_1 + V_2 + V_3$

(Where the voltage is distributed across the resistors)

- To determine the current intensity (I): $I = \frac{V}{\grave{R}} = \frac{V_1}{R_1} = \frac{V_2}{R_2} = \frac{V_3}{R_3}$

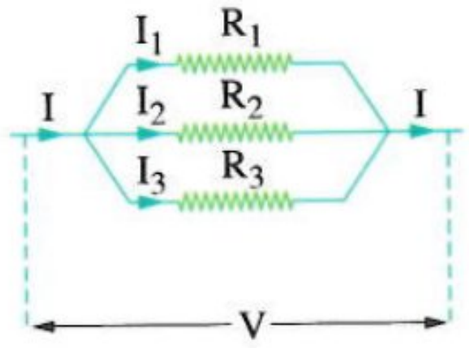
(Where the current intensity through all resistors is the same)

⇒ Resistors in parallel:

- To determine the equivalent resistance ($\grave{R}$):

$$\frac{1}{\grave{R}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

- If N equal resistors each of resistance R are connected in parallel, then: $\grave{R} = \frac{R}{N}$



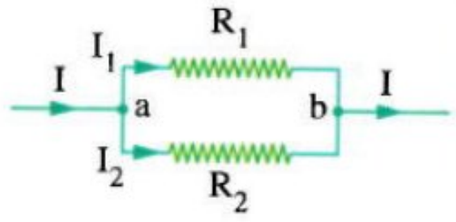
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- If two different resistors R_1 and R_2 are connected in parallel; $\grave{R} = \frac{R_1 R_2}{R_1 + R_2}$
- To determine the potential difference (V): $V = I\grave{R} = I_1R_1 = I_2R_2 = I_3R_3$
  (Where the voltage is the same across each resistor)
- To determine the total current (I): $I = I_1 + I_2 + I_3$
  (Where the current is distributed on the resistors)

- To calculate the current intensity in a branch:

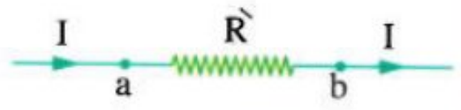
$$\grave{R} = \frac{R_1 R_2}{R_1 + R_2} \ , \ V_1 = V_2 = V_{ab} \ , \ I_1 R_1 = I_2 R_2 = I\grave{R}$$

$$I_1 = \frac{V_{ab}}{R_1} \ , \ I_2 = \frac{V_{ab}}{R_2}$$

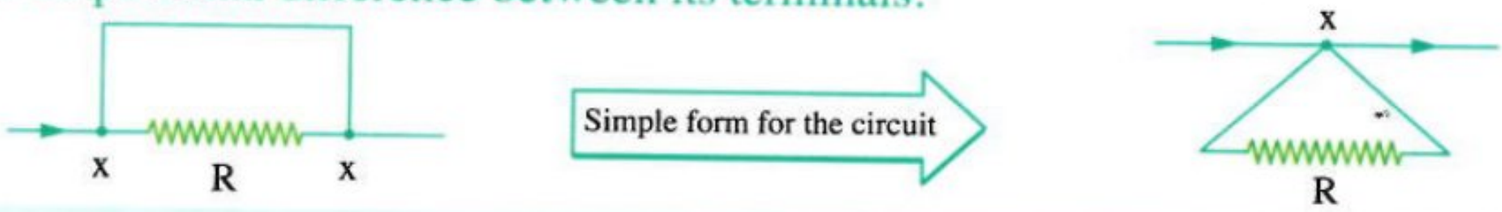


- In the case of having two branches only :

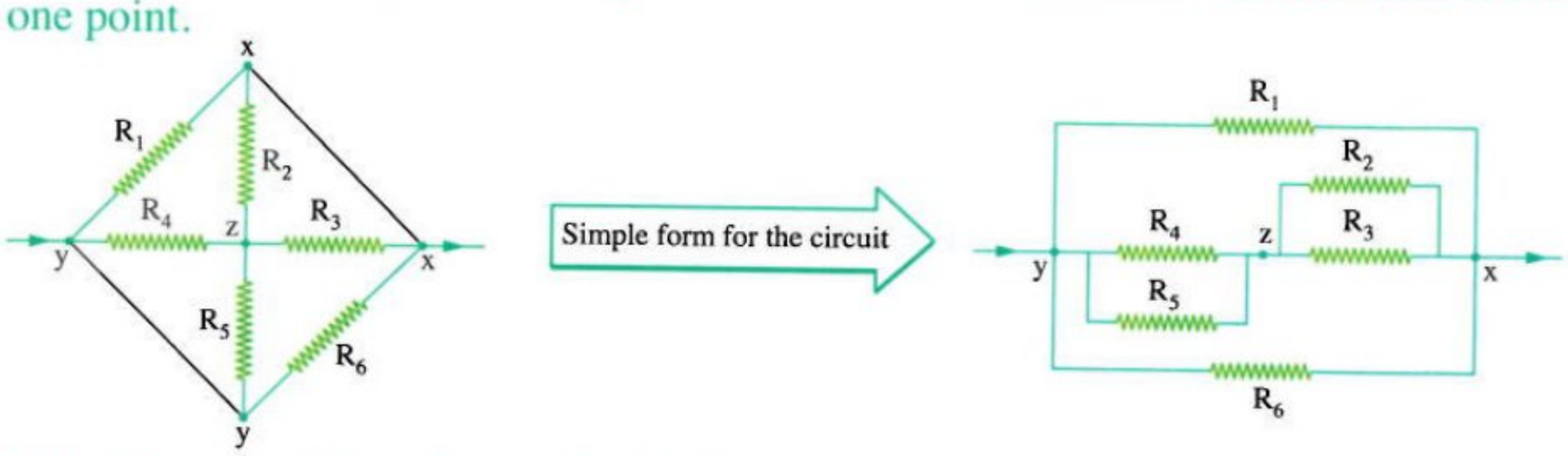
$$I_1 = I\frac{R_2}{(R_1 + R_2)} \ , \ I_2 = I\frac{R_1}{(R_1 + R_2)}$$



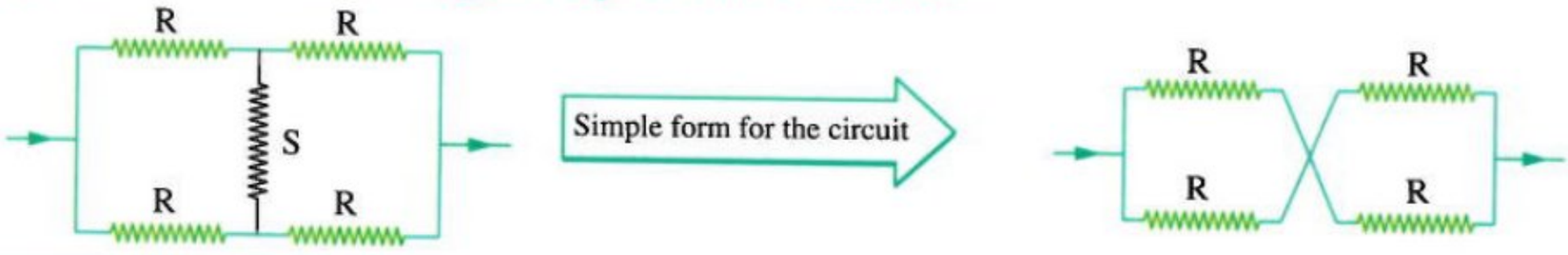
- In the case where the terminals of a resistor are connected together by a conducting wire, its resistance is ignored when calculating the equivalent resistance because there is no potential difference between its terminals.



- In the case of having a connecting wire of no resistance, its ends can be considered as one point.



- When the potentials at the terminals of a resistor are equal, the value of this resistor is ignored when calculating the equivalent resistance.



For comparing the consumed power in two resistors

at constant electric current	at constant potential difference
$\frac{(P_w)_1}{(P_w)_2} = \frac{R_1}{R_2}$	$\frac{(P_w)_1}{(P_w)_2} = \frac{R_2}{R_1}$

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First Multiple choice questions

1 Three resistors are connected in parallel, if one of them equals one ohm, then their equivalent resistance is ..... one ohm.

(a) less than
(b) greater than
(c) equal to
(d) the answer can't be determined unless knowing the values of the other two resistors

2 ✱ Three resistors have resistances of 40 Ω, 60 Ω, 120 Ω, so their total equivalent resistance when connecting them:

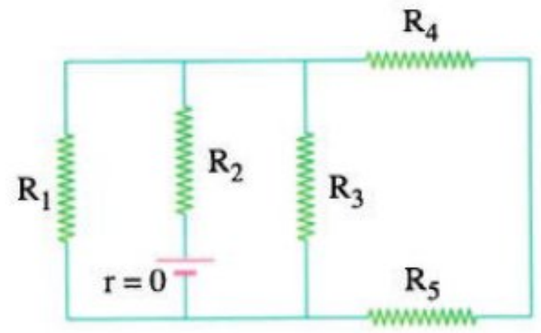
(i) In series will be ..... .

(a) 520 Ω (b) 220 Ω (c) 110 Ω (d) 34 Ω

(ii) In parallel will be ..... .

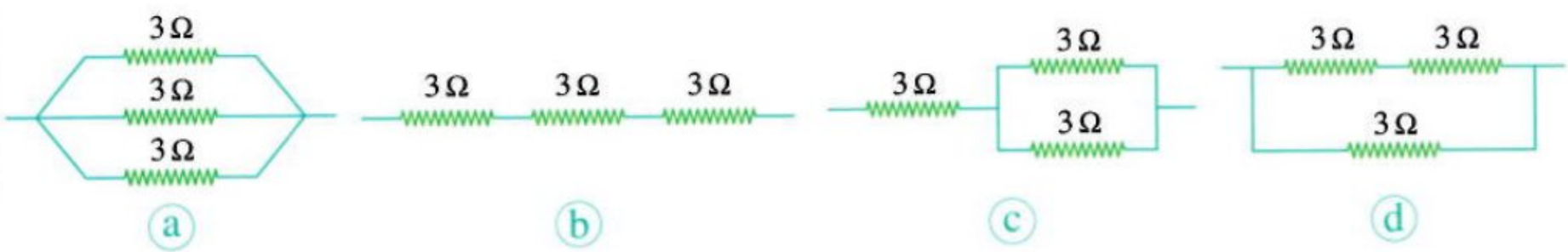
(a) 10 Ω (b) 20 Ω (c) 30 Ω (d) 40 Ω

3 Which of the following statements is correct for the opposite electric circuit?



(a) Resistors R_2 and R_3 are connected in series.
(b) Resistors R_1 and R_3 are connected in parallel.
(c) Resistors R_4 and R_5 are connected in parallel.
(d) Resistors R_1 and R_2 are connected in parallel.

4 ✱ You have three resistors, the resistance of each is 3 Ω. The resistors are connected in four different methods as shown in the following figures (a), (b), (c) and (d);



(a) (b) (c) (d)

so, the method of connection in which the equivalent resistance equals:

(i) 4.5 Ω is ..... .

(a) (a) (b) (b) (c) (c) (d) (d)

(ii) 2 Ω is ..... .

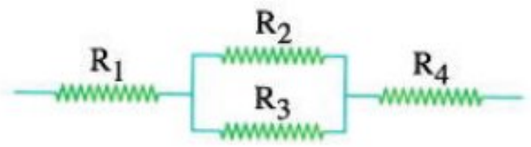
(a) (a) (b) (b) (c) (c) (d) (d)

(iii) 1 Ω is ..... .

(a) (a) (b) (b) (c) (c) (d) (d)

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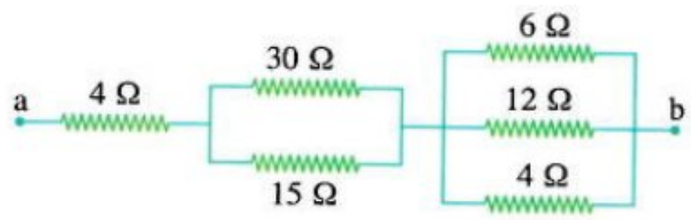
5 The opposite figure represents a part of an electric circuit, which of the resistors whose removal from the circuit leads to the increase of the value of the equivalent resistance?



(a) R_1 (b) R_2
(c) R_4 (d) R_1, R_2, R_3

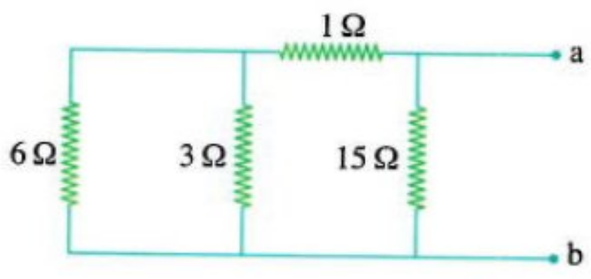
6 What is the equivalent resistance between points a and b in each of the following cases?

(i)



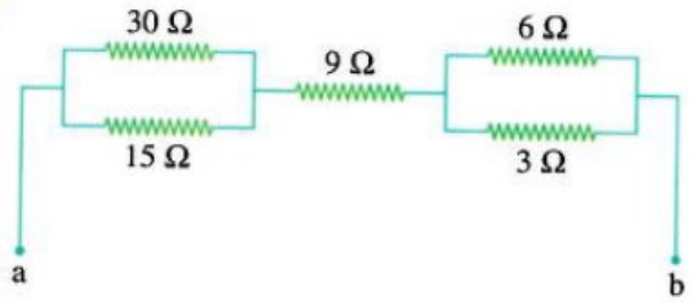
(a) 15 Ω (b) 16 Ω
(c) 17 Ω (d) 18 Ω

(ii)



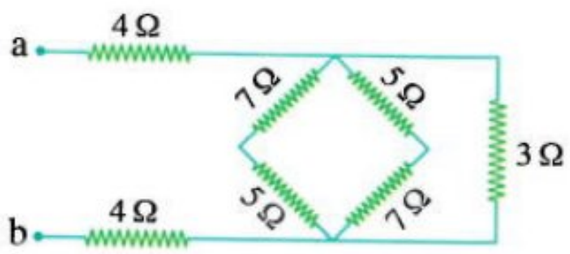
(a) 1 Ω (b) 1.5 Ω
(c) 2 Ω (d) 2.5 Ω

(iii)



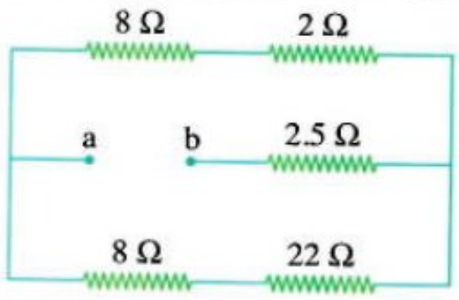
(a) 15 Ω (b) 21 Ω
(c) 24 Ω (d) 30 Ω

(iv)



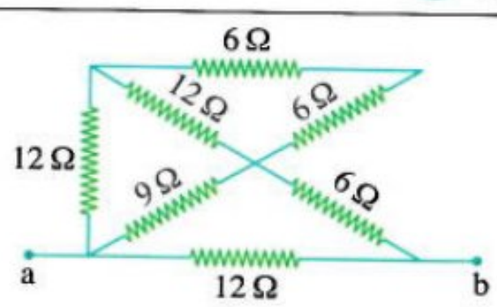
(a) 5 Ω (b) 10 Ω
(c) 15 Ω (d) 20 Ω

(v)



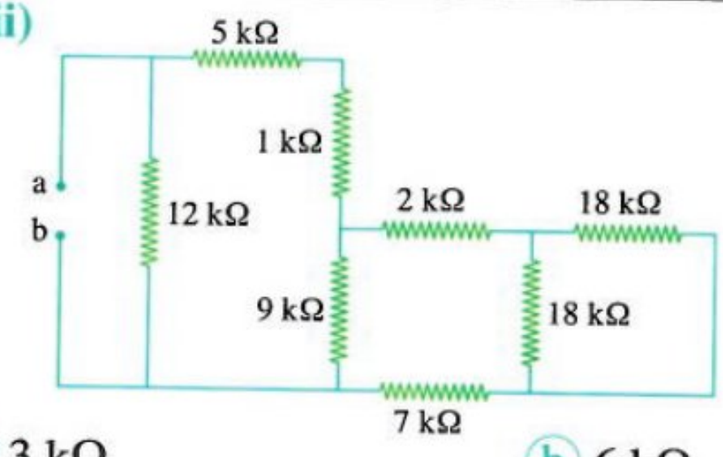
(a) 2.5 Ω (b) 5 Ω
(c) 7.5 Ω (d) 10 Ω

(vi)



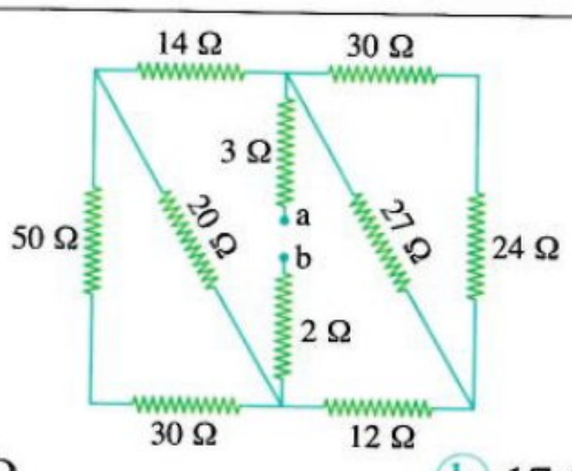
(a) 2 Ω (b) 4 Ω
(c) 6 Ω (d) 8 Ω

(vii)



(a) 3 kΩ (b) 6 kΩ
(c) 9 kΩ (d) 12 kΩ

(viii)

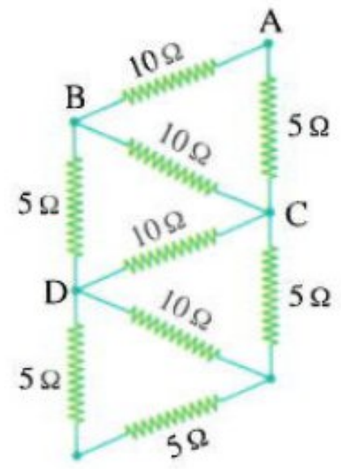


(a) 13 Ω (b) 17 Ω
(c) 20 Ω (d) 25 Ω

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7 The equivalent resistance for the opposite electric circuit when connecting an electric source to the circuit between the points:



(i) A and B equals ..... .

(a) 20 Ω (b) 17 Ω

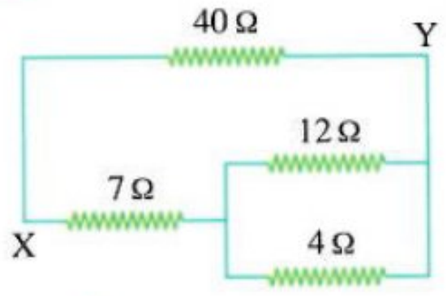
(c) 9 Ω (d) 5 Ω

(ii) B and C equals ..... .

(a) 1.25 Ω (b) 2.5 Ω (c) 3.75 Ω (d) 6.3 Ω

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8 In the opposite figure:



(i) The equivalent resistance when a battery of negligible internal resistance is connected between X and Y is ..... .

(a) 2 Ω (b) 4 Ω (c) 6 Ω (d) 8 Ω

(ii) If a battery is connected instead of the 7 Ω resistor, the equivalent resistance of the circuit becomes ..... .

(a) 40 Ω (b) 41 Ω (c) 42 Ω (d) 43 Ω

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9 Three resistors (x), (y) and (z) are connected together with a battery and a variable resistor (S) as shown in figure (1). Figure (2) graphically represents the relation between the potential difference (V) across each of the three resistors and the current intensity (I) flowing through each of them:

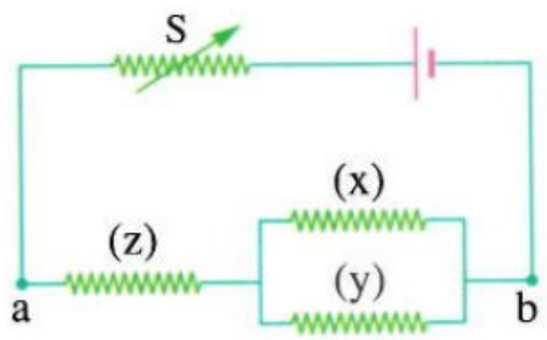


Figure (1)

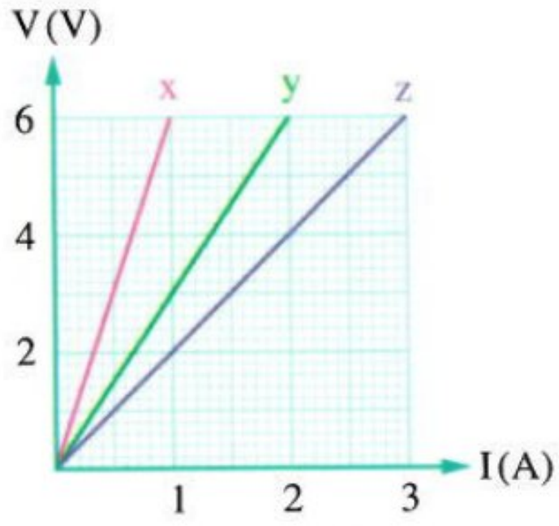


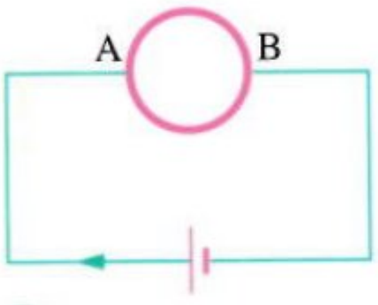
Figure (2)

Then the equivalent resistance between points (a) and (b) shown in figure (1) is equal to ..... .

(a) 1 Ω (b) 3 Ω (c) 4 Ω (d) 11 Ω

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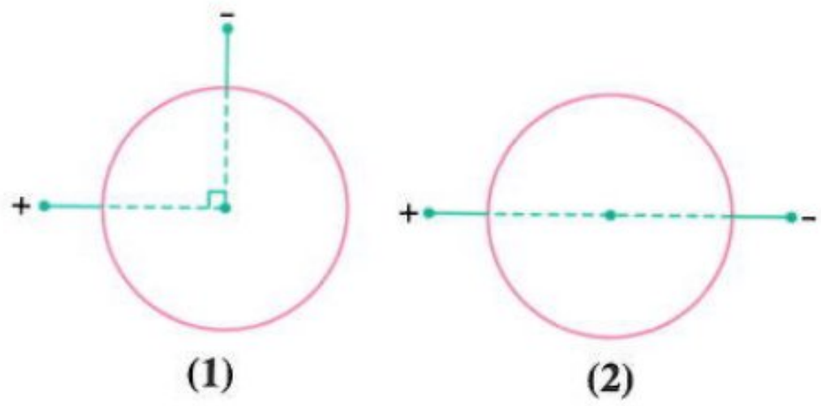
10 A wire of resistance 48 Ω is reshaped as a closed circular ring, then a battery is connected between two points on the circumference of the ring and along its diameter as in the figure. The equivalent resistance between points A, B is ..... .



(a) 96 Ω (b) 48 Ω (c) 24 Ω (d) 12 Ω

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11 A straight wire of resistance R is wrapped in the form of a circular ring then connected in two different methods to an electric source as in the opposite figures (1) and (2), so the ratio between the equivalent resistance in the two figures $\left(\frac{R_1}{R_2}\right)$ equals ..... .



(1) (2)

(a) $\frac{3}{4}$ (b) $\frac{4}{3}$ (c) $\frac{2}{1}$ (d) $\frac{1}{2}$

12 ✳ A wire of uniform cross-section carries a current of intensity 0.1 A when the potential difference between its terminals becomes 1.2 V. If the wire is bent into the shape of a square of consecutive vertices abcd, so the equivalent resistance of the wire if:

(i) the terminals of the source are connected to points a and c equals ..... .

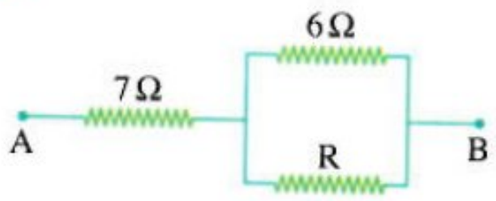
(a) 2 Ω (b) 3 Ω (c) 4 Ω (d) 6 Ω

(ii) the terminals of the source are connected to points a and d equals ..... .

(a) 6 Ω (b) 4.5 Ω (c) 2.25 Ω (d) 1.5 Ω

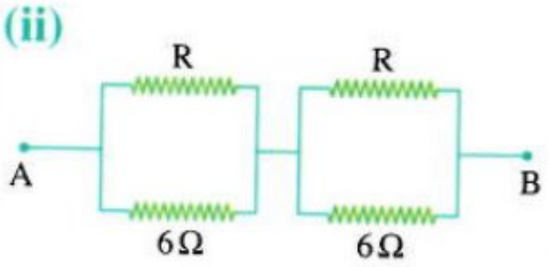
13 Each of the following figures illustrates a section of an electrical circuit in which the equivalent resistance between points A and B is equal to 9 Ω. So, what is the value of R in each of them?

(i)



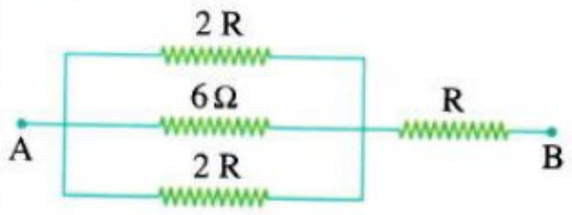
(a) 1 Ω (b) 2 Ω
(c) 3 Ω (d) 6 Ω

(ii)



(a) 4 Ω (b) 6 Ω
(c) 12 Ω (d) 18 Ω

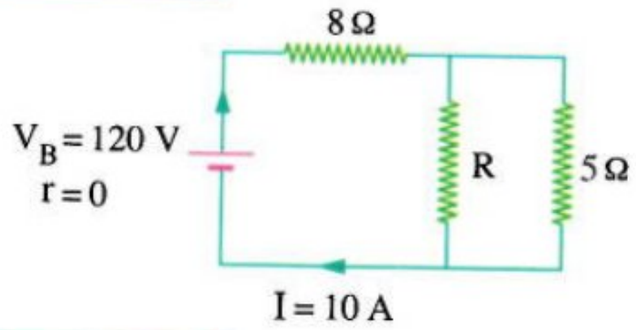
(iii)



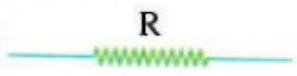
(a) 8 Ω (b) 6 Ω
(c) 5 Ω (d) 3 Ω

14 In the electric circuit shown in the figure, the value of R equals ..... .

(a) 20 Ω (b) 40 Ω
(c) 60 Ω (d) 80 Ω



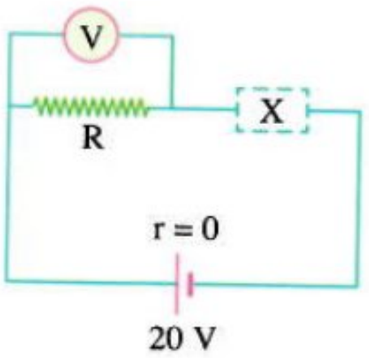
15 The opposite figure represents a closed electric circuit in which there is an unknown component X, if the reading of the voltmeter is 5 V, the component X is represented by the form ..... .

(a) 

(b) 

(c) 

(d) 



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16 ✱ A number of resistors (N), each of 40 ohms, are connected together either in series or in parallel with an electric source of potential difference 120 V, so an electric current of intensity 15 amperes passes through the circuit, so the number of resistors (N) equals ..... .

(a) 3 (b) 4 (c) 5 (d) 6

17 ✱ An electric circuit in a house consists of an electric source that has a terminal potential difference of 110 V, a fuse whose wire can't withstand a current greater than 5 A and other components of equivalent resistance that equals 2 Ω. What is the maximum number of similar lamps that each has a resistance of 620 Ω, where they can be turned on at the same time without damaging the fuse wire?

(a) 10 (b) 27 (c) 31 (d) 40

18 A number of equal resistors are connected in series, so their equivalent resistance was 100 Ω and when they are connected in parallel, their equivalent resistance becomes 4 Ω, then the resistance of each resistor of them is ..... .

(a) 20 Ω (b) 25 Ω (c) 100 Ω (d) 104 Ω

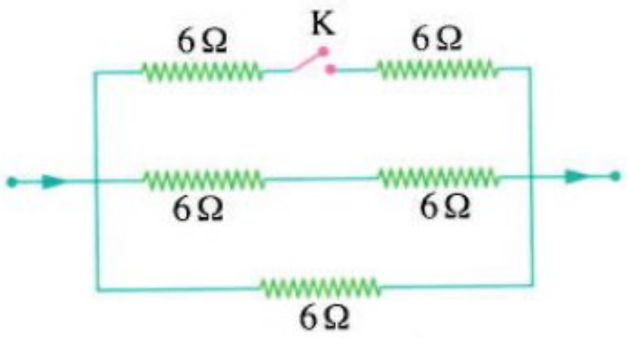
19 ✱ When two resistors R_1 and R_2 are connected in parallel, their equivalent resistance equals 6 Ω and when they are connected in series, their equivalent resistance equals 27 Ω, so the value of one of them is ..... .

(a) 8 Ω (b) 12 Ω (c) 15 Ω (d) 18 Ω

20 A set of n identical resistors are connected in series so that their equivalent resistance equals R_1 and when the same set of resistors are connected in parallel, their equivalent resistance equals R_2. The number of resistors (n) is equal to ..... .

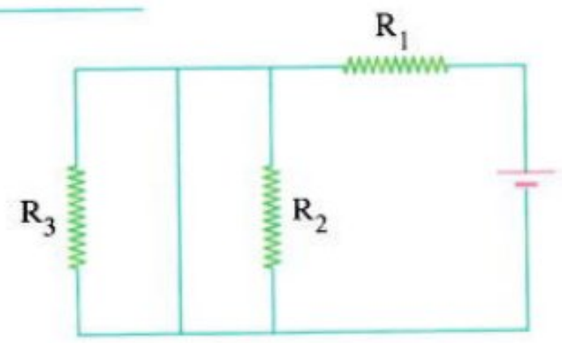
(a) $\frac{R_1}{R_2}$ (b) $\left(\frac{R_1}{R_2}\right)^2$ (c) $\sqrt{\frac{R_1}{R_2}}$ (d) $\sqrt{\frac{R_2}{R_1}}$

21 In the opposite electric circuit, the difference between the equivalent resistance of the circuit in the cases of opening and closing the switch (K) equals ..... .



(a) 1 Ω (b) 2 Ω (c) 4 Ω (d) 6 Ω

22 From the opposite circuit, through which resistor does an electric current pass?

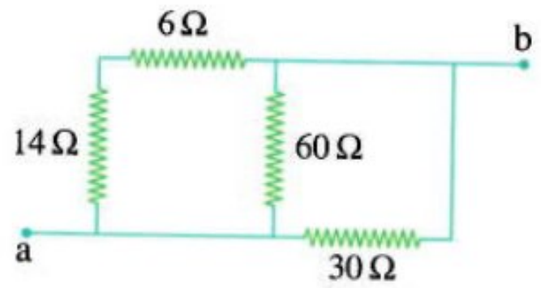


(a) R_1 only (b) R_1, R_2 (c) R_1, R_3 (d) R_1, R_2, R_3

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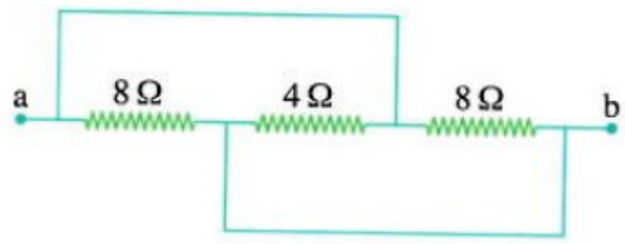
23 What is the equivalent resistance between points a and b in each of the following cases?

(i)



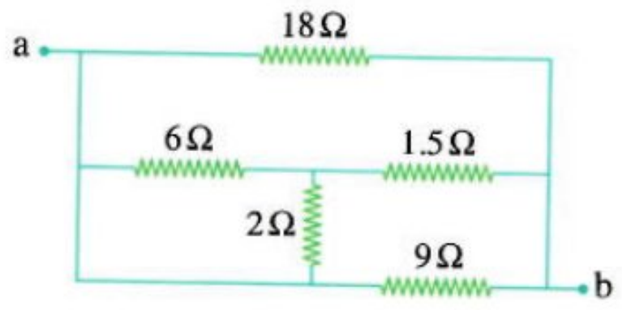
(a) 5 Ω (b) 10 Ω
(c) 30 Ω (d) 60 Ω

(ii)



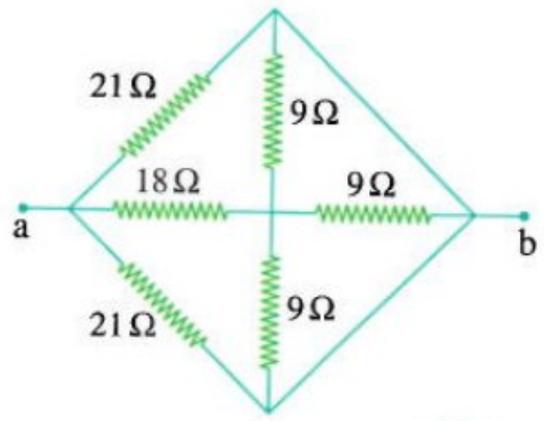
(a) 1 Ω (b) 2 Ω
(c) 3 Ω (d) 4 Ω

(iii)



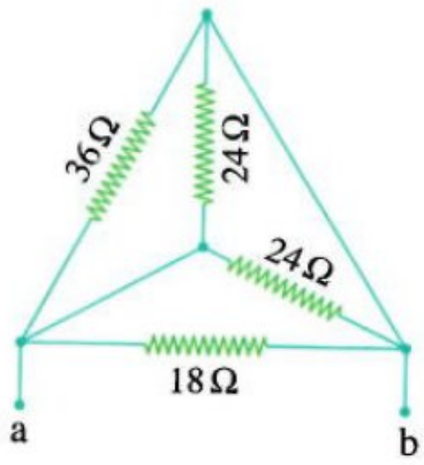
(a) 9 Ω (b) 6 Ω
(c) 3 Ω (d) 2 Ω

(iv)



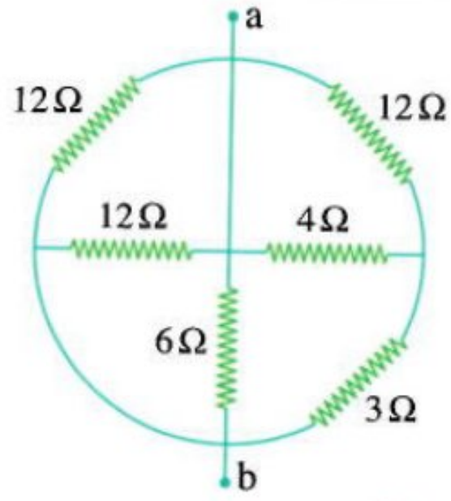
(a) 3 Ω (b) 7 Ω
(c) 9 Ω (d) 11 Ω

(v)



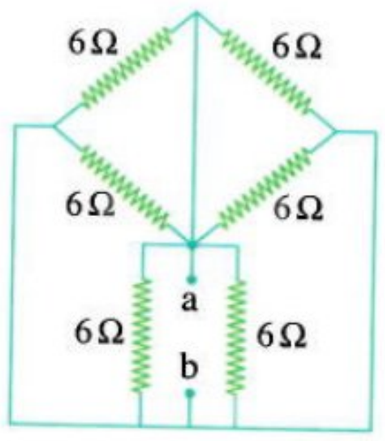
(a) 3 Ω (b) 5 Ω
(c) 6 Ω (d) 8 Ω

(vi)



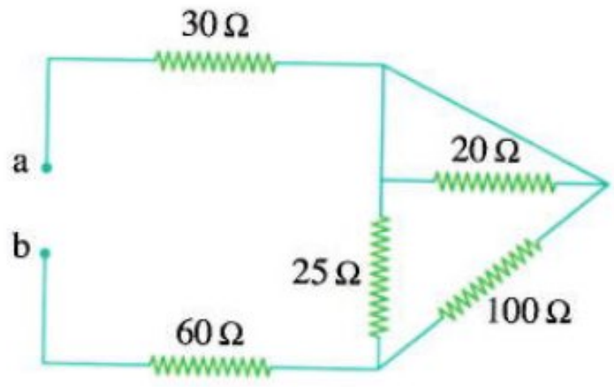
(a) 1 Ω (b) 1.5 Ω
(c) 2 Ω (d) 2.5 Ω

(vii)



(a) 1 Ω (b) 3 Ω
(c) 6 Ω (d) 12 Ω

(viii)

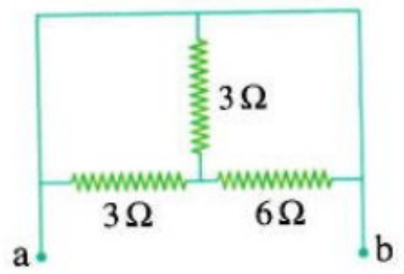


(a) 45 Ω (b) 72 Ω
(c) 90 Ω (d) 110 Ω

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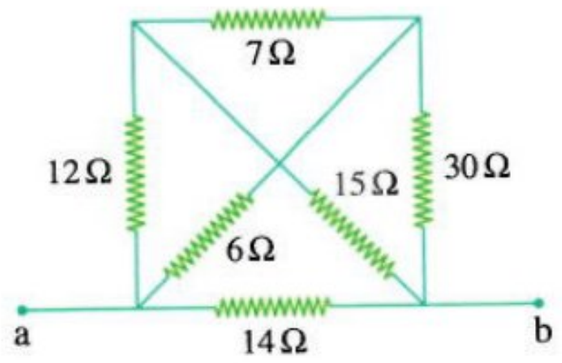
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(ix)



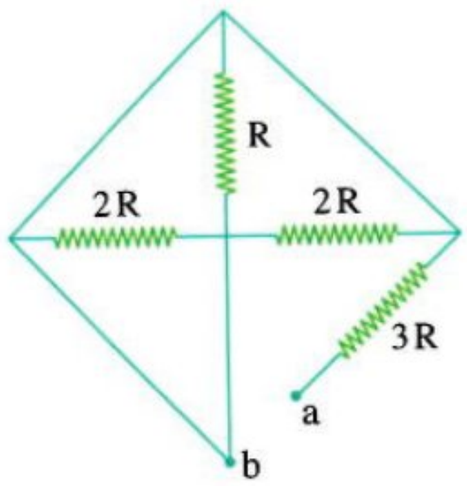
(a) 0 Ω (b) 1 Ω
(c) 5 Ω (d) 2 Ω

(x)



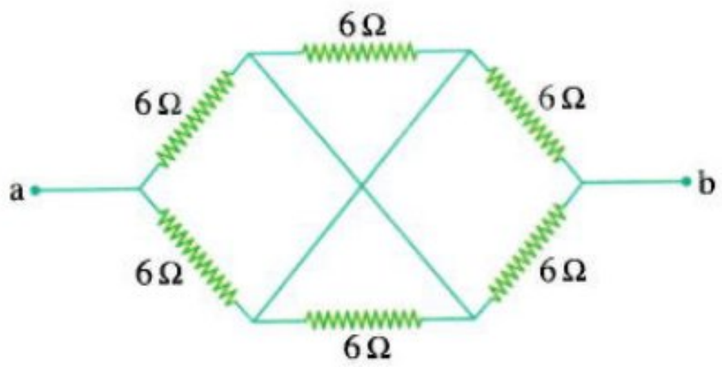
(a) 4 Ω (b) 7 Ω
(c) 14 Ω (d) 15 Ω

(xi)



(a) R (b) 2 R
(c) 3 R (d) 6 R

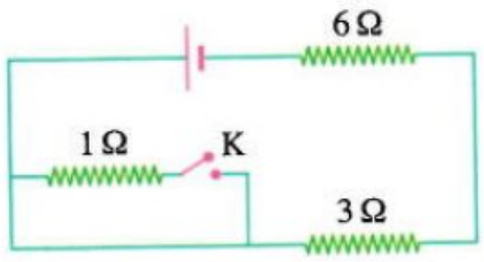
(xii)



(a) 18 Ω (b) 12 Ω
(c) 9 Ω (d) 6 Ω

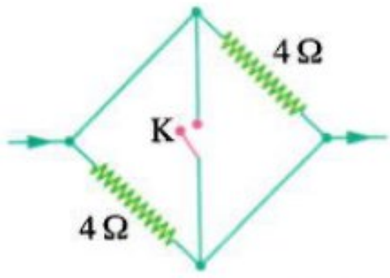
24 What is the equivalent resistance before and after closing the switch K respectively in each of the following cases?

(i)



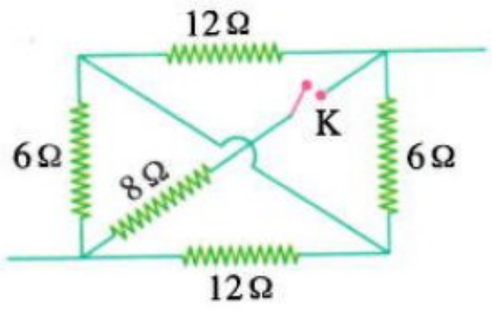
(a) 9 Ω, 10 Ω (b) 9 Ω, 9 Ω
(c) 6 Ω, 3 Ω (d) 6 Ω, 6 Ω

(ii)



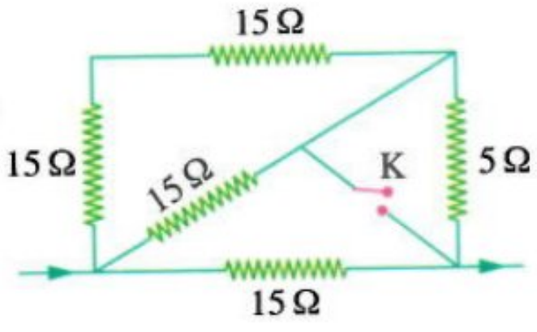
(a) 0, 2 Ω (b) 2 Ω, 0
(c) 8 Ω, 0 (d) 4 Ω, 0

(iii)



(a) 8 Ω, 2 Ω (b) 9 Ω, 4 Ω
(c) 8 Ω, 4 Ω (d) 36 Ω, 6 Ω

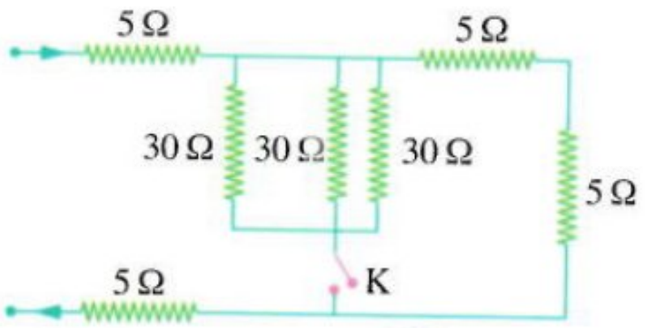
(iv)



(a) 7.5 Ω, 6 Ω (b) 3 Ω, 6 Ω
(c) 15 Ω, 7.5 Ω (d) 7.5 Ω, 3.5 Ω

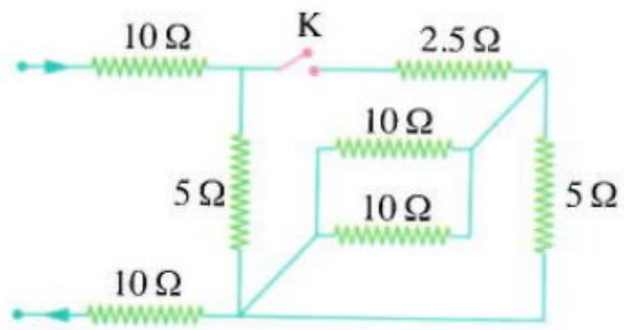
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(v)



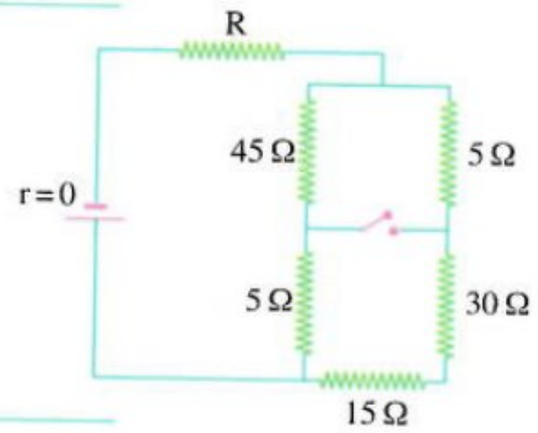
(a) 10 Ω, 20 Ω
(b) 15 Ω, 20 Ω
(c) 20 Ω, 15 Ω
(d) 20 Ω, 10 Ω

(vi)



(a) 10 Ω, 22.5 Ω
(b) 25 Ω, 22.5 Ω
(c) 25 Ω, 7.5 Ω
(d) 30 Ω, 15 Ω

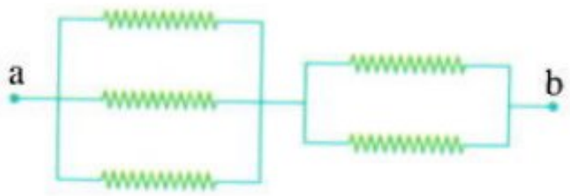
25 ✳ In the opposite circuit, when the switch is closed, the equivalent resistance of the circuit decreases to half its initial value, hence the resistance of R equals ..... .

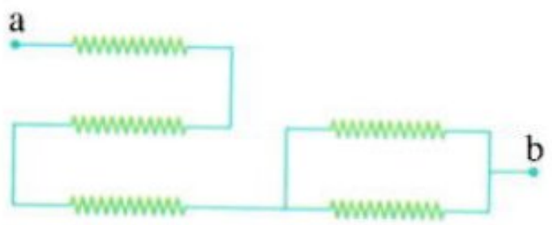


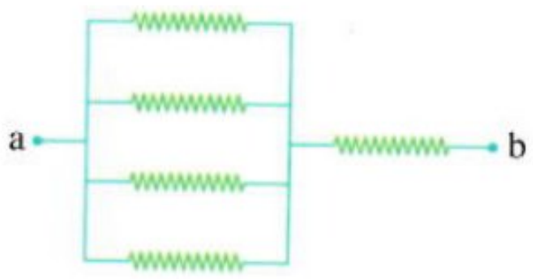
(a) 3 Ω
(b) 5 Ω
(c) 7 Ω
(d) 11 Ω

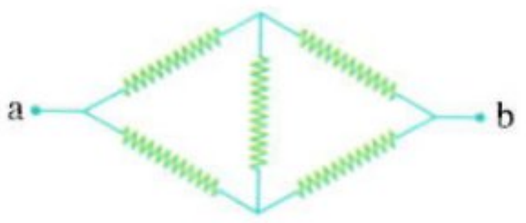
26 Five identical resistors are connected together as shown in the opposite figure so that their equivalent resistance is found to be R_o. If the five resistors are reconnected in different ways, which of the following connection ways would make the equivalent resistance of the group equal to $\frac{1}{2} R_o$?



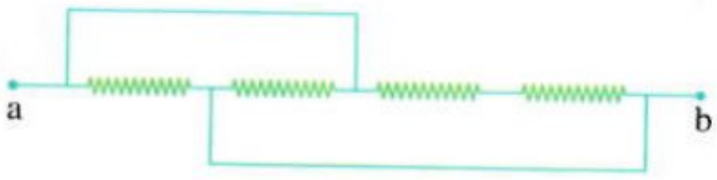
(a)


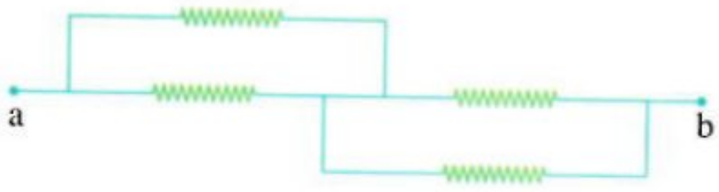
(b)


(c)


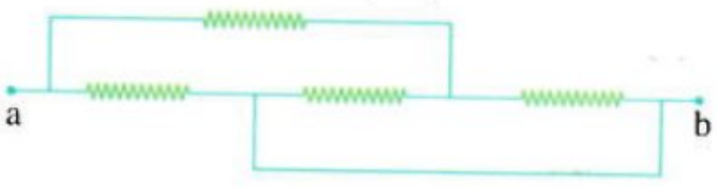
(d)


27 ✳ Four identical resistors are connected in different ways as shown in the following figures. In which of these figures is the equivalent resistance between points a and b the largest?

(a)


(b)


(c)

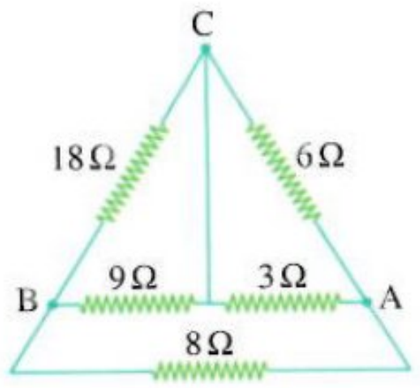

(d)


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28 The equivalent resistance of the set of resistors shown in the opposite figure when the electric source is connected between:



(i) points A and B equals ..... .

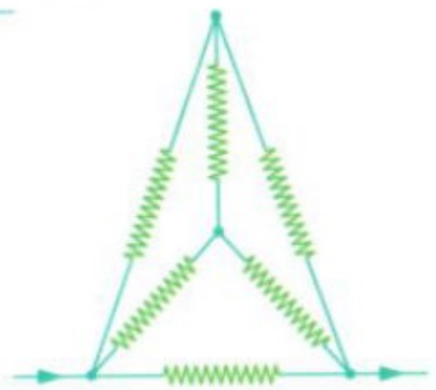
(a) 2 Ω (b) 4 Ω

(c) 6 Ω (d) 8 Ω

(ii) points C and B equals ..... .

(a) 3.75 Ω (b) 6 Ω (c) 7.5 Ω (d) 10 Ω

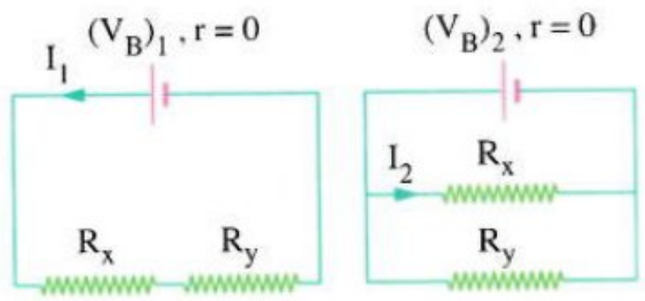
29 ✱ In the electric circuit shown in the figure, if the resistance of each resistor = R, the equivalent resistance of the circuit = ...



(a) 6 R (b) 3 R

(c) $\frac{R}{2}$ (d) $\frac{R}{6}$

30 In the two opposite electric circuits, the current intensities I_1 and I_2 are equal, so the ratio between the two electromotive forces of the two batteries $\left(\frac{(V_B)_1}{(V_B)_2}\right)$ is equal to ..... .

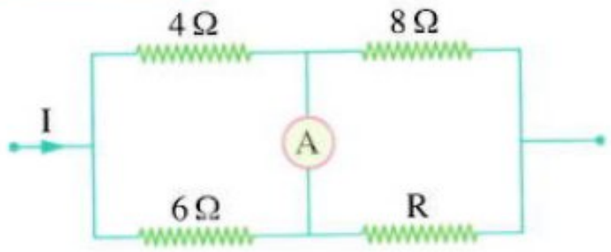


Circuit (1) Circuit (2)

(a) $\frac{R_y}{R_x}$ (b) $\frac{R_x}{R_y}$

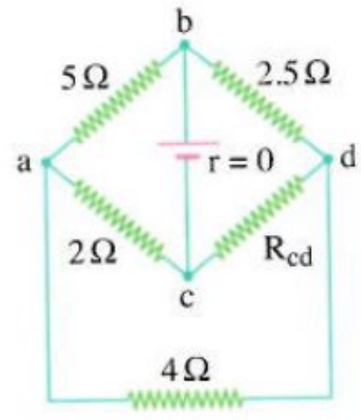
(c) $\frac{R_x + R_y}{R_y}$ (d) $\frac{R_x + R_y}{R_x}$

31 ✱ The opposite figure represents a part of an electric circuit, if the ammeter reading equals zero, the resistance of resistor R equals ..... .



(a) 8 Ω (b) 9 Ω (c) 12 Ω (d) 24 Ω

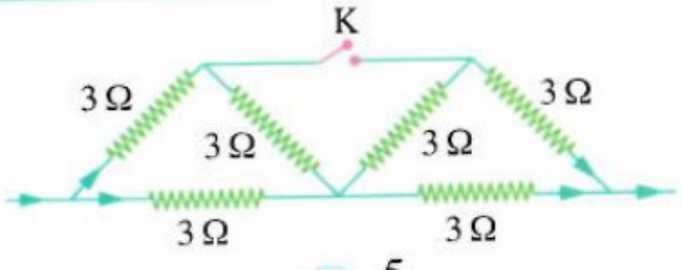
32 In the opposite electric circuit, what is the value of resistance R_{cd} if the potential difference across the 4 Ω resistance is zero?



(a) 0.5 Ω (b) 1 Ω

(c) 2 Ω (d) 4 Ω

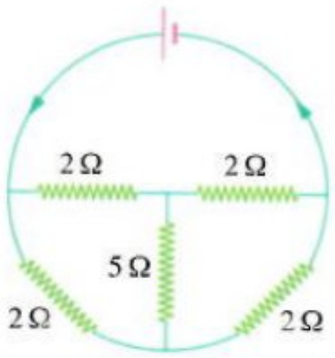
33 In the opposite electric circuit, the ratio between the equivalent resistance of the circuit in the cases of opening and closing the switch (K) equals ..... .



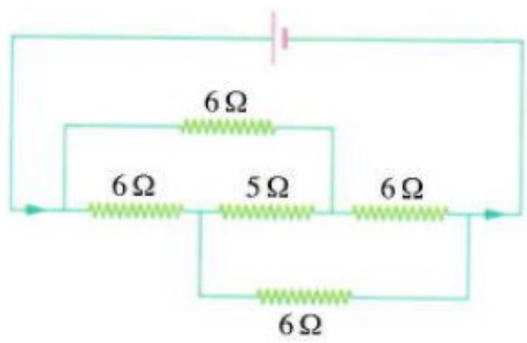
(a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{4}{3}$ (d) $\frac{5}{2}$

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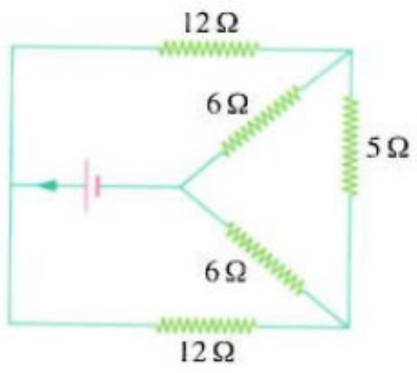
34 The following figures represent four electric circuits. So, the circuits in which the resistor of the value 5 Ω carries no current are ..... .



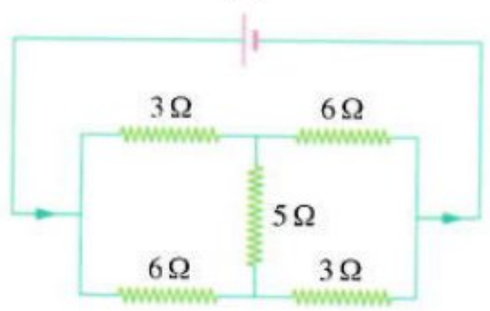
(1)



(2)



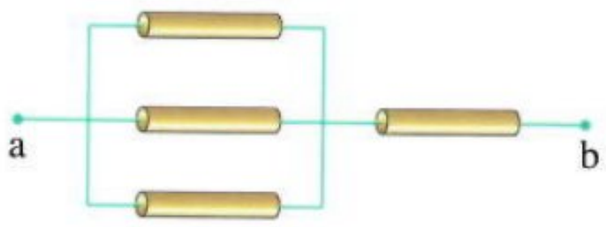
(3)



(4)

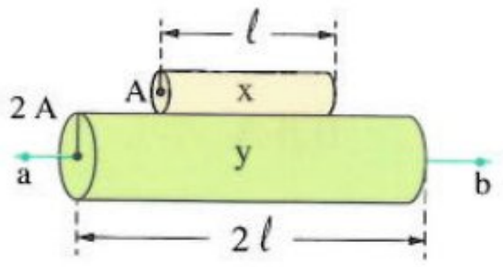
a (1) and (2) only
b (3) and (4) only
c (1), (2) and (3) only
d (2), (3) and (4) only

35 A metal wire of uniform cross-section and electrical resistance R was divided into four identical parts, and these parts were connected together as shown in the opposite figure, so the equivalent resistance of the group between points (a) and (b) equals ..... .



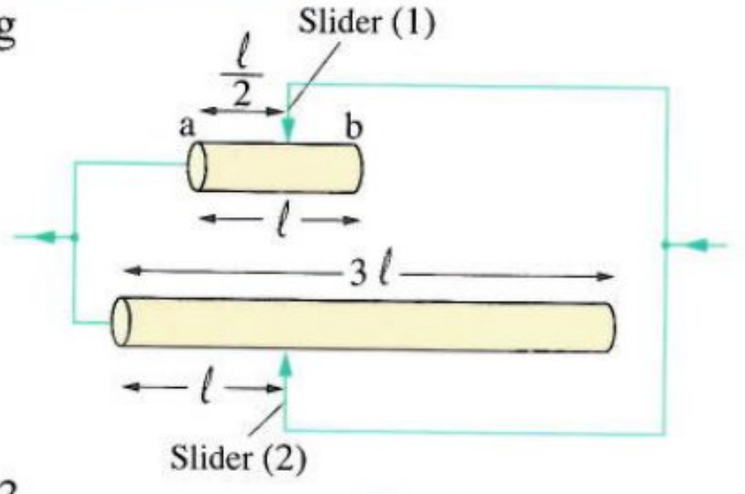
a $\frac{R}{12}$
b $\frac{R}{6}$
c $\frac{R}{4}$
d $\frac{R}{3}$

36 Two non-insulated conductors x and y of uniform cross-section, have lengths of ℓ and 2ℓ, cross-sectional areas of A and 2A, and electrical conductivities 8σ and σ respectively. The conductors are placed in perfect contact as shown in the opposite figure, so if the electrical resistance of conductor x is 5 Ω, the equivalent resistance of them when connecting an electrical source between points a and b is equal to ..... .



a 7.5 Ω
b 14 Ω
c 20 Ω
d 24 Ω

37 Two conductors are made of the same material having uniform cross-sections of the same cross-sectional area. They are connected together using two sliders (1) and (2) in an electric circuit as shown in the opposite figure so that their equivalent resistance equals R. Then, when moving slider (1) to position (b), the equivalent resistance becomes ..... .



a $\frac{R}{2}$
b $\frac{2}{3}$ R
c $\frac{3}{2}$ R
d 2 R

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38 * In figure (1), a wire of uniform cross-section has length (ℓ) and resistance (R). If it is cut into two halves and one of the halves is drawn and then the two halves are reconnected together so that the total length of the wire becomes (3ℓ) as shown in figure (2). Then the total resistance of the wire in figure (2) is equal to ..... .

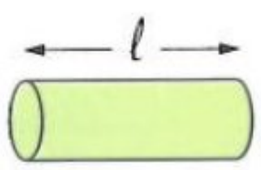


Figure (1)

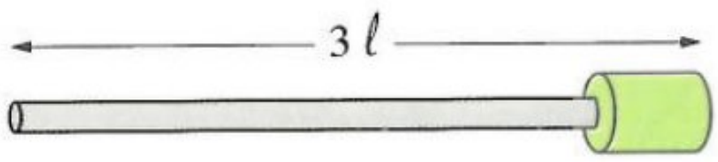
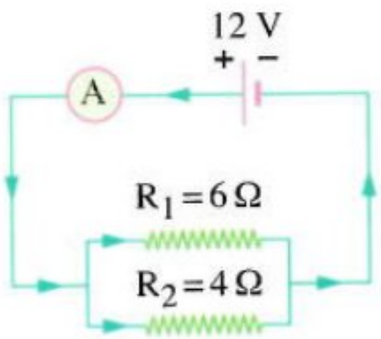


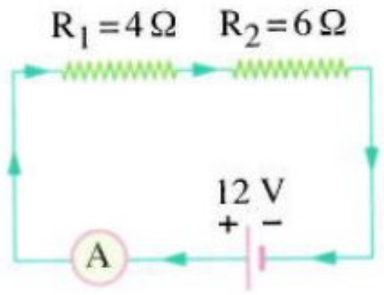
Figure (2)

a 12.5 R b 13 R c 25 R d 25.5 R

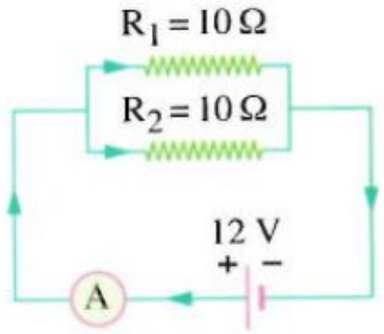
39 In which of the following electric circuits the intensity of the electric current which passes in one of the two resistors differs from the other resistor?



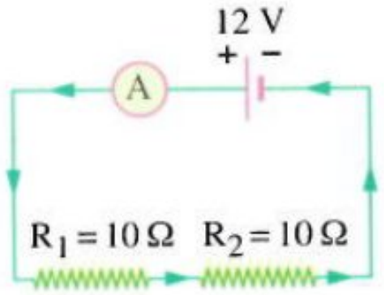
a



b

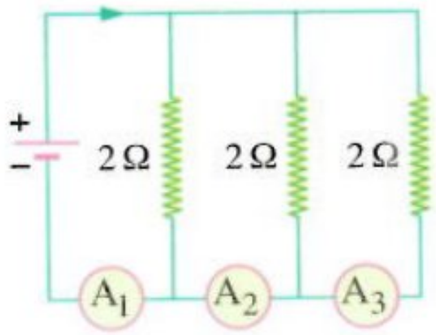


c



d

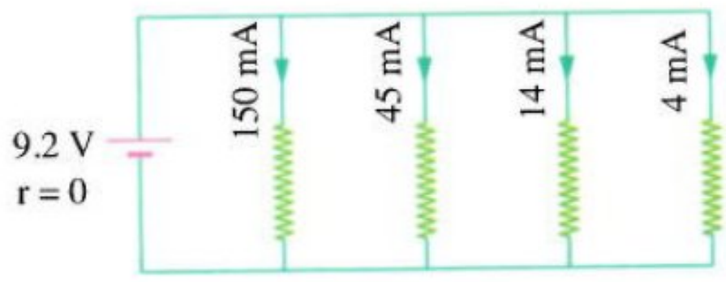
40 In the electric circuit shown in the figure, if the reading of the ammeter A_1 equals 1.2 A, the reading of the ammeter A_2 equals ..... .



a 0.2 A b 0.4 A

c 0.6 A d 0.8 A

41 Four resistors are connected in parallel to a battery of emf 9.2 V, hence the current intensities through them are 4 mA, 14 mA, 45 mA and 150 mA as shown in the opposite circuit, so:



(i) The total resistance of the circuit equals ..... .

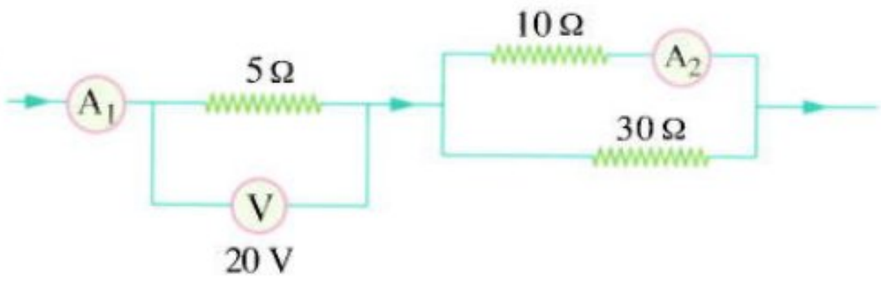
a 9.71 Ω b 13.35 Ω c 29.53 Ω d 43.19 Ω

(ii) If the largest resistance is replaced with another of double its value, the total current intensity through the battery, will equal ..... .

a 105.5 mA b 211 mA c 422 mA d 633 mA

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42 In the electric circuit shown in the figure:



(i) The reading of ammeter A_1 equals ..... .

(a) 2 A (b) 4 A

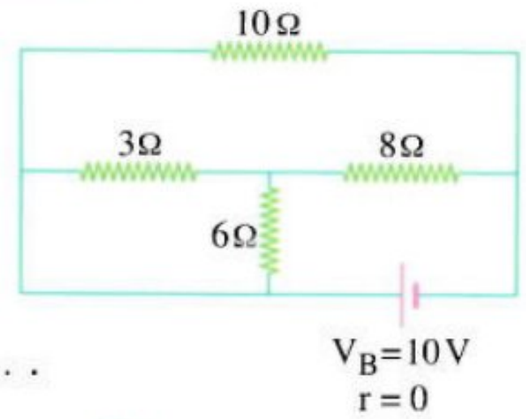
(c) 6 A (d) 8 A

(ii) The reading of ammeter A_2 equals ..... .

(a) 1 A (b) 2 A (c) 3 A (d) 4 A

---

43 ✳ In the opposite circuit:



(i) The equivalent resistance of the circuit equals ..... .

(a) 5 Ω (b) 9 Ω

(c) 11 Ω (d) 13 Ω

(ii) The total current intensity passing in the circuit equals ..... .

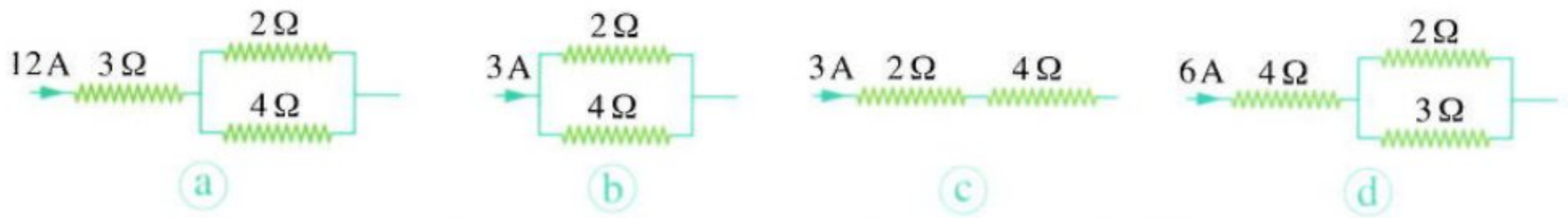
(a) 0.5 A (b) 2 A (c) 3.5 A (d) 4 A

(iii) The current intensity passing through the 6 Ω resistor equals ..... .

(a) 0.13 A (b) 0.91 A (c) 0.33 A (d) 0.45 A

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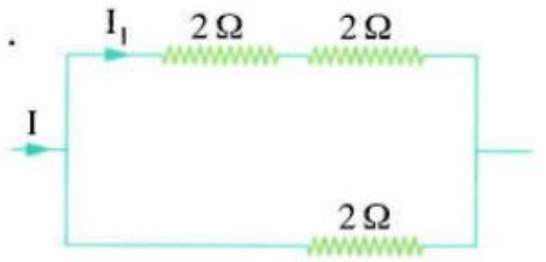
44 In which of the following figures, the potential difference across the 4 Ω resistor equals 4 V?



(a) (b) (c) (d)

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45 In the opposite electric circuit, the ratio of $\left(\frac{I_1}{I}\right)$ equals ..... .



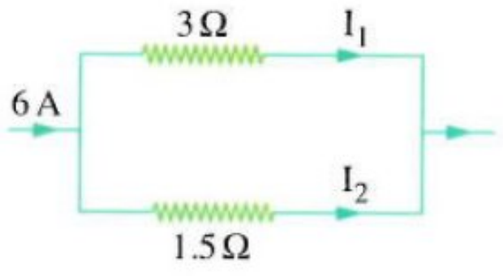
(a) $\frac{1}{2}$ (b) $\frac{1}{3}$

(c) $\frac{1}{4}$ (d) $\frac{1}{6}$

---

46 Each of the following figures represents section of electric circuit. What are the intensities of currents I_1 and I_2 respectively?

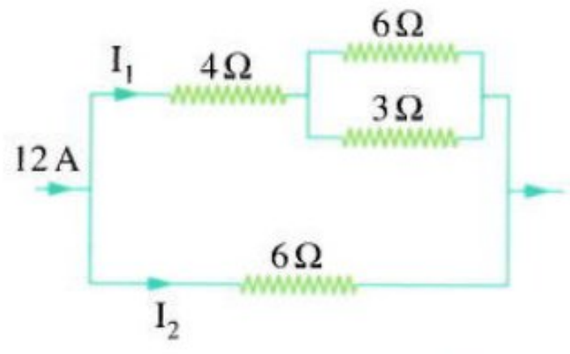
(i)



(a) 1 A, 5 A (b) 2 A, 4 A

(c) 4 A, 2 A (d) 5 A, 1 A

(ii)



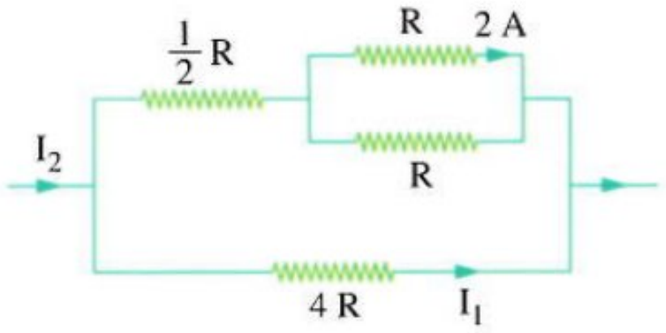
(a) 3 A, 6 A (b) 3 A, 9 A

(c) 6 A, 6 A (d) 6 A, 9 A

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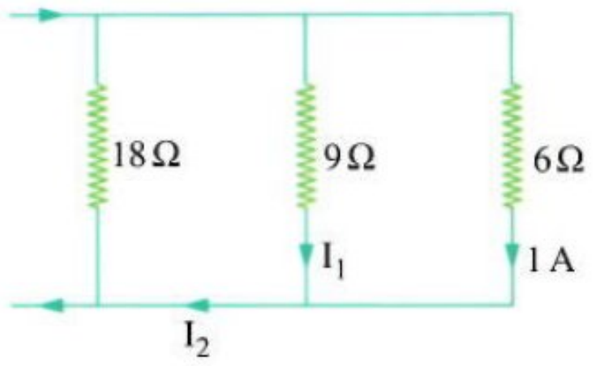
(iii)



(a) 4 A, 6 A (b) 4 A, 3 A

(c) 1 A, 4 A (d) 1 A, 5 A

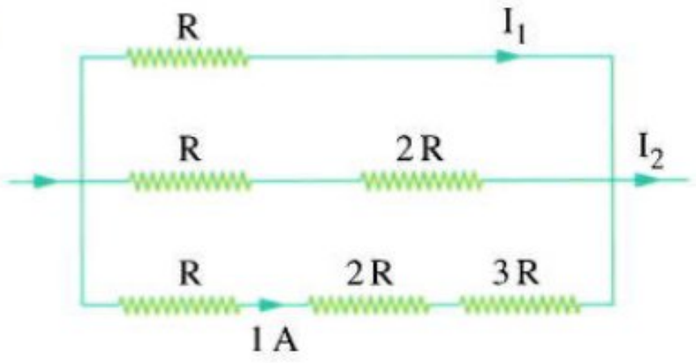
(iv)



(a) $\frac{1}{2}$ A, $\frac{3}{2}$ A (b) $\frac{1}{2}$ A, $\frac{5}{3}$ A

(c) $\frac{2}{3}$ A, $\frac{5}{3}$ A (d) $\frac{2}{3}$ A, $\frac{1}{2}$ A

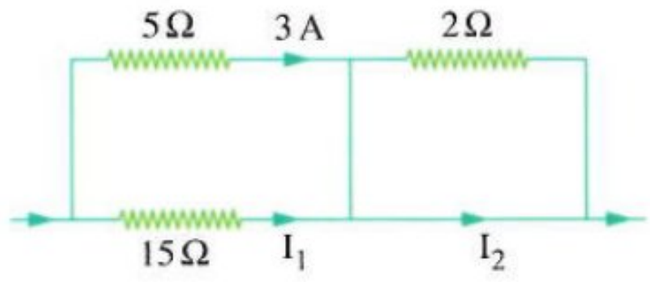
(v)



(a) 6 A, 5 A (b) 6 A, 9 A

(c) 12 A, 5 A (d) 12 A, 9 A

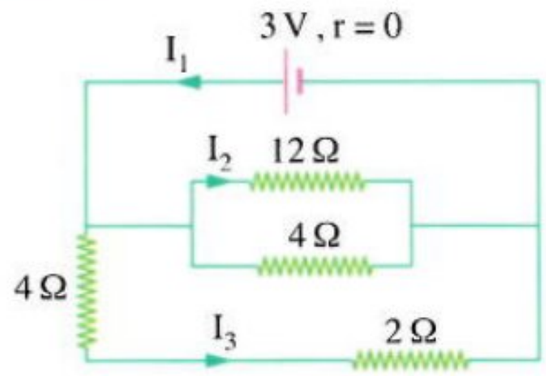
(vi)



(a) 0.5 A, 2 A (b) 1 A, 2 A

(c) 0.5 A, 4 A (d) 1 A, 4 A

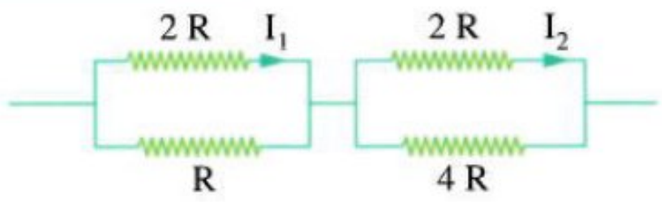
47 In the opposite electric circuit, the ratio of $\left(\frac{I_2}{I_3}\right)$ equals ..... .



(a) $\frac{1}{4}$ (b) $\frac{1}{3}$

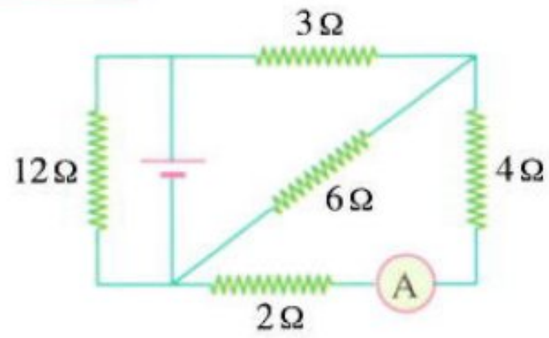
(c) $\frac{1}{2}$ (d) $\frac{3}{4}$

48 The opposite figure represents a part of an electric circuit, then the ratio between the two currents $\left(\frac{I_1}{I_2}\right)$ equals ..... .



(a) $\frac{1}{1}$ (b) $\frac{2}{1}$ (c) $\frac{1}{2}$ (d) $\frac{1}{4}$

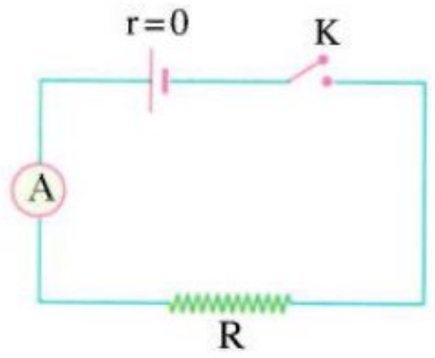
49 In the opposite figure, if the ammeter reading is 1 A, then the electric current intensity through the 12 Ω resistor equals ..... .



(a) 0.5 A (b) 1 A

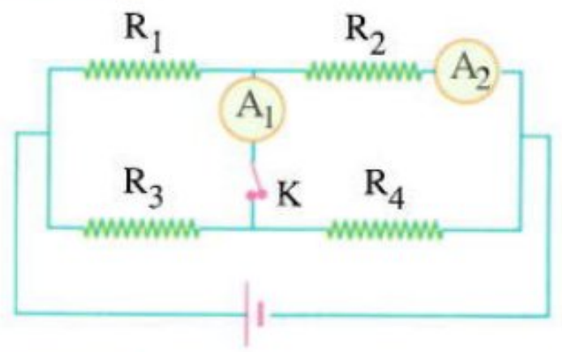
(c) 1.5 A (d) 2 A

50 In the opposite simple electric circuit, the ammeter reads 5 A when the switch (K) is closed but when a 2 Ω resistor is connected in series with resistor R in this circuit, the ammeter reading gets decreased to 4 A. So, the value of the resistor R is ..... .



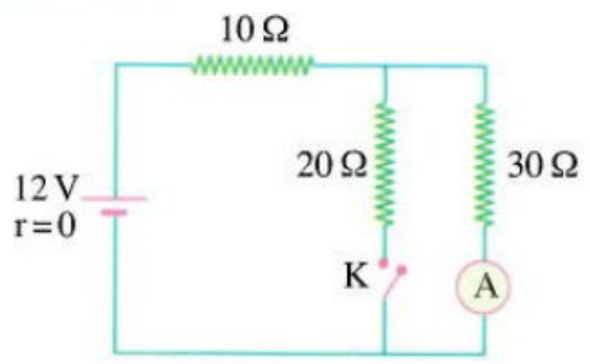
(a) 4 Ω (b) 8 Ω (c) 12 Ω (d) 16 Ω

51 In the opposite circuit if the reading of the ammeter A_1 equals zero, then when the switch K is opened the reading of the ammeter A_2 ..... .



(a) increases (b) decreases
(c) remains unchanged (d) becomes zero

52 * From the opposite circuit:



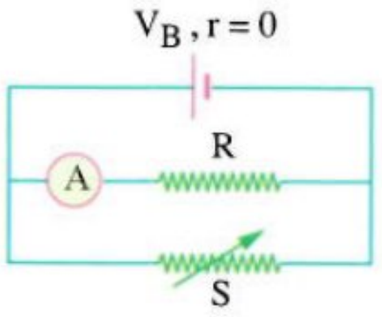
(i) What is the reading of the ammeter when the switch K is open?

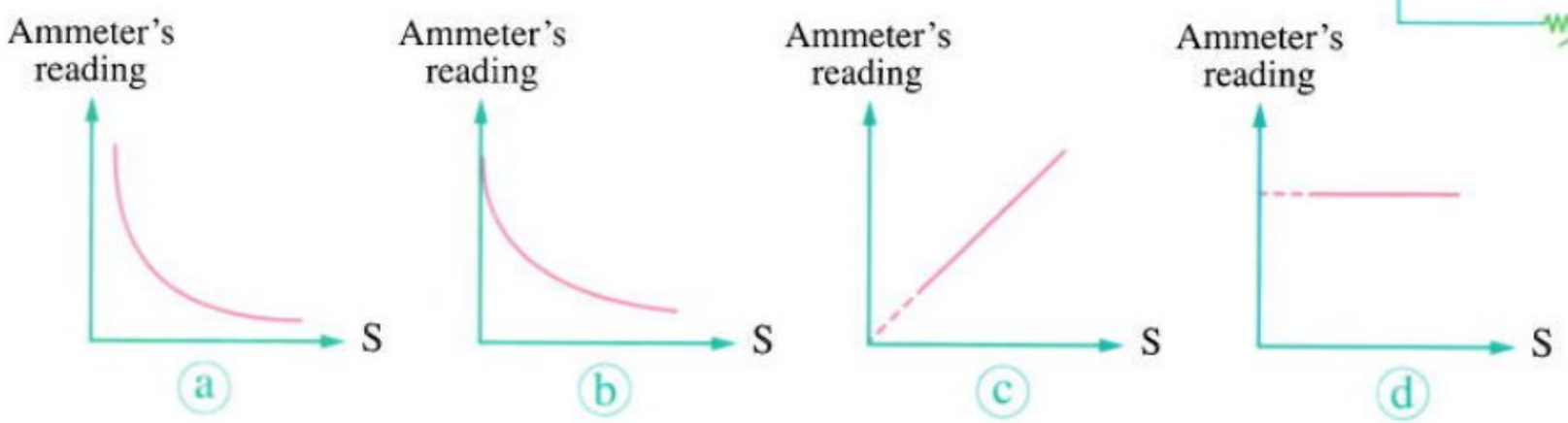
(a) 0.21 A (b) 0.3 A
(c) 0.43 A (d) 1.6 A

(ii) What is the reading of the ammeter when the switch K is closed?

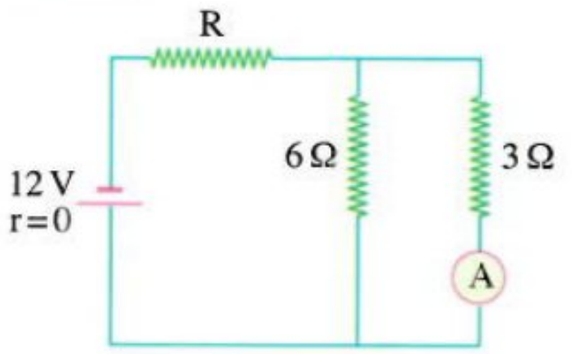
(a) 5.4 A (b) 1.52 A (c) 0.34 A (d) 0.22 A

53 Which of the following graphs represents the relation between the ammeter's reading and the value of variable resistance S?





54 * If the ammeter reading in the opposite circuit is 2 A, so:



(i) The current intensity that passes in the circuit equals ..... .

(a) 6 A (b) 9 A
(c) 5 A (d) 3 A

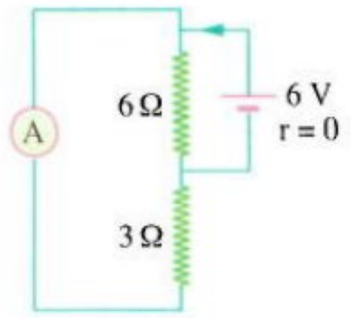
(ii) The resistance of R equals ..... .

(a) 4 Ω (b) 2 Ω (c) 1 Ω (d) 0.5 Ω

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55 In the opposite electric circuit, the reading of the ammeter (A) equals ..... .



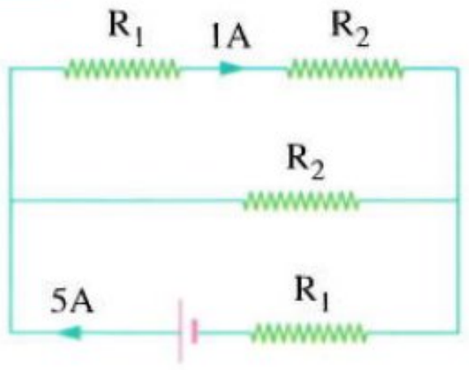
(a) 1 A (b) 2 A

(c) 3 A (d) 4 A

56 ✳ Three resistors of 1 Ω, 3 Ω and 6 Ω are connected to an electric source where the electric current intensity through them are 0.3 A, 0.2 A, 0.1 A respectively, so the equivalent resistance of the circuit equals ..... .

(a) 1 Ω (b) 2 Ω (c) 3 Ω (d) 4 Ω

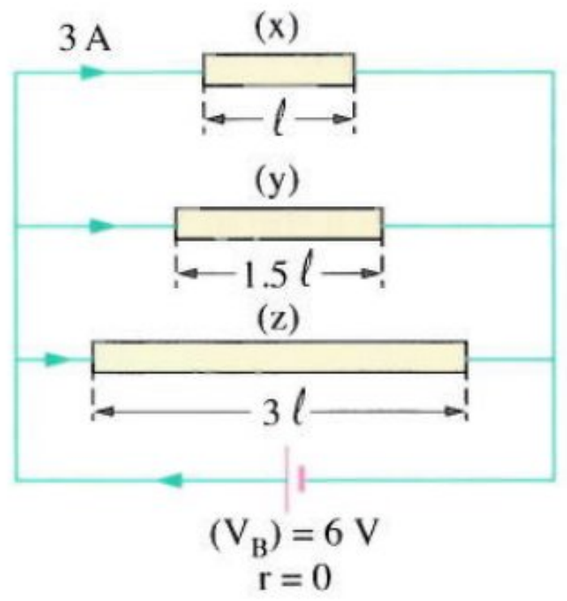
57 In the opposite circuit, the ratio $\left(\frac{R_1}{R_2}\right)$ is ..... .



(a) $\frac{3}{2}$ (b) $\frac{3}{1}$

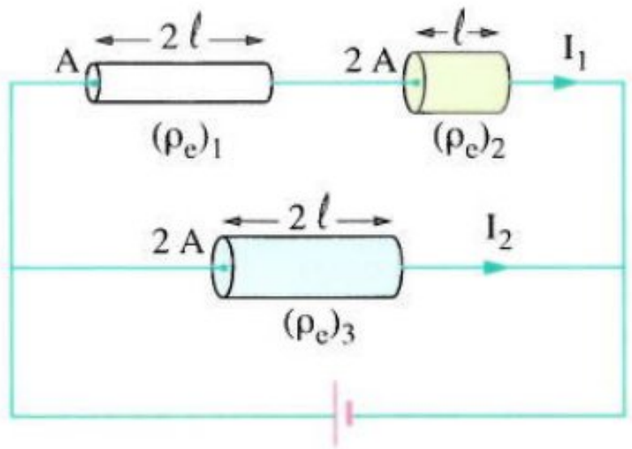
(c) $\frac{4}{1}$ (d) $\frac{4}{3}$

58 Three conductors (x), (y) and (z) made of the same metal and having equal cross-sectional areas but different lengths are connected together in parallel and then to a battery with an electromotive force of 6 V and negligible internal resistance, as shown in the opposite circuit. If the current flowing through conductor (x) is 3A, then the current flowing through the circuit if the conductors are connected in series and then to the same battery is equal to ..... .



(a) $\frac{2}{11}$ A (b) $\frac{3}{11}$ A (c) $\frac{6}{11}$ A (d) $\frac{9}{11}$ A

59 In the opposite electric circuit, if $I_1 = I_2$, then .....



(a) $(\rho_e)_3 = (\rho_e)_1 + (\rho_e)_2$

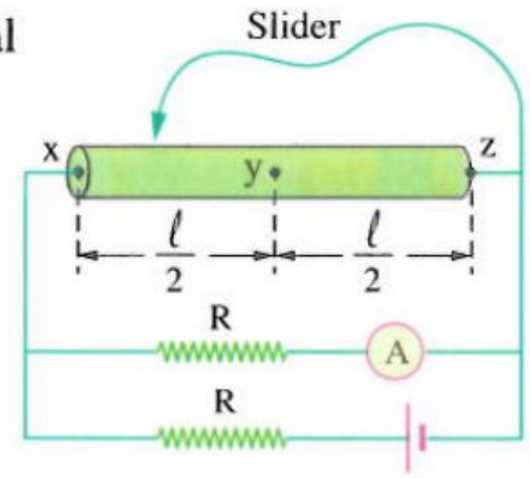
(b) $(\rho_e)_3 = (\rho_e)_1 + 2\,(\rho_e)_2$

(c) $(\rho_e)_3 = 2\,(\rho_e)_1 + \frac{(\rho_e)_2}{2}$

(d) $(\rho_e)_3 = \sqrt{(\rho_e)_1 + (\rho_e)_2}$

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60 A metal rod (xyz) of uniform cross-section, length ℓ and electrical resistance R is connected in an electrical circuit as shown in the opposite figure. At which position on the rod should the slider be placed to get the largest possible ammeter reading?



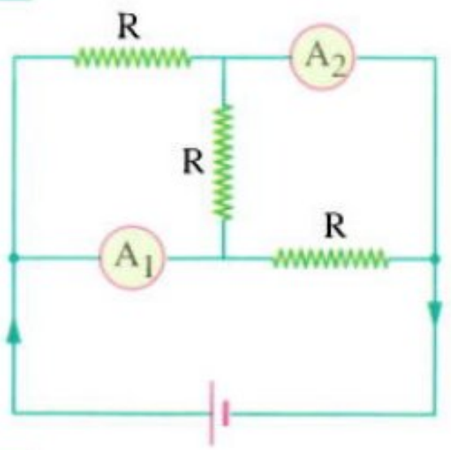
(a) At position (x)

(b) At position (y)

(c) At position (z)

(d) The ammeter readings are equal at all three positions.

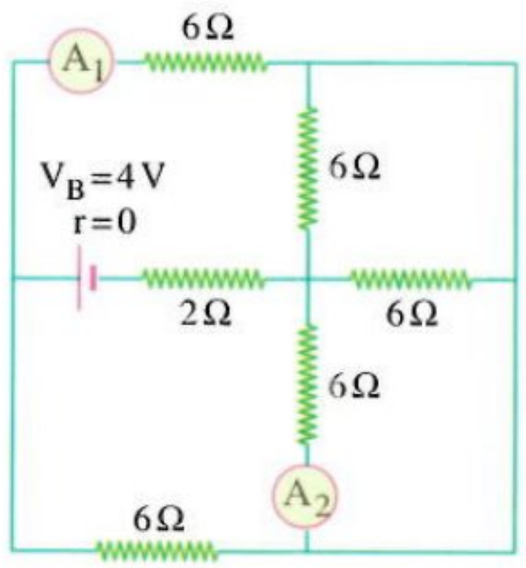
61 In the opposite circuit, the ratio between the readings of ammeters $\left(\frac{A_1}{A_2}\right)$ is ..... .



(a) $\frac{1}{2}$ (b) $\frac{2}{3}$

(c) $\frac{1}{1}$ (d) $\frac{3}{2}$

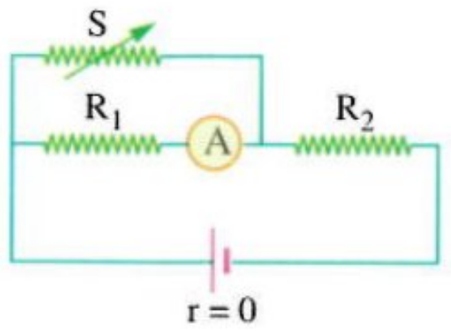
62 The ratio between the readings of the two ammeters $\left(\frac{A_1}{A_2}\right)$ in the opposite electrical circuit equals ..... .



(a) $\frac{1}{1}$ (b) $\frac{2}{1}$

(c) $\frac{3}{1}$ (d) $\frac{3}{2}$

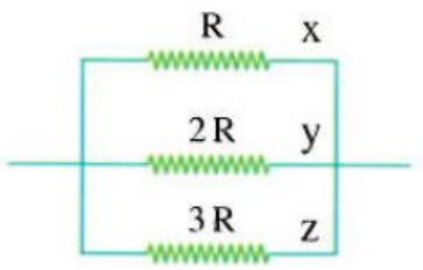
63 In the opposite electric circuit, when the value of the variable resistor S increases, the ammeter reading ..... .



(a) increases (b) vanishes

(c) remains constant (d) decreases but does not vanish

64 The opposite figure represents three resistors x, y and z that are connected in parallel, so the ratio of the current intensity that passes in each of them respectively is ..... .



(a) 6 : 3 : 2 (b) 3 : 2 : 1

(c) 2 : 3 : 1 (d) 2 : 3 : 6

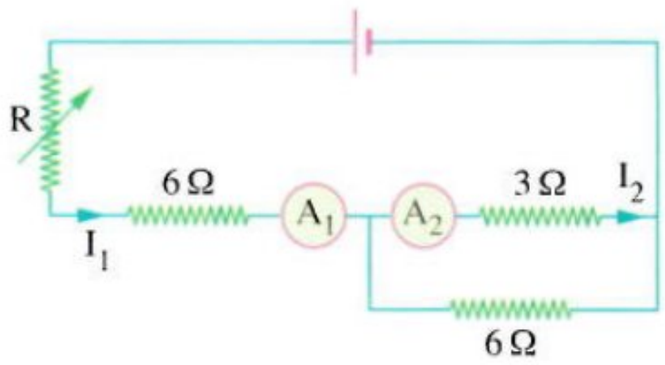
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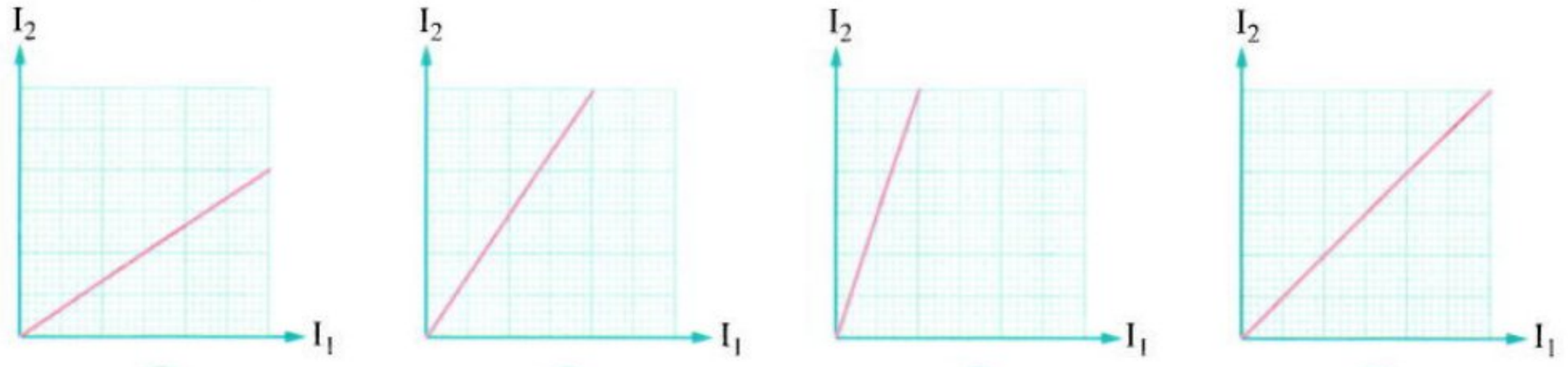
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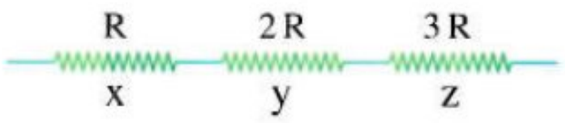
65 * Which of the following graphs represents the relation between the readings of the two ammeters A_1 and A_2 when changing the resistance of rheostat (R)?

(Notice that: I_1, I_2 are drawn with the same scale)



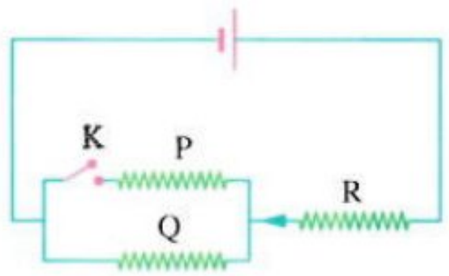


66 The opposite figure represents three resistors x, y and z that are connected in series, the ratio between the potential differences across each of them respectively equals ..... .



(a) 3 : 2 : 1 (b) 2 : 3 : 1 (c) 1 : 1 : 1 (d) 1 : 2 : 3

67 In the opposite electric circuit, three identical resistors are connected to an electric source, so when switch K is closed, ...



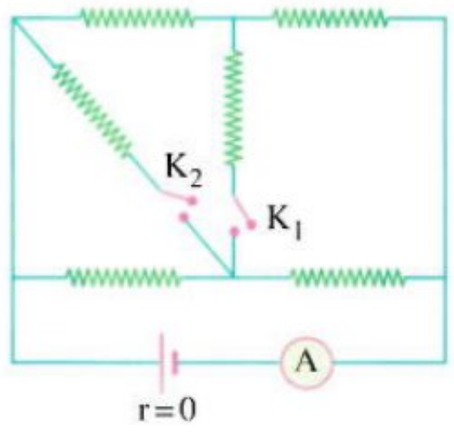
(a) the current in R decreases and in Q increases

(b) the current in R decreases and in Q decreases

(c) the current in R increases and in Q decreases

(d) the current in R increases and in Q increases

68 In the opposite electric circuit, if the electrical resistors are identical, what happens to the ammeter reading when closing:



	switch K_1 only?	switch K_2 only?
(a)	Decreases	Increases
(b)	Decreases	Doesn't change
(c)	Doesn't change	Increases
(d)	Doesn't change	Doesn't change

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69 ✳ Wire ab carries an electric current of intensity 3 mA. The wire is connected in parallel to another wire of the same material and the same length but its diameter is three times as large as the first wire (ab), so the total current intensity which is required to be passed to keep the potential difference between the terminals of wire ab constant is ..... .

(a) 0.02 A (b) 0.1 A (c) 0.03 A (d) 0.5 A

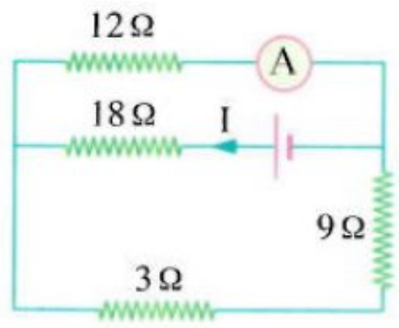
70 ✳ Two wires A and B which have the same length are made of the same material, the cross-sectional area of wire A is double that of B. The two wires are connected together in parallel in an electric circuit and when the circuit is closed, the intensity of the passing current in the circuit becomes 3 A, so the current intensity through each of them; I_A and I_B, is ..... respectively.

(a) 2 A, 1 A (b) 2 A, 3 A (c) 3 A, 3 A (d) 2 A, 2 A

71 ✳ An electric current of intensity 8 mA was passing through a thin metallic wire and when another wire of the same metal and the same length is connected in parallel with the first wire, the current has increases to 10 mA. To keep the potential difference across the wire unchanged, the ratio between the radii of the two wires $\left(\frac{r_1}{r_2}\right)$ must be ..... .

(a) $\frac{1}{2}$ (b) $\frac{3}{2}$ (c) $\frac{2}{1}$ (d) $\frac{5}{3}$

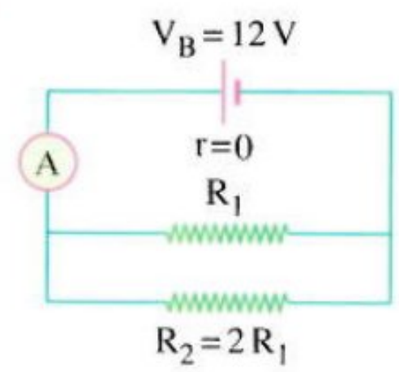
72 In the opposite electric circuit, the ammeter reading equals ..... .



(a) I (b) $\frac{I}{2}$

(c) $\frac{I}{3}$ (d) $\frac{I}{6}$

73 In the opposite figure, if the intensity of the passing electric current in resistor R_1 is 2 A, the equivalent resistance of the circuit is ..... .

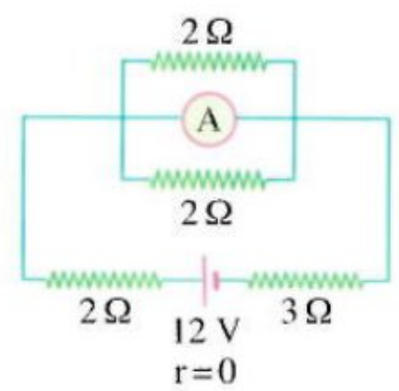


(a) 3 Ω (b) 4 Ω

(c) 6 Ω (d) 12 Ω

74 The reading of the ammeter in the opposite electric circuit is ..... .

(Knowing that the resistance of the ammeter is negligible)



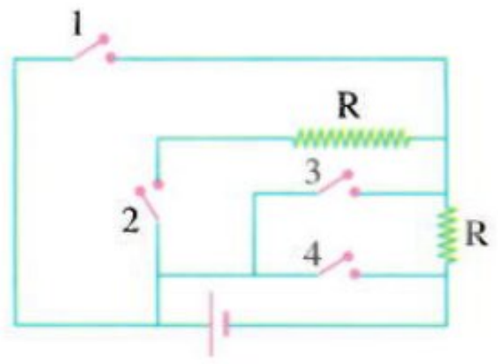
(a) 1 A (b) 1.2 A

(c) 2 A (d) 2.4 A

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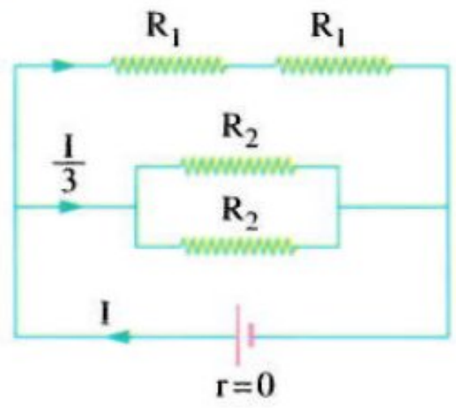
75 In the opposite electric circuit, the electric current through the battery becomes minimum value when closing switch ..... .



(a) 1 (b) 2

(c) 3 (d) 4

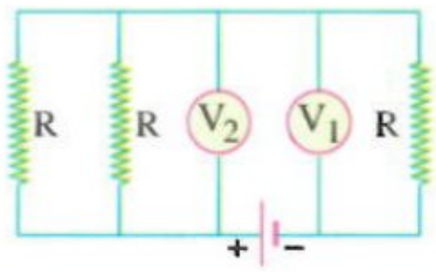
76 In the opposite electric circuit, the ratio between the values of the two resistors $(\frac{R_1}{R_2})$ equals ..... .



(a) $\frac{1}{2}$ (b) $\frac{1}{4}$

(c) $\frac{1}{8}$ (d) $\frac{1}{16}$

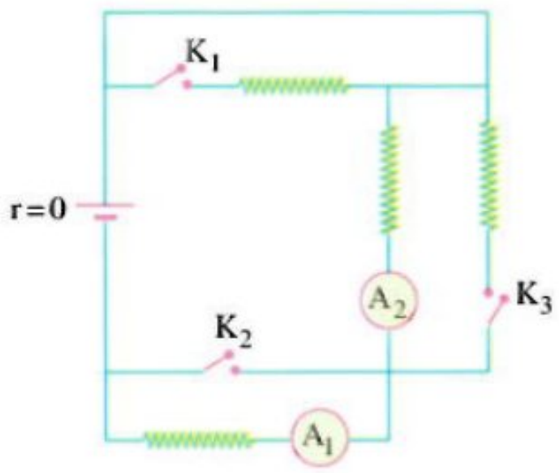
77 ✳ From the opposite figure, the ratio between the voltmeter readings $(\frac{V_1}{V_2})$ equals ..... .



(a) $\frac{1}{2}$ (b) $\frac{1}{3}$

(c) $\frac{2}{1}$ (d) $\frac{3}{1}$

78 In the opposite electric circuit, what happens to the readings of the two ammeters A_1 and A_2, respectively, when closing:



(i) switch K_1 only?

(a) Does not change, increases.

(b) Decreases, does not change.

(c) Does not change, does not change.

(d) Increases, decreases.

(ii) switch K_2 only?

(a) Increases, vanishes. (b) Decreases, vanishes.

(c) Vanishes, decreases. (d) Vanishes, increases.

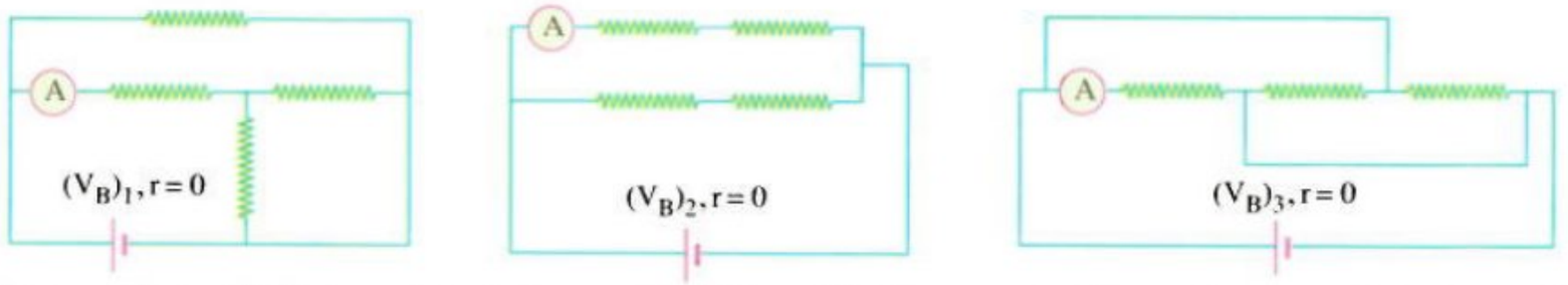
(iii) switch K_3 only?

(a) Increases, decreases. (b) Increases, does not change.

(c) Does not change, increases. (d) Decreases, increases.

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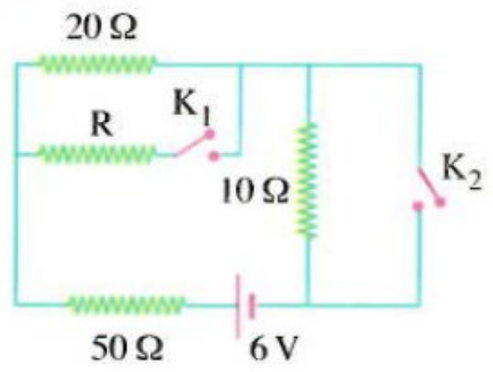
79 Each of the following electric circuits consists of a battery with negligible internal resistance, a number of identical resistors, and an ammeter:



If the reading of the ammeter in each of them is the same, then the correct order of the electromotive forces of these batteries is .....

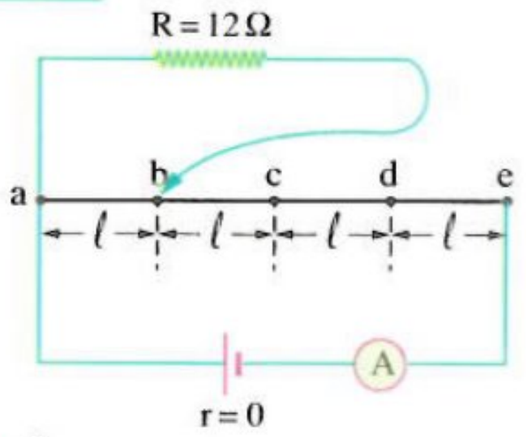
(a) $(V_B)_1 > (V_B)_2 > (V_B)_3$
(b) $(V_B)_3 > (V_B)_2 > (V_B)_1$
(c) $(V_B)_2 > (V_B)_1 > (V_B)_3$
(d) $(V_B)_2 > (V_B)_3 > (V_B)_1$

80 ✳ In the opposite electric circuit, the intensity of current passing in resistance 20 Ω doesn't change in the case of closing switches K_1 and K_2 together or opening them together, so the resistance of R equals ..... .



(a) 50 Ω
(b) 100 Ω
(c) 150 Ω
(d) 200 Ω

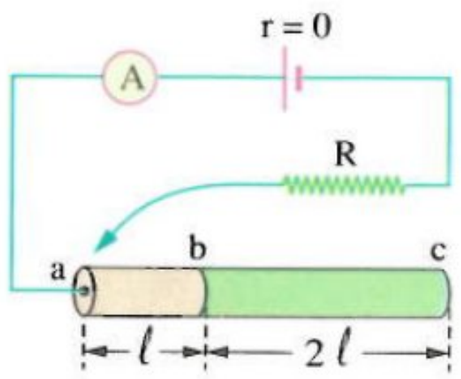
81 In the opposite electric circuit, a wire ae has a uniform cross-section and a resistance of 24 Ω. If the ammeter reading in this case equals I_1 and when the slider is moved from position b to position d, it becomes I_2, so the ratio $\left(\frac{I_1}{I_2}\right)$ equals ..... .



(a) $\frac{1}{2}$
(b) $\frac{2}{1}$
(c) $\frac{3}{5}$
(d) $\frac{5}{3}$

82 Two non-insulated metal conductors (ab) and (bc) of equal cross-sectional areas have lengths ℓ and 2ℓ, and resistivities ρ_e and $2\rho_e$ respectively. The two conductors are connected together in an electrical circuit as shown in the following figure, and the following table shows the change in the ammeter reading with the change in the slider position in the circuit,

Slider position	Ammeter reading
a	I
b	$\frac{1}{2}$I
c	?



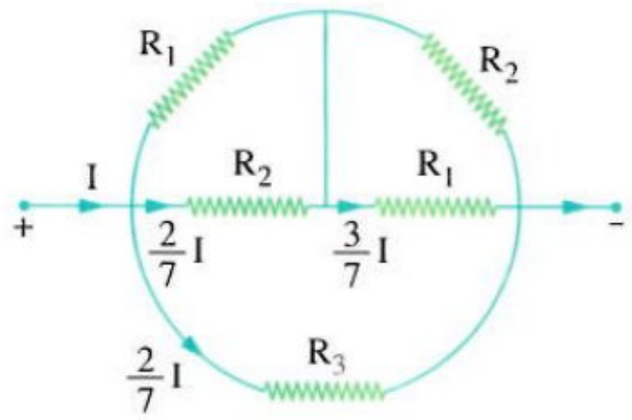
So, the ammeter reading when the slider becomes at position (c) will be ..... .

(a) $\frac{1}{3}$I
(b) $\frac{1}{4}$I
(c) $\frac{1}{6}$I
(d) $\frac{1}{8}$I

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83 ✻ Several electrical resistors are connected together where the electric currents that pass through them have intensities as shown in the opposite figure, so the correct order of resistors R_1, R_2, and R_3 concerning their values is .....

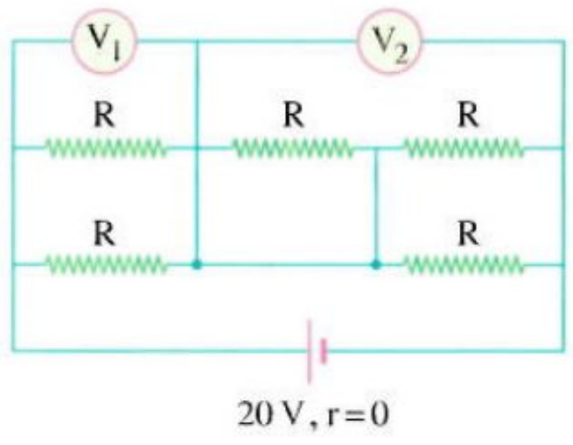


(a) $R_3 < R_2 < R_1$ (b) $R_1 < R_2 < R_3$

(c) $R_3 < R_2 = R_1$ (d) $R_2 = R_3 < R_1$

84 What are the readings of the two voltmeters V_1 and V_2 respectively in each of the following figures?

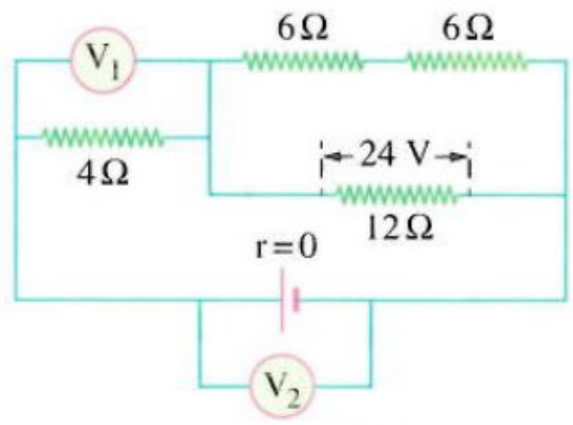
(i)



(a) 10 V, 10 V (b) 10 V, 20 V

(c) 20 V, 10 V (d) 20 V, 20 V

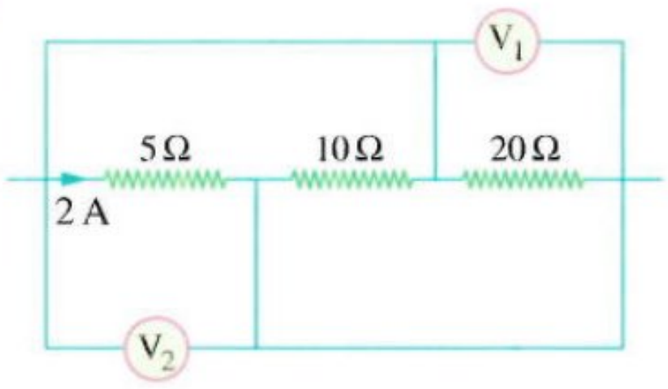
(ii)



(a) 16 V, 36 V (b) 16 V, 40 V

(c) 36 V, 36 V (d) 36 V, 40 V

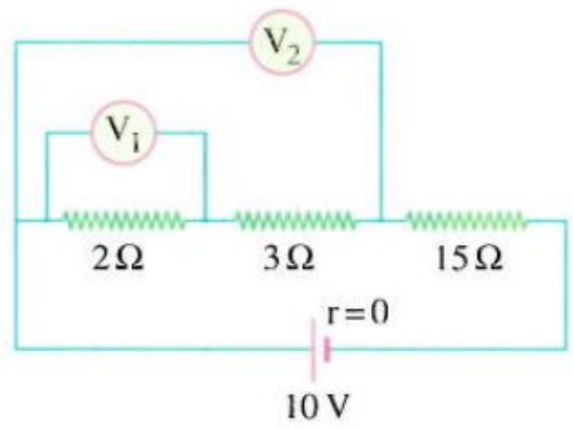
(iii)



(a) 10 V, 10 V (b) 10 V, 40 V

(c) 40 V, 10 V (d) 40 V, 40 V

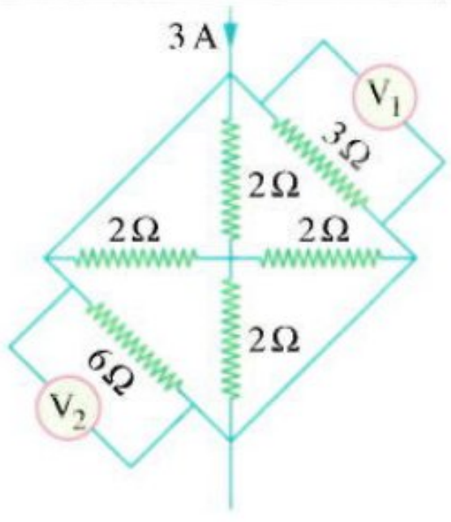
(iv)



(a) 1 V, 1.5 V (b) 1 V, 2.5 V

(c) 2 V, 1.5 V (d) 2 V, 2.5 V

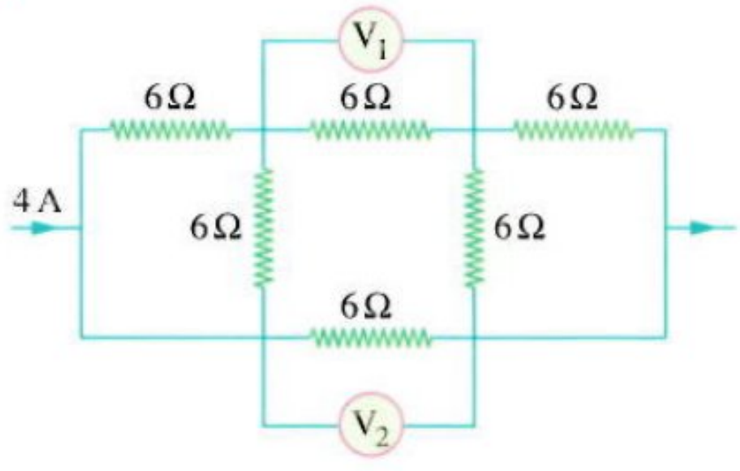
(v)



(a) 1.5 V, 3 V (b) 1.5 V, 1.5 V

(c) 3 V, 6 V (d) 3 V, 3 V

(vi)

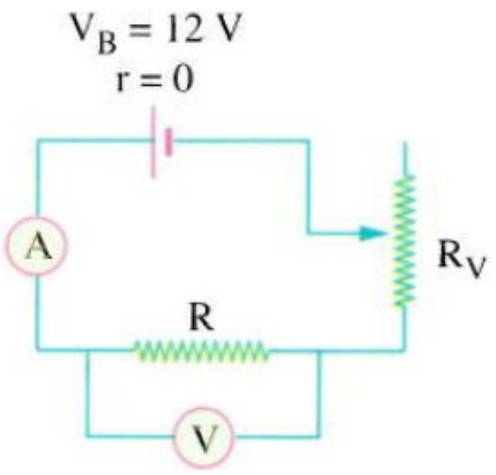


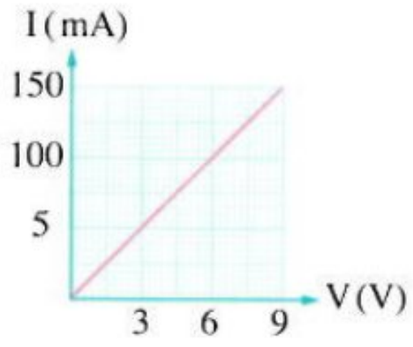
(a) 12 V, 12 V (b) 4 V, 12 V

(c) 16 V, 16 V (d) 8 V, 16 V

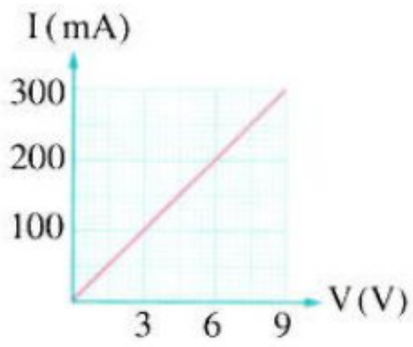
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85 In one of the experiments to verify Ohm's law practically using the opposite electrical circuit, when the resistance taken from the rheostat was 20 Ω, the ammeter reading was 150 mA. Which of the following graphs correctly represents the relation between the ammeter reading (I) and the voltmeter reading (V) in this circuit?

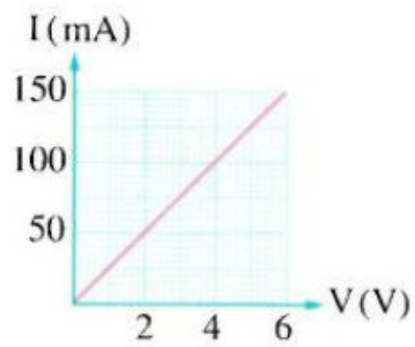




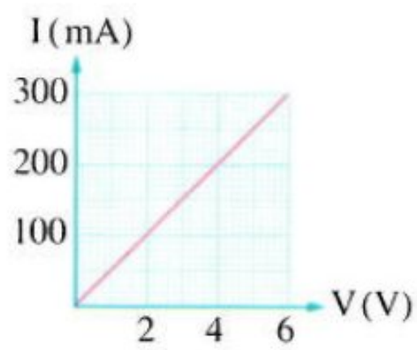
a



b

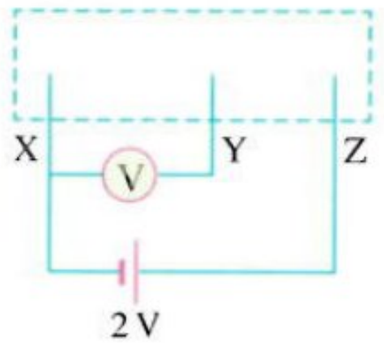


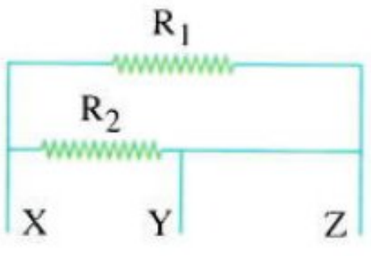
c



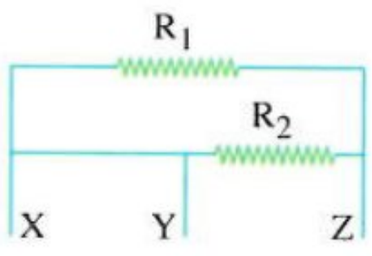
d

86 The opposite figure represents a part of an electric circuit that consists of a battery of electromotive force 2 V and negligible internal resistance and a voltmeter, which of the following figures complete that circuit at points X, Y and Z in order for the voltmeter to read 1.5 V?
(Consider the resistance of the voltmeter to be infinite)

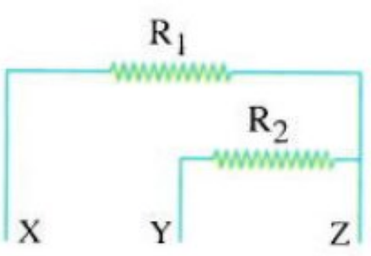




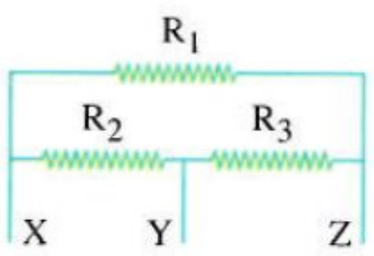
a



b



c

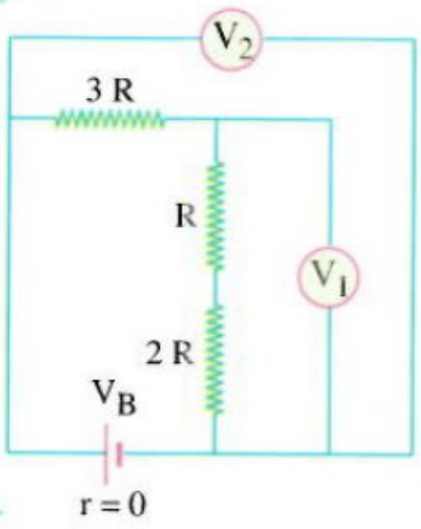


d

87 In the opposite electric circuit, the ratio between the readings of $(\frac{V_1}{V_2})$ is ..... .

a $\frac{1}{1}$ b $\frac{1}{2}$

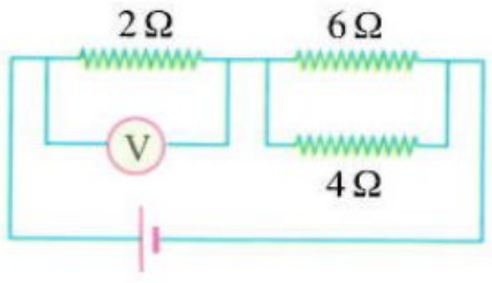
c $\frac{1}{3}$ d $\frac{1}{6}$



88 In the opposite electric circuit, if the reading of the voltmeter is 4 V, the electric current intensity that passes through the 6 Ω resistor equals ..... .

a 0.8 A b 1 A

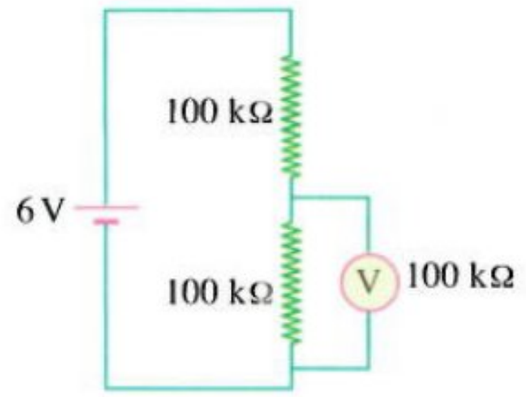
c 1.2 A d 2 A



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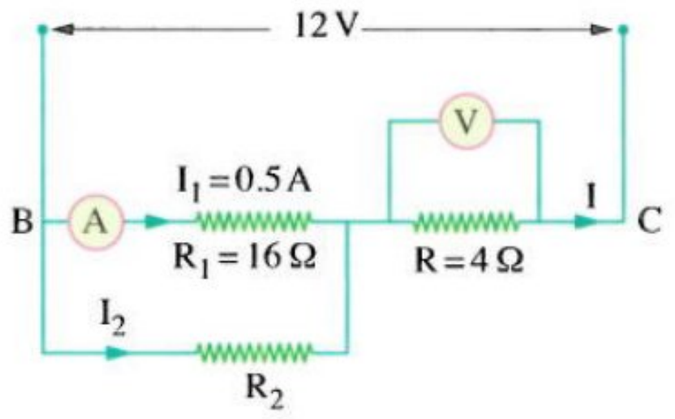
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89 In the opposite figure, if the resistance of the voltmeter is 100 kΩ, how much does it read?



(a) 0 (b) 2 V
(c) 3 V (d) 4 V

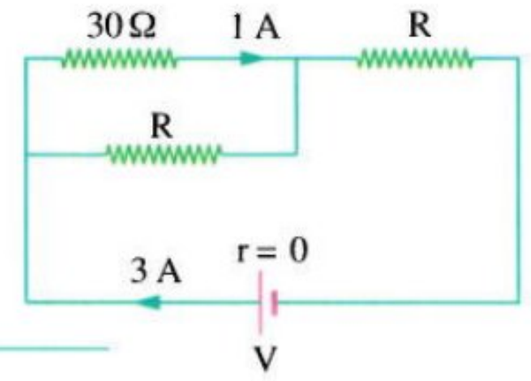
90 The opposite figure represents a part of an electric circuit, so:



(i) The voltmeter reading (V) equals ..... .
(a) 8 V (b) 4 V
(c) 2.5 V (d) 1.3 V

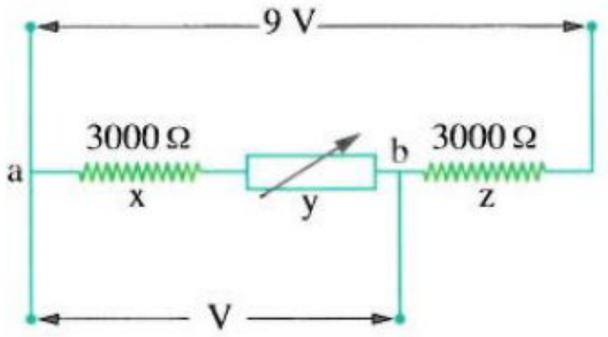
(ii) The resistance of R_2 equals ..... .
(a) 3 Ω (b) 5 Ω (c) 9 Ω (d) 16 Ω

91 In the opposite circuit, the potential difference between the terminals of the source (V) is ..... .



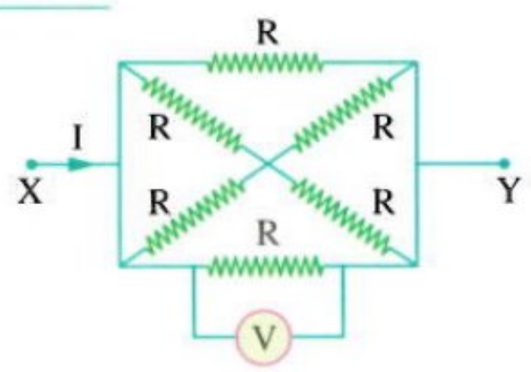
(a) 25 V (b) 50 V
(c) 75 V (d) 100 V

92 ✳ In the opposite figure:



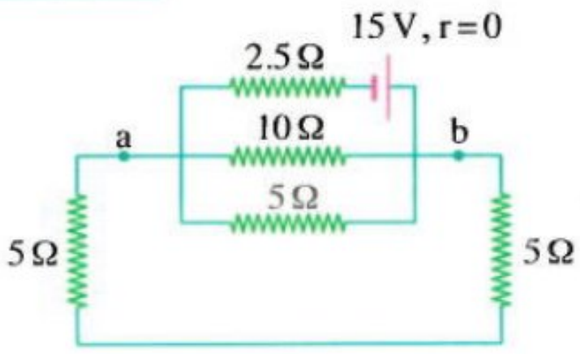
There are two fixed resistors x and z and a variable resistor y changes from zero to 3000 Ω, so the range of potential differences between the two points a and b is from ..... .
(a) 0 to 6 V (b) 3 V to 6 V
(c) 4.5 V to 6 V (d) 4.5 V to 9 V

93 The opposite figure represents a part of an electric circuit. If the voltmeter reads 1 V, then the potential difference between the two points X and Y equals ..... .



(a) 1 V (b) 2 V
(c) 3 V (d) 4 V

94 ✳ In the opposite electric circuit:



(i) The total resistance in the circuit equals ..... .
(a) 3 Ω (b) 5 Ω
(c) 9 Ω (d) 11 Ω

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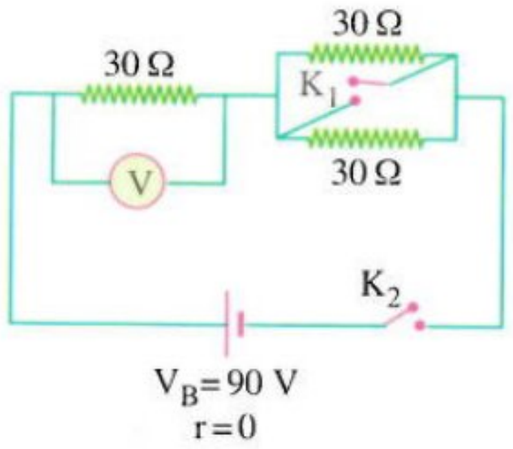
(ii) The total current intensity passing in the circuit equals ..... .

(a) 12 A (b) 5 A (c) 3 A (d) 2 A

(iii) The potential difference between the two points a and b equals ..... .

(a) 2.5 V (b) 5 V (c) 6 V (d) 7.5 V

95 ✱ In the opposite circuit, the reading of the voltmeter when:



(i) Switch K_2 is closed and switch K_1 is opened equals ..... .

(a) 20 V (b) 30 V

(c) 60 V (d) 40 V

(ii) Switch K_2 is closed and switch K_1 is closed equals ..... .

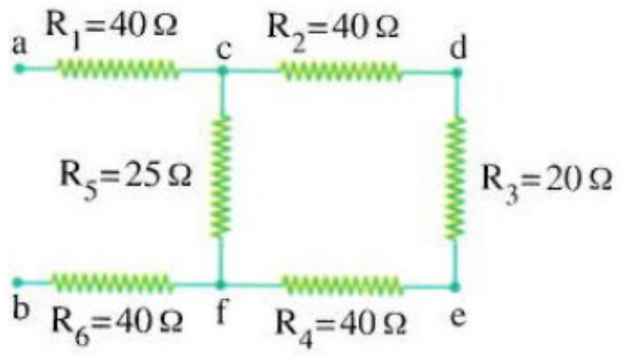
(a) 30 V (b) 40 V

(c) 90 V (d) 60 V

(iii) Switch K_2 is opened and switch K_1 is closed equals ..... .

(a) 0 (b) 20 V (c) 40 V (d) 60 V

96 The opposite figure represents a part of an electric circuit, so if the potential difference V_{ab} equals 200 V;



(i) The equivalent resistance of this part of the circuit equals ..... .

(a) 100 Ω (b) 80 Ω

(c) 75 Ω (d) 54 Ω

(ii) The current intensity passing through resistance R_1 equals ..... .

(a) 5 A (b) 3.5 A (c) 2 A (d) 1.6 A

(iii) The current intensity through resistance R_5 equals ..... .

(a) 0.1 A (b) 0.4 A (c) 1.4 A (d) 1.6 A

(iv) The potential difference between the two points e and d equals ..... .

(a) 16 V (b) 12 V (c) 10 V (d) 8 V

97 ✱ Three resistors of 60 Ω, 40 Ω and 20 Ω are connected to an electric source where the potential difference across them are 30 V, 20 V and 50 V respectively, so the equivalent resistance of the circuit equals ..... .

(a) 16.67 Ω (b) 12.23 Ω (c) 10.53 Ω (d) 9.75 Ω

98 ✱ A voltmeter of 500 Ω resistance is connected in parallel to an unknown resistance, then an ammeter of negligible resistance is connected to them in series and when the terminals of this circuit are connected to an electric cell, the ammeter reading becomes 0.01 A and the voltmeter reading becomes 3 V, so the value of the unknown resistance is ..... .

(a) 900 Ω (b) 800 Ω (c) 750 Ω (d) 620 Ω

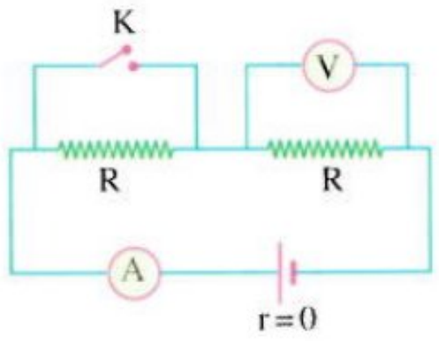
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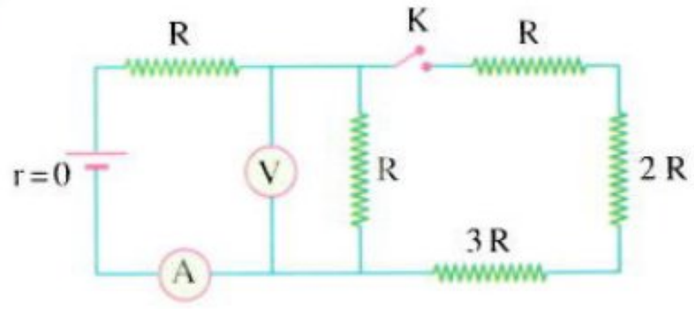
99 What happens to the readings of the ammeter and voltmeter, respectively, when the switch K is closed in each of the following cases?

(i)



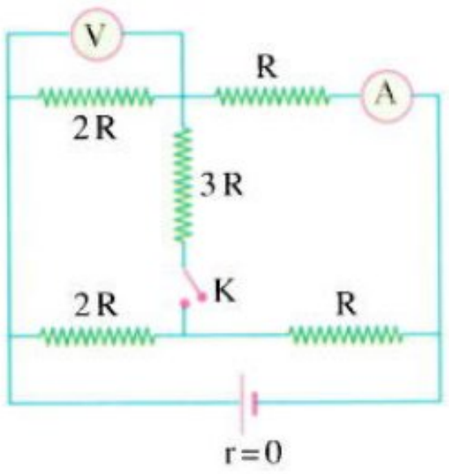
(a) Increases, increases.
(b) Decreases, decreases.
(c) Decreases, increases.
(d) Increases, decreases.

(ii)



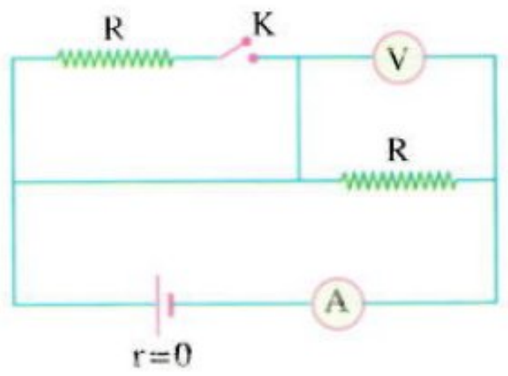
(a) Increases, increases.
(b) Decreases, doesn't change.
(c) Increases, decreases.
(d) Doesn't change, doesn't change.

(iii)



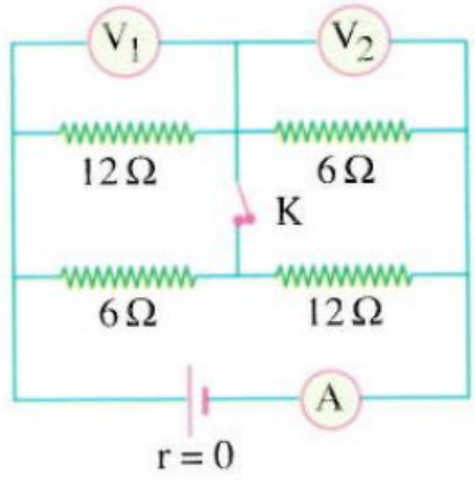
(a) Increases, increases.
(b) Decreases, doesn't change.
(c) Doesn't change, decreases.
(d) Doesn't change, doesn't change.

(iv)



(a) Increases, increases.
(b) Decreases, doesn't change.
(c) Increases, decreases.
(d) Doesn't change, doesn't change.

100 In the opposite electrical circuit, what happens to the reading of each of the ammeter (A) and the voltmeters (V_1) and (V_2) when the switch (K) is opened ?



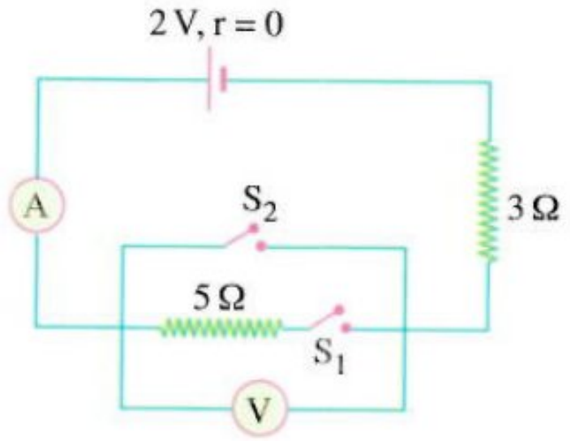
	Ammeter reading (A)	Voltmeter reading (V_1)	Voltmeter reading (V_2)
(a)	Decreases	Increases	Increases
(b)	Decreases	Increases	Decreases
(c)	Increases	Increases	Increases
(d)	Increases	Decreases	Decreases

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101 In the opposite circuit, the readings of the ammeter and the voltmeter at:

(i) Closing the two switches S_1 and S_2 together equals ..... .

	The ammeter reading	The voltmeter reading
a	0.67 A	0.2 V
b	0	0.3 V
c	0.67 A	0
d	0.5 A	0.3 V

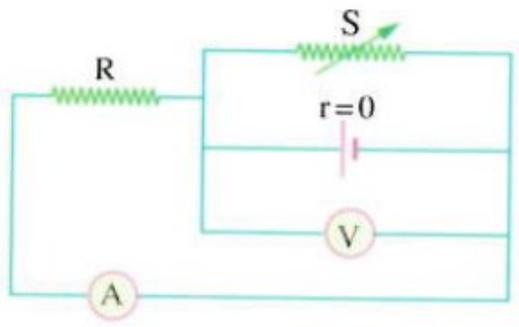


(ii) Closing switch S_1 and opening switch S_2 equals ..... .

	The ammeter reading	The voltmeter reading
a	3.2 A	1.2 V
b	0.25 A	1.25 V
c	1.25 A	3.2 V
d	0.25 A	2.05 V

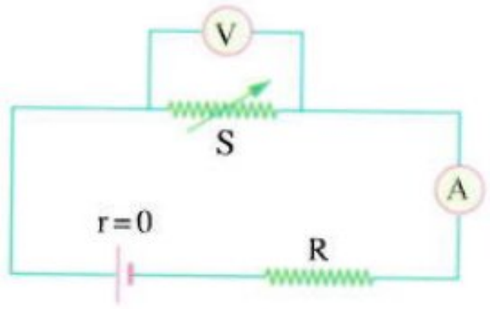
102 What happens to the readings of the ammeter and voltmeter, respectively, when increasing the value of variable resistance S in each of the following cases?

(i)



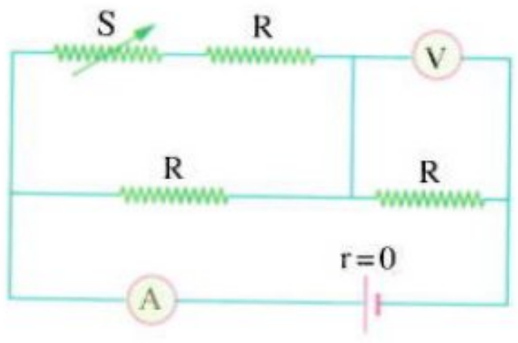
a Decreases, decreases.
b Decreases, doesn't change.
c Doesn't change, increases.
d Doesn't change, doesn't change.

(ii)



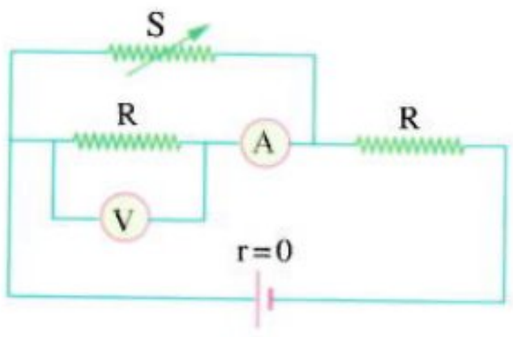
a Decreases, decreases.
b Decreases, increases.
c Increases, decreases.
d Increases, increases.

(iii)



a Decreases, decreases.
b Decreases, increases.
c Increases, decreases.
d Increases, increases.

(iv)

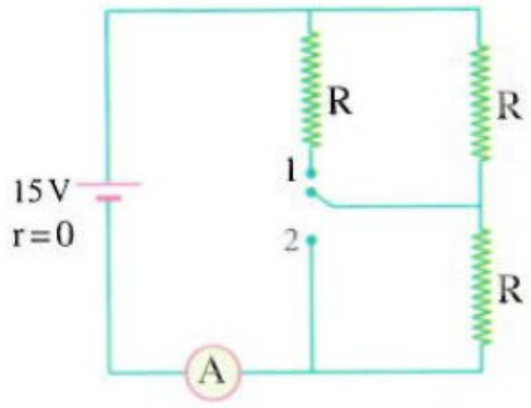


a Decreases, decreases.
b Decreases, increases.
c Increases, decreases.
d Increases, increases.

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103 In the opposite electric circuit:



(i) When closing the switch in direction (1), a current of intensity 2 A passes in the ammeter, thus the value of the resistance of R is ..... .

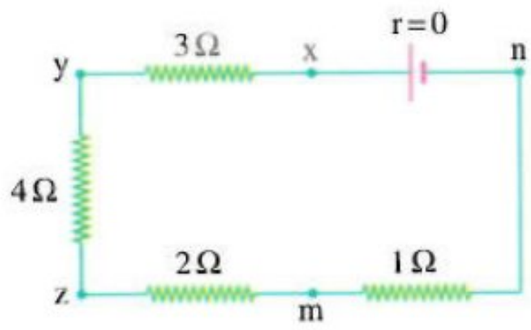
(a) 30 Ω (b) 5 Ω

(c) 7.5 Ω (d) 2.5 Ω

(ii) When closing the switch in direction (2), a current of intensity ..... passes in the ammeter.

(a) 1 A (b) 2 A (c) 3 A (d) 4 A

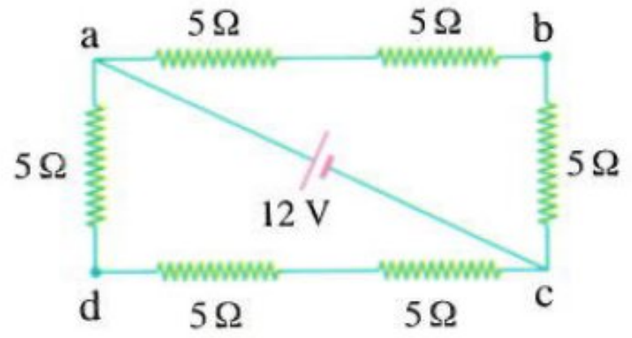
104 In the opposite electric circuit, what are the two points between which a free electron requires the greatest amount of work to be moved?



(a) x, y (b) y, z

(c) z, m (d) z, n

105 ✱ In the shown electric circuit, the potential difference between the two points b and d equals ..... .

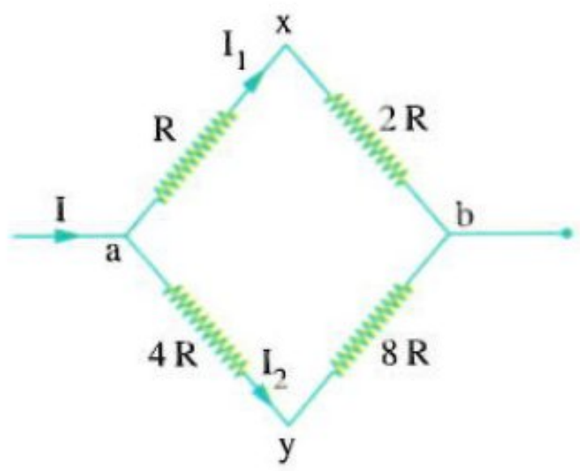


(a) 2 V (b) 4 V

(c) 6 V (d) 8 V

106 ✱ In the opposite figure:

The potential difference between x and y equals ..... .



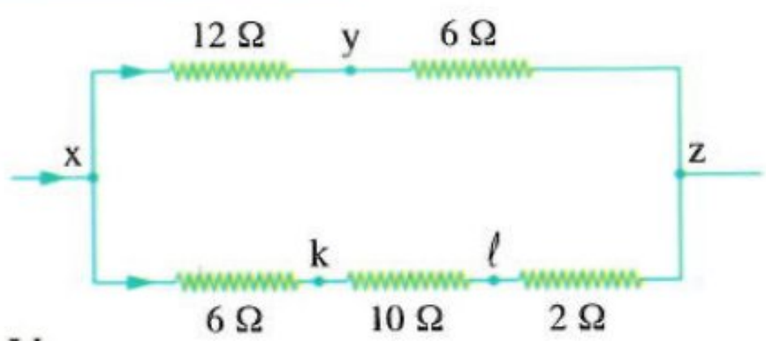
(a) 0

(b) 3 IR

(c) 6 IR

(d) IR

107 The opposite figure represents a part of an electric circuit, which of the following is correct regarding the ratio of the potential differences between the shown points?

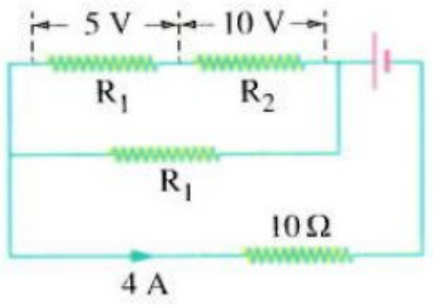


(a) $V_{k\ell} > V_{xy}$ (b) $V_{yz} > V_{xk}$

(c) $V_{xy} = V_{kz}$ (d) $V_{xz} = 2\,V_{\ell z}$

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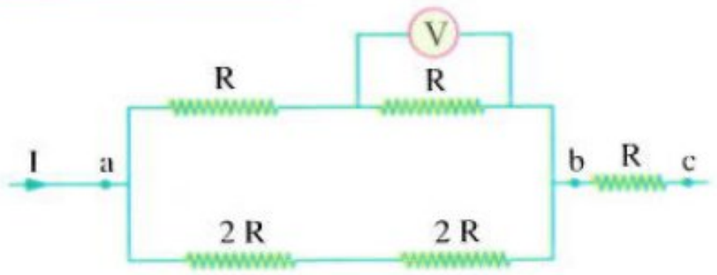
108 In the opposite electrical circuit, the values of R_1 and R_2, respectively, are ..... .



(a) 2.5 Ω, 5 Ω (b) 2.5 Ω, 10 Ω

(c) 5 Ω, 2.5 Ω (d) 5 Ω, 10 Ω

109 ✱ In the opposite figure, if the reading of the voltmeter is 4 V:



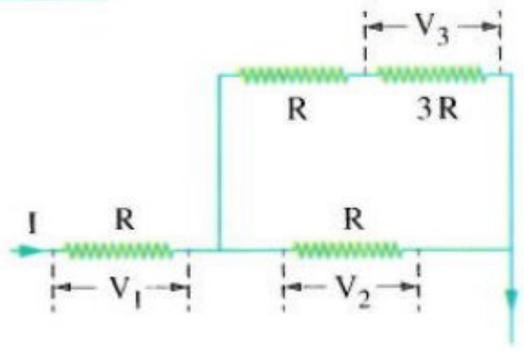
(i) What will be its reading when it is connected between the two points b and c?

(a) 10 V (b) 8 V (c) 6 V (d) 2 V

(ii) What will be its reading when it is connected between the two points a and c?

(a) 14 V (b) 15 V (c) 16 V (d) 17 V

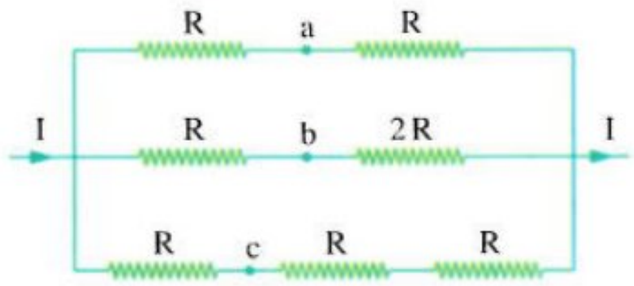
110 The opposite figure shows a section of an electric circuit, so the correct mathematical relationship for the potential differences V_1, V_2, and V_3 is .....



(a) $V_1 = V_2 > V_3$ (b) $V_1 = V_2 < V_3$

(c) $V_1 > V_3 > V_2$ (d) $V_1 > V_2 > V_3$

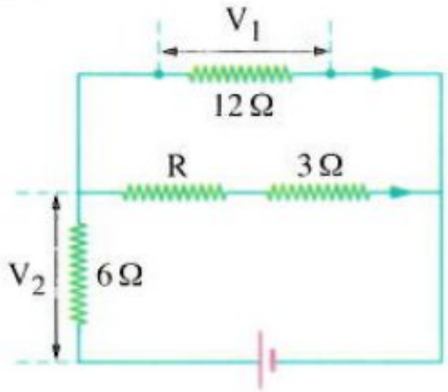
111 The opposite figure represents a section of an electrical circuit. Which of the following mathematical relationships relates the values of electrical potential at points a, b, and c correctly?



(a) $V_a > V_b > V_c$ (b) $V_a < V_b < V_c$

(c) $V_a > V_b = V_c$ (d) $V_a < V_b = V_c$

112 The opposite diagram represents a closed electric circuit, if $V_1 = V_2$, the resistance of resistor R equals ..... .



(a) 3 Ω (b) 9 Ω

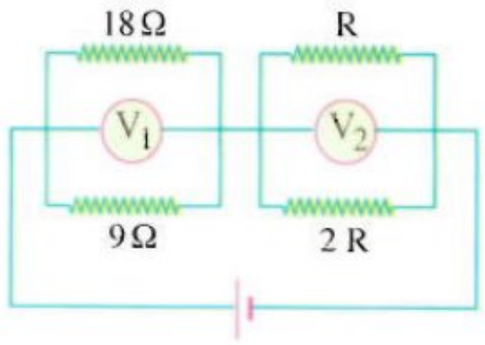
(c) 12 Ω (d) 15 Ω

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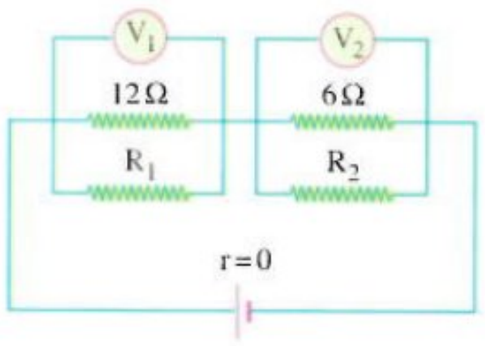
113 In the opposite circuit, if $\frac{V_1}{V_2} = \frac{3}{4}$, the value of the resistor R equals ..... .



(a) 6 Ω (b) 12 Ω

(c) 18 Ω (d) 24 Ω

114 In the opposite electric circuit, the readings of the two voltmeters V_1 and V_2 are equal, so the relation between R_1 and R_2 is ..... .

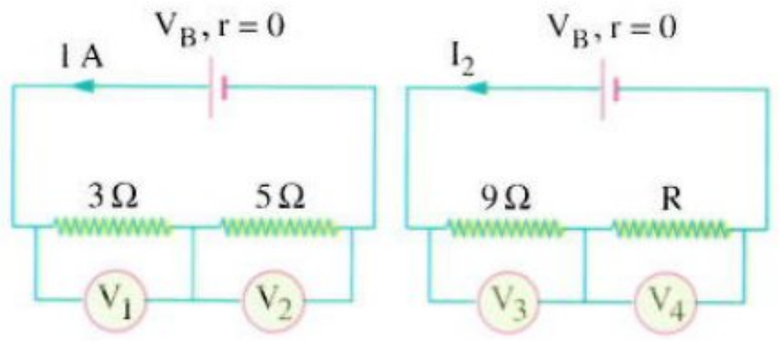


(a) $R_1 = R_2$ (b) $R_1 > R_2$

(c) $R_1 < R_2$ (d) indeterminable

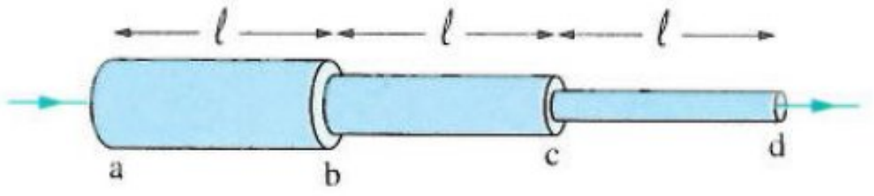
115 In the two opposite circuits, if the readings of voltmeters V_1 and V_3 are equal, and the readings of voltmeters V_2 and V_4 are equal, then ..... .

(Assuming that all voltmeters are ideal)



	R	I_2
(a)	6 Ω	$\frac{2}{3}$ A
(b)	6 Ω	$\frac{1}{3}$ A
(c)	15 Ω	$\frac{2}{3}$ A
(d)	15 Ω	$\frac{1}{3}$ A

116 Three solid conductors ab, bc and cd are made of the same metal and had the same length (ℓ). Each has a regular circular cross-section with cross-sectional areas 3A, 2A and A respectively, as shown in the opposite figure, then the ratio between:



(i) the potential differences $(\frac{V_{cd}}{V_{bc}})$ equals ..... .

(a) $\frac{1}{2}$ (b) $\frac{2}{1}$ (c) $\frac{2}{3}$ (d) $\frac{3}{2}$

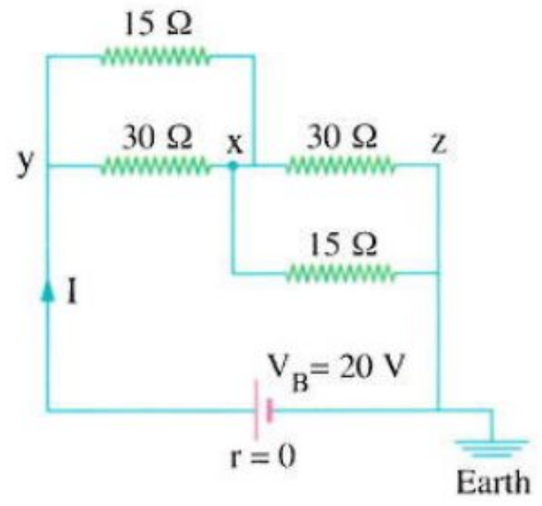
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(ii) the potential differences $\left(\frac{V_{bc}}{V_{ad}}\right)$ equals ..... .

(a) $\frac{11}{3}$ (b) $\frac{3}{11}$ (c) $\frac{5}{22}$ (d) $\frac{22}{5}$

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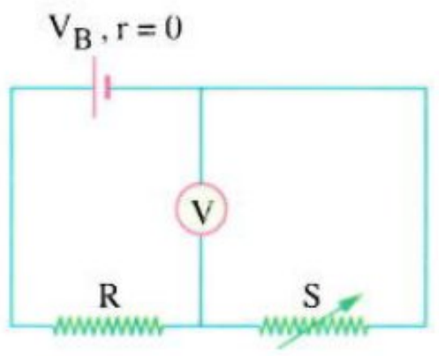
117 In the opposite electric circuit, if you know that the electric potential of the point connected to the Earth is zero, ...



	The value of I	The electric potential of point x
(a)	$\frac{1}{2}$ A	10 V
(b)	$\frac{1}{2}$ A	5 V
(c)	1 A	5 V
(d)	1 A	10 V

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118 The opposite diagram represents a closed electric circuit, so when the variable resistance (S) is doubled, the voltmeter reading ..... .



(a) increases but doesn't reach the double of its initial value

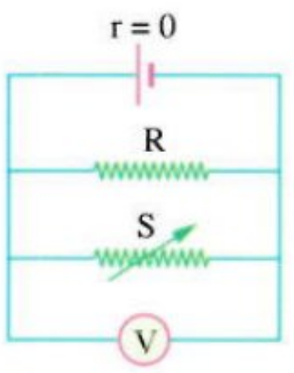
(b) is doubled

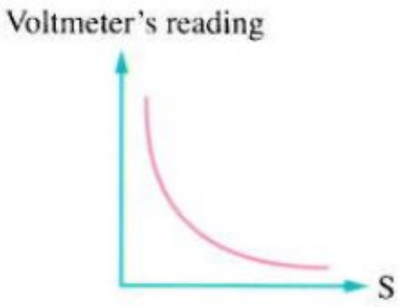
(c) decreases but doesn't reach half its initial value

(d) decreases to half its initial value

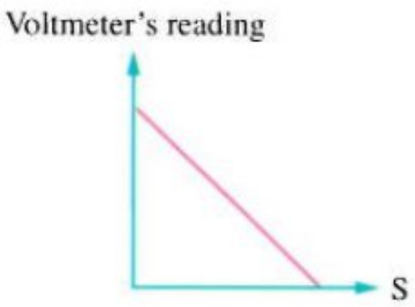
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119 Which of the following graphs represents the relation between the reading of the voltmeter and the variable resistance S?

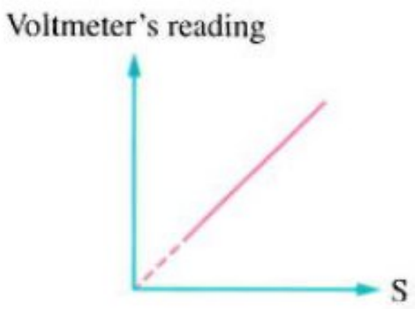




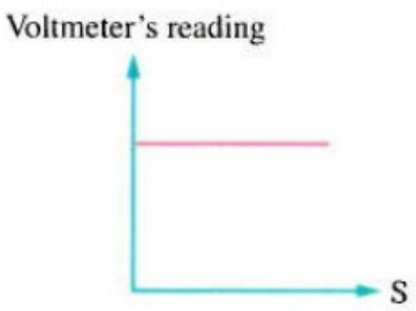
(a)



(b)



(c)



(d)

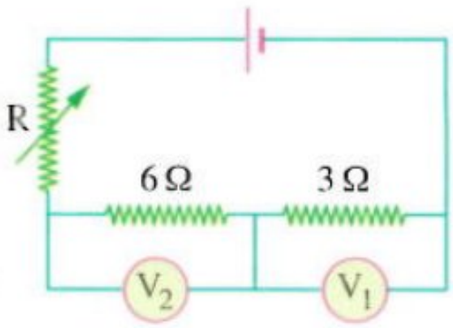
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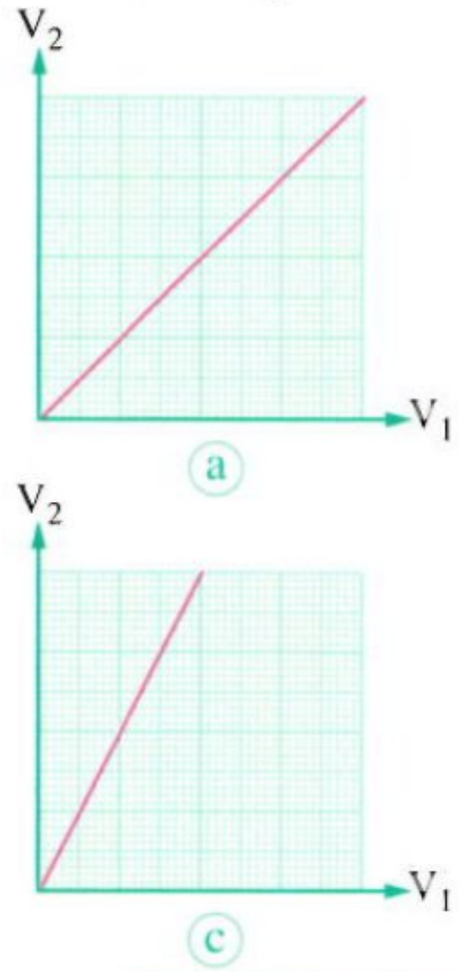
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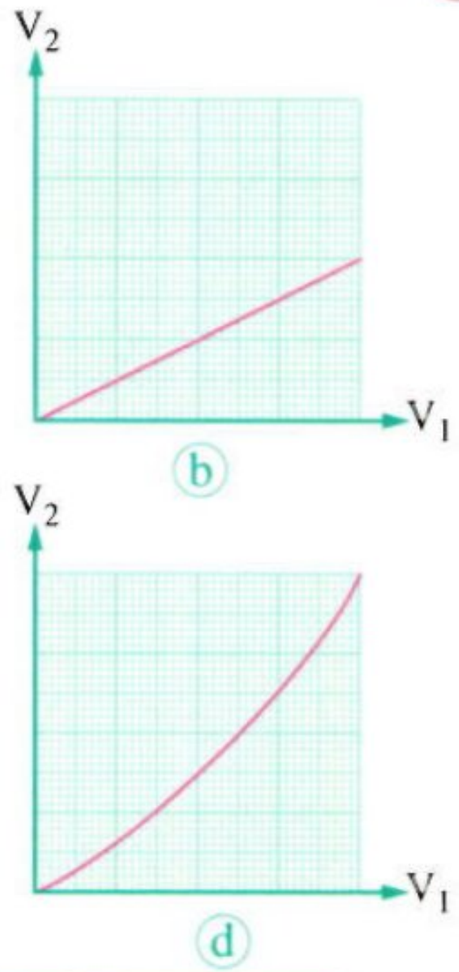
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120 Which of the following graphs represents the relation between the reading of the voltmeter V_1 and the reading of the voltmeter V_2 when changing the value of the variable resistance R?

(Noticing that: V_1 and V_2 are represented with the same drawing scale)







121 Figure (1) represents the graph of the ammeter reading (A) in the electrical circuit shown in figure (2) versus the reading voltmeter (V_1) in it.

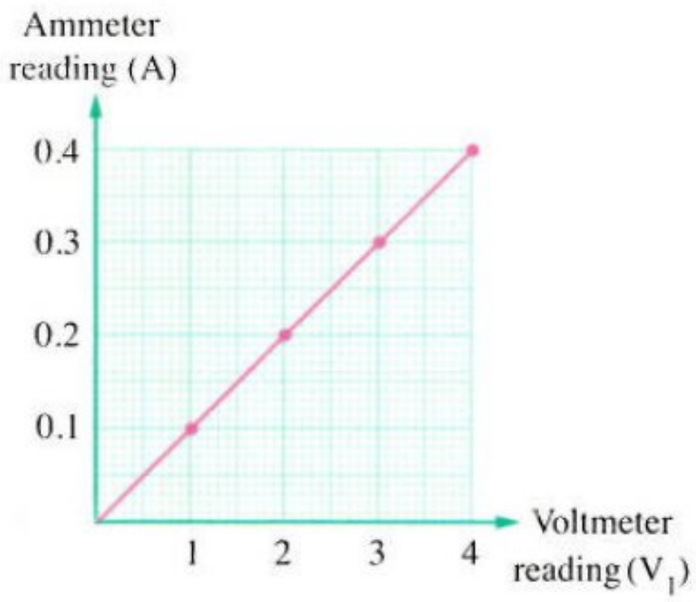


Figure (1)

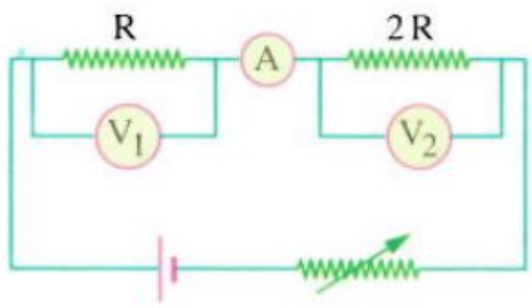


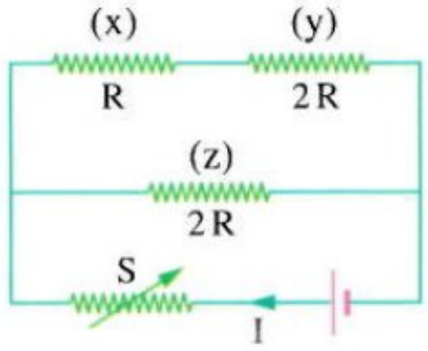
Figure (2)

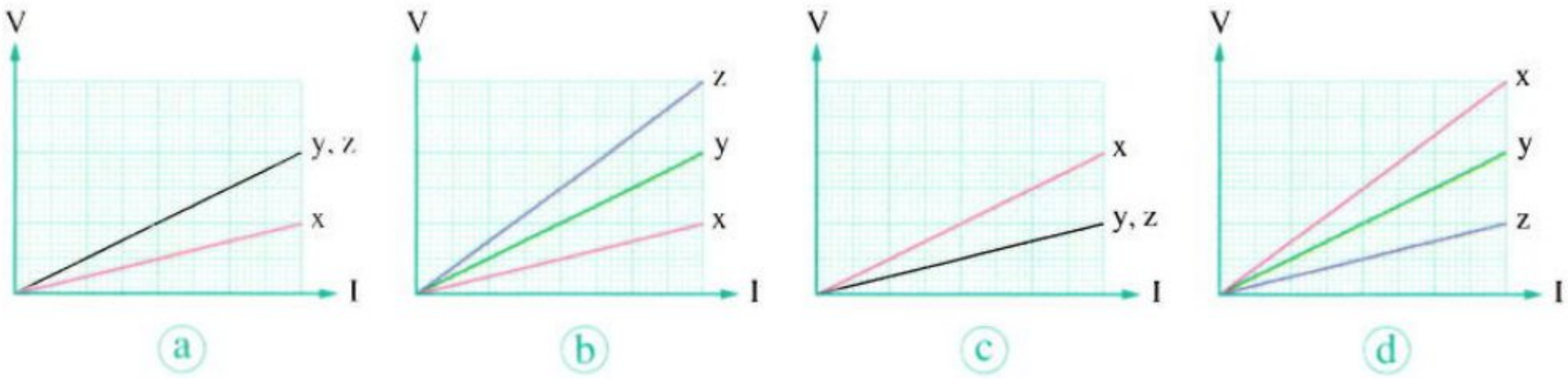
Which of the following can represent possible values for the readings of the ammeter (A) and the voltmeter (V_2) at different values of the variable resistance?

	Ammeter reading (A)	Voltmeter reading (V_2)
a	1 A	2 V
b	0.8 A	8 V
c	0.6 A	3 V
d	0.5 A	10 V

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122 ✱ Which of the following graphs represents the relation between the total current (I) through the circuit shown in the opposite figure and the potential difference (V) across each of the three resistors x, y, z?





123 ✱ Figure (1) shows a graph that represents the potential differences across the parts of the electric circuit in figure (2),

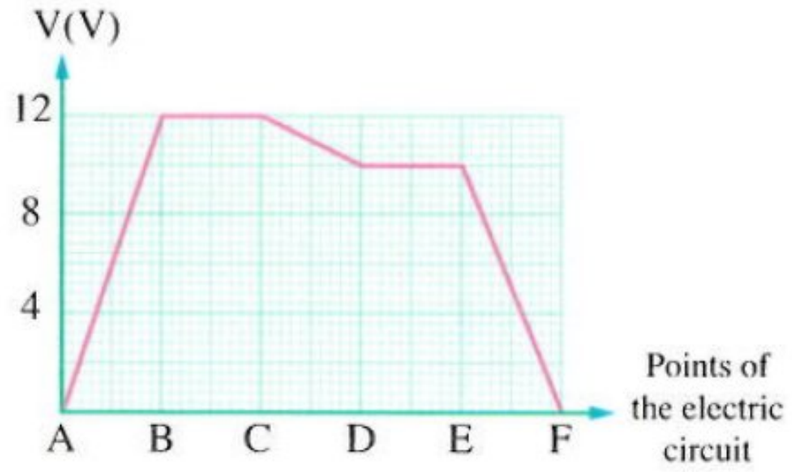


Figure (1)

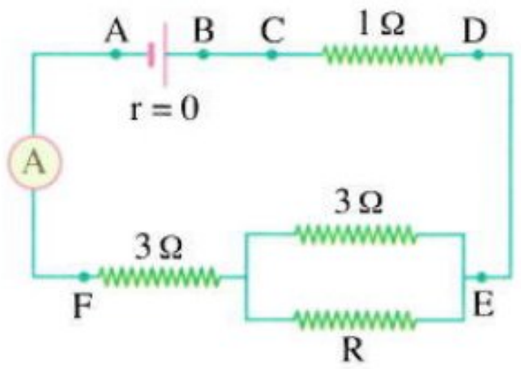


Figure (2)

so, from studying the two figures (1) and (2):

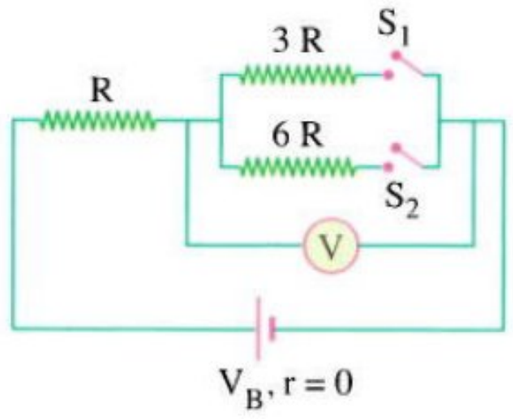
(i) The ammeter reading equals ..... .

(a) 1 A (b) 2 A (c) 3 A (d) 4 A

(ii) The resistance of R is ..... .

(a) 10 Ω (b) 9 Ω (c) 6 Ω (d) 4 Ω

124 ✱ In the opposite electric circuit when switch S_1 only is closed, the reading of the voltmeter becomes V_1, when switch S_2 only is closed, the reading of the voltmeter becomes V_2 and when both switches S_1 and S_2 are closed together, the reading of the voltmeter becomes V_3, then .....



(a) $V_1 < V_2 < V_3$ (b) $V_3 < V_1 < V_2$

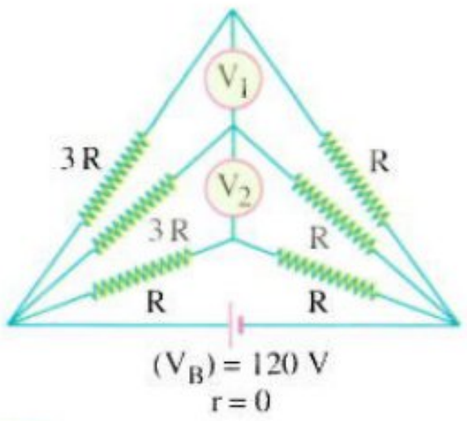
(c) $V_2 < V_1 < V_3$ (d) $V_3 < V_2 < V_1$

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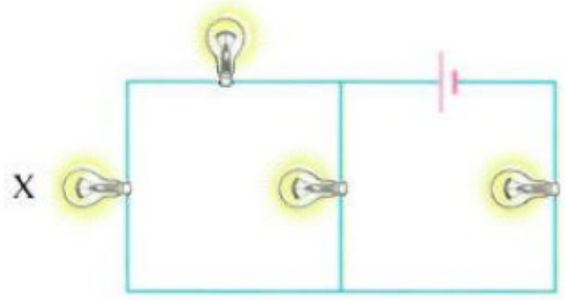
125 ✳ In the opposite electrical circuit, the readings of the voltmeters (V_1) and (V_2) respectively are ..... .



(a) 0, 0 (b) 0, 30 V

(c) 30 V, 30 V (d) 30 V, 60 V

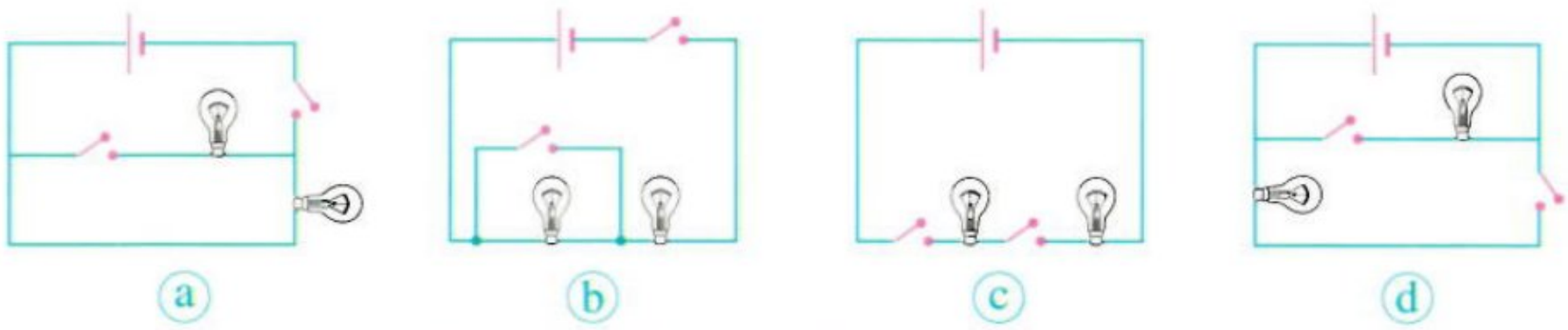
126 The opposite electric circuit shows four lit lamps, if the lamp X is blown up, how many lamps remain lighted?



(a) 0 (b) 1

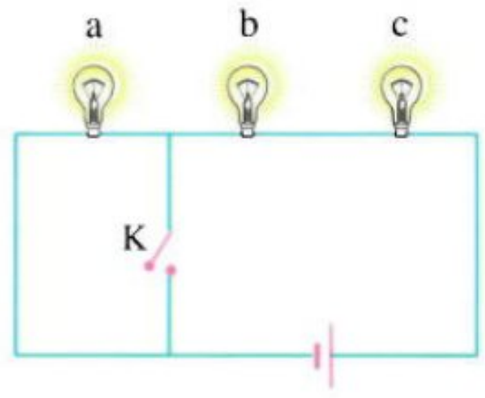
(c) 2 (d) 3

127 In which of the following electric circuits does one of the two bulb light up when the two switches are closed?



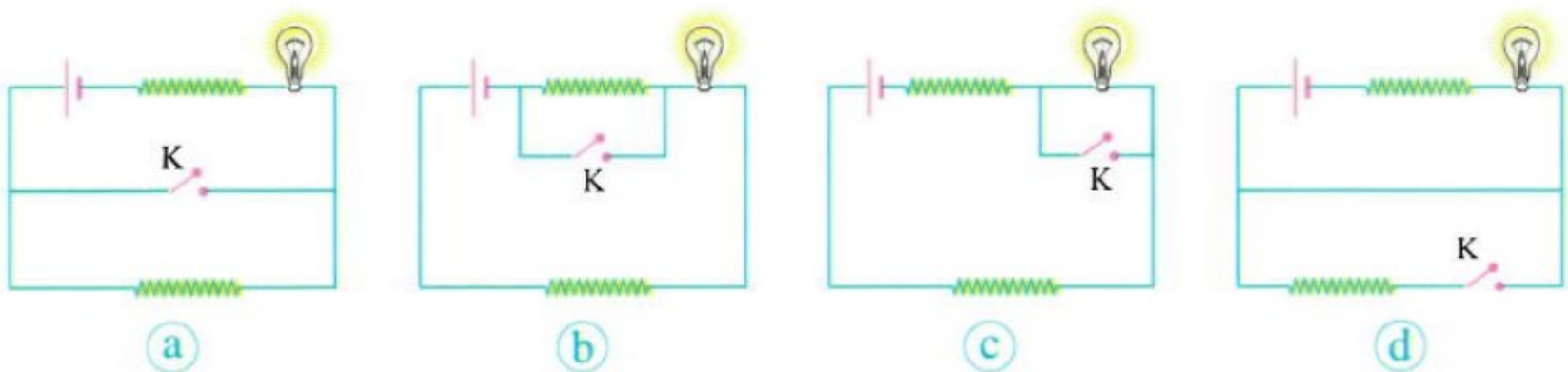
(a) (b) (c) (d)

128 ✳ In the opposite electric circuit, when closing the switch K, what will happen to the brightness of the two bulbs a and b ..... .



	Bulb a	Bulb b
(a)	increases	decreases
(b)	vanishes	increases
(c)	increases	increases
(d)	vanishes	decreases

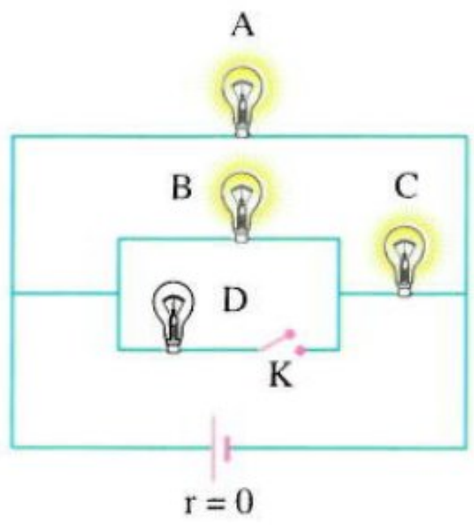
129 In which of the following electric circuits does closing the switch K not affect the brightness of the lamp?



(a) (b) (c) (d)

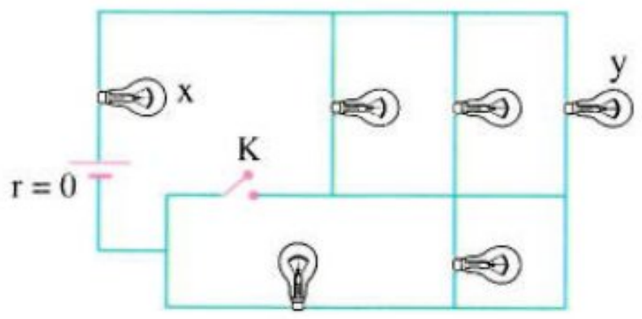
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130 ✱ In the opposite circuit, if the four bulbs A, B, C and D are identical, what happens to the illumination intensity of the bulbs A and B when closing the switch K?



	Brightness of bulb A	Brightness of bulb B
a	Increases	Decreases
b	Remains constant	Increases
c	Remains constant	Decreases
d	Decreases	Increases

131 Six identical bulbs, a DC source, and a switch (K) are connected as shown in the opposite figure, so:



(i) the number of lighted bulbs when:

1- the switch (K) is open is ..... .

(a) 3 (b) 4 (c) 5 (d) 6

2- the switch (K) is closed is ..... .

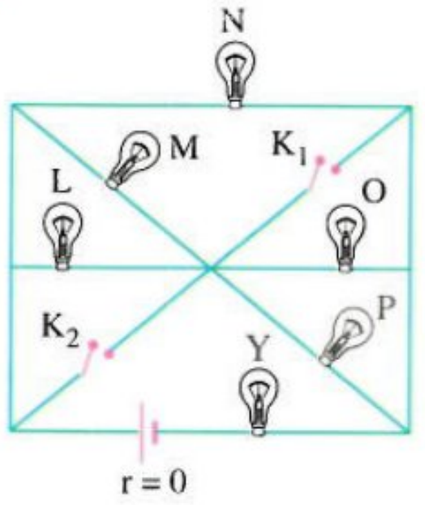
(a) 3 (b) 4 (c) 5 (d) 6

(ii) What happens to the brightness of bulbs x and y, respectively, when the switch (K) is closed?

(a) Increases, does not change. (b) Increases, increases.

(c) Decreases, increases. (d) Does not change, decreases.

132 In the opposite electric circuit, there are six identical bulbs:



(i) How many of these bulbs will remain lit when switches K_1 and K_2 are closed?

(a) 1 (b) 2 (c) 4 (d) 6

(ii) What happens to the brightness of bulbs O and M, respectively, when only switch K_1 is closed?

(a) Increases, decreases. (b) Increases, vanishes.

(c) Vanishes, decreases. (d) Vanishes, increases.

(iii) Which bulb, P or N, will have a higher brightness when only switch K_2 is closed?

(a) Bulb P (b) Bulb N

(c) Both bulbs will have the same brightness.

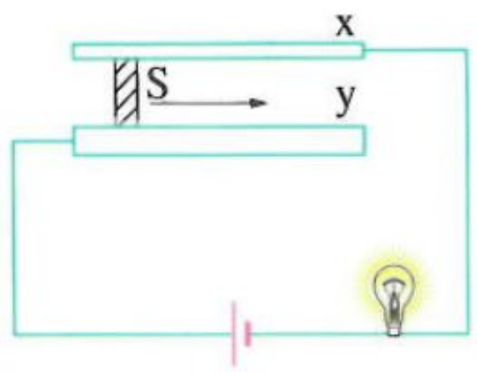
(d) The answer cannot be determined.

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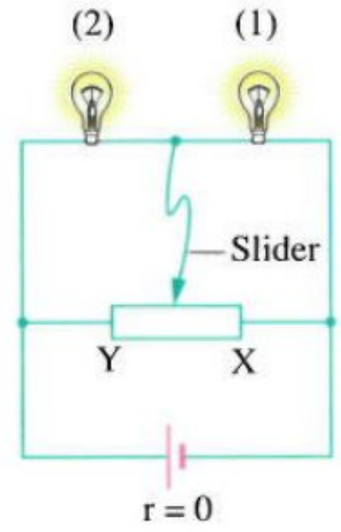
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133 In the opposite electric circuit two rods x and y of the same metal have the same length but the cross-sectional area of y is double that of x and they are connected together via a copper slider (S), if the slider is moved in the shown direction, the brightness of the lamp ..... .



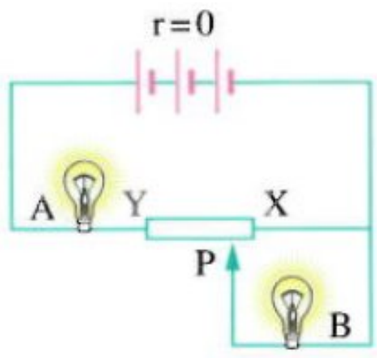
(a) increases
(b) decreases till it vanishes
(c) doesn't change
(d) decreases but doesn't vanish

134 ✱ In the opposite electric circuit, the two lamps are identical and when the slider is positioned in the middle of the resistance wire XY, the brightness of the two identical lamps (1) and (2) become equal, so if the slider is moved slightly towards X, what will happen to the brightness of the two lamps?



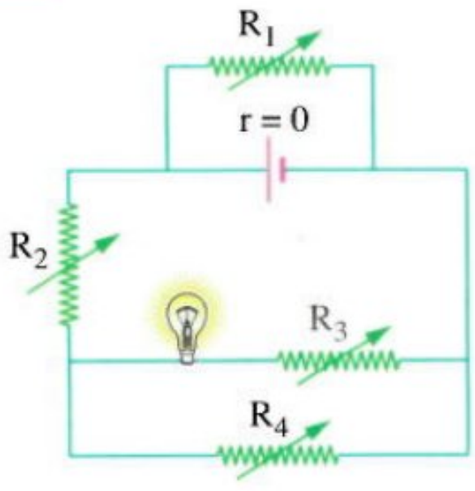
	Brightness of lamp (1)	Brightness of lamp (2)
(a)	Increases	Increases
(b)	Increases	Decreases
(c)	Decreases	Increases
(d)	Decreases	Decreases

135 ✱ In the opposite circuit, rod XY of resistance 10 Ω and two identical lamps A and B each of resistance 10 Ω, what happens to the light of the bulbs A and B when the slider P moves from its position towards point Y?



	Bulb A	Bulb B
(a)	Does not change	Increases
(b)	Increases	Increases
(c)	Decreases	Does not change
(d)	Increases	Decreases

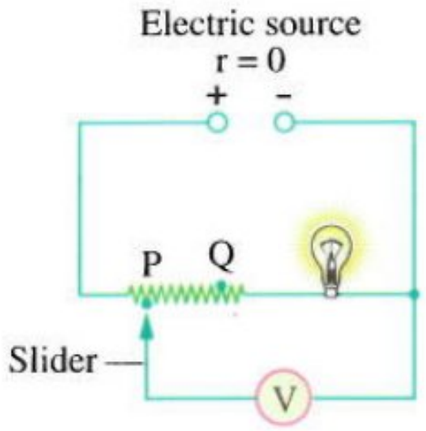
136 In the opposite electrical circuit, which of the shown variable resistors, when its value is increased, leads to an increase in the brightness of the lamp?



(a) Resistor R_1
(b) Resistor R_2
(c) Resistor R_3
(d) Resistor R_4

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137 ✳ In the opposite electric circuit, what happens to the brightness of the bulb and the reading of the voltmeter when the slider moves from P to Q?



	The brightness of the bulb	The voltmeter reading
a	Increases	Decreases
b	Increases	Increases
c	Does not change	Decreases
d	Does not change	Increases

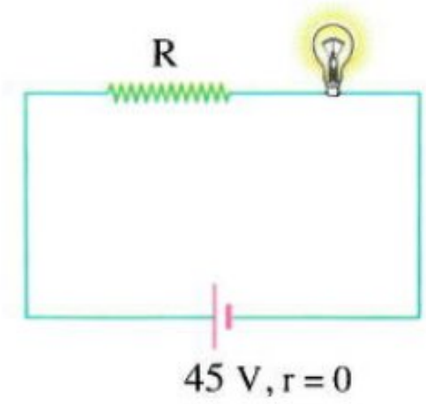
138 Two electric bulbs 1 and 2 of resistances R_1, R_2 respectively are connected in series with an electric source, if $R_1 < R_2$, then ..... .

a the brightness of bulb 1 is greater
b the brightness of bulb 2 is greater
c the two bulbs have the same brightness
d the answer is indeterminable

139 Two electric bulbs 1 and 2 of resistances R_1, R_2 respectively are connected in parallel with an electric source, if $R_1 > R_2$, then ..... .

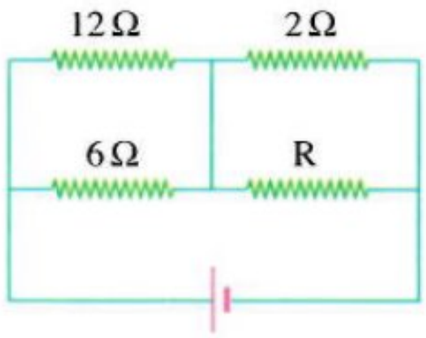
a the brightness of bulb 1 is greater
b the brightness of bulb 2 is greater
c the two bulbs have the same brightness
d the answer is indeterminable

140 ✳ In the opposite electric circuit, the electric bulb is labelled with (30 V, 45 W), so the resistance of R that makes the bulb glows according to its labelled specifications of voltage and power equals ..... .



a 1.5 Ω
b 10 Ω
c 15 Ω
d 67.5 Ω

141 In the opposite electrical circuit, if the electrical powers consumed in the 12 Ω and 2 Ω resistors are 48 W and 32 W, respectively, then the electrical power consumed in resistor R equals ..... .



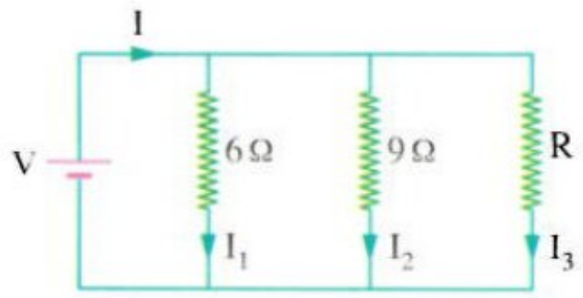
a 16 W
b 64 W
c 75 W
d 81 W

142 Three resistors x, y and z whose values are R, 2 R and 0.5 R respectively that are connected in series in a closed electric circuit, the ratio of the consumed electric power in each of them is ..... respectively.

a 4 : 2 : 1
b 2 : 3 : 1
c 2 : 4 : 1
d 2 : 3 : 4

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143 ✱ In the opposite electric circuit, if the consumed electric power in resistor R is 12 W and the value of $I_1 = 2$ A, so:



(i) The potential difference between the terminals of the battery equals ..... .

(a) 24 V (b) 12 V (c) 6 V (d) 3 V

(ii) The total current intensity in the circuit equals ..... .

(a) $\frac{4}{3}$ A (b) $\frac{13}{3}$ A (c) $\frac{7}{3}$ A (d) 1 A

(iii) The resistance of R equals ..... .

(a) 24 Ω (b) 18 Ω (c) 12 Ω (d) 9 Ω

144 When two resistors R, 4 R are connected in parallel with a battery, then the consumed power in resistor R is ..... the consumed power in resistor 4 R.

(a) four times (b) double (c) equal to (d) quarter

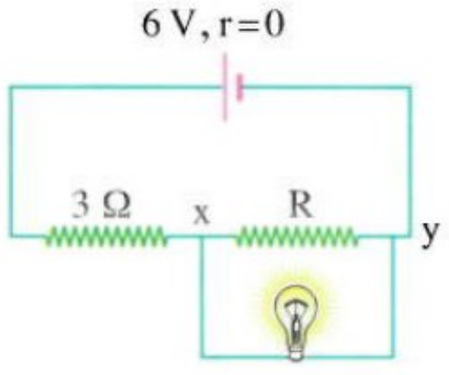
145 ✱ Two resistors have resistances of 10 Ω and R, the consumed power through them when they are connected in parallel with a battery of negligible internal resistance is four times the power consumed through them when they are connected in series with the same battery, then the value of R equals ..... .

(a) 10 Ω (b) 7.5 Ω (c) 5 Ω (d) 2.5 Ω

146 Two different conductors that are made of the same material are connected together in parallel with a potential difference that equals 220 V. If the length of the first conductor is double that of the second conductor and the diameter of the first conductor is double that of the second conductor, then the ratio between the consumed electric powers in the two conductors $\left(\frac{(P_w)_1}{(P_w)_2}\right)$ equals ..... .

(a) $\frac{4}{1}$ (b) $\frac{2}{1}$ (c) $\frac{1}{2}$ (d) $\frac{1}{4}$

147 ✱ In the opposite electric circuit, if the potential difference between the terminals of the bulb is 1.5 V, the equivalent resistance between the two points x and y equals ..... .

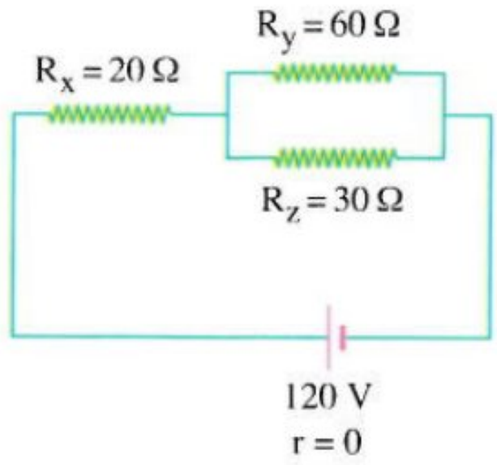


(a) 0.45 Ω (b) 1 Ω

(c) 5 Ω (d) 3 Ω

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148 From the opposite circuit:



(i) The total consumed power in the circuit equals ..... .

(a) 120 W (b) 240 W

(c) 360 W (d) 480 W

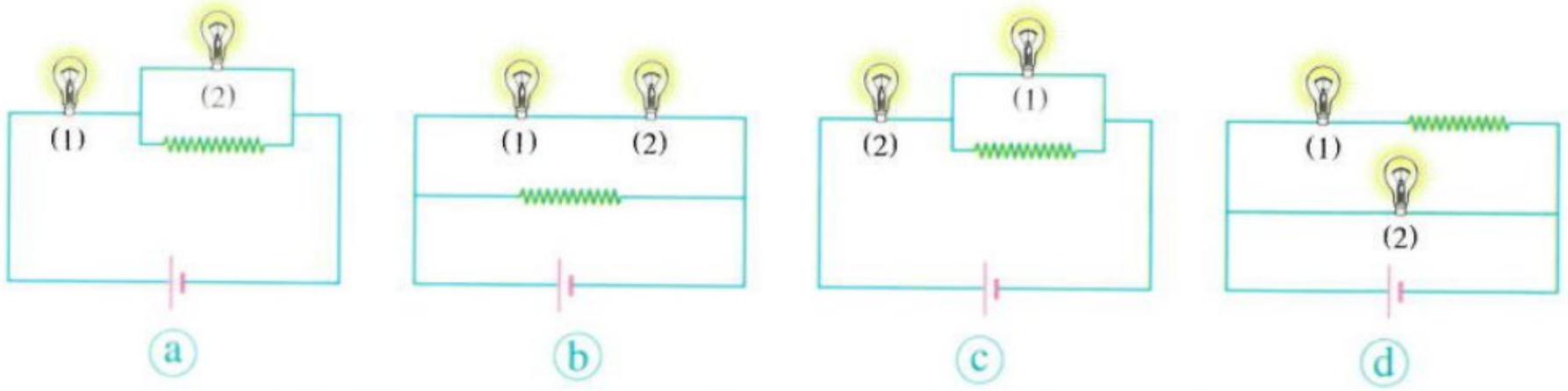
(ii) The ratio of the consumed power in R_y to that in R_x $\left(\frac{(P_W)_y}{(P_W)_x}\right)$ equals ..... .

(a) $\frac{1}{2}$ (b) $\frac{1}{3}$ (c) $\frac{2}{3}$ (d) $\frac{3}{1}$

(iii) Which of the following represents the ratios of the consumed power in the three resistors x, y and z respectively?

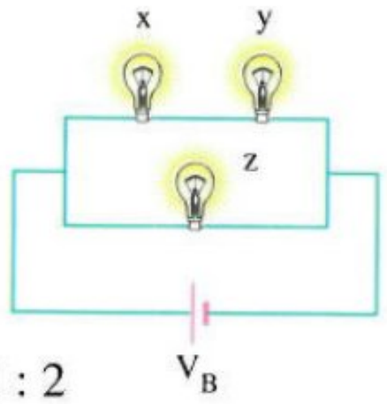
(a) 3 : 1 : 2 (b) 1 : 1 : 1 (c) 2 : 3 : 1 (d) 3 : 2 : 1

149 Two lamps, (1) and (2), are connected together with a resistor and a battery in several ways as shown in the following electrical circuits. If lamp (1) is labelled (120 V - 60 W) and lamp (2) is labelled (120 V - 120 W), which of the following circuits allows the lamps to operate at their labelled powers?



(a) (b) (c) (d)

150 ✳ In the opposite electric circuit, three identical electric bulbs x, y and z are connected together with a battery of negligible internal resistance, then the ratio between the consumed power in the three bulbs $(P_w)_x : (P_w)_y : (P_w)_z$ respectively is .....

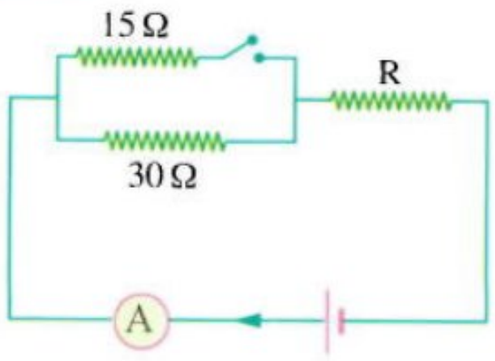


(a) 1 : 1 : 4 (b) 1 : 1 : 1 (c) 4 : 4 : 1 (d) 1 : 1 : 2

151 ✳ Three identical bulbs are connected once in series and another in parallel to the same electric source, so the ratio $\left(\frac{(P_W)_{series}}{(P_W)_{parallel}}\right)$ equals ..... .

(a) $\frac{4}{3}$ (b) $\frac{9}{5}$ (c) $\frac{1}{9}$ (d) $\frac{3}{4}$

152 ✳ In the opposite electric circuit, if you know that when closing the switch, the consumed electric power in the circuit increases to the double, so the resistance of R is ..... .



(a) 3 Ω (b) 6 Ω

(c) 8 Ω (d) 10 Ω

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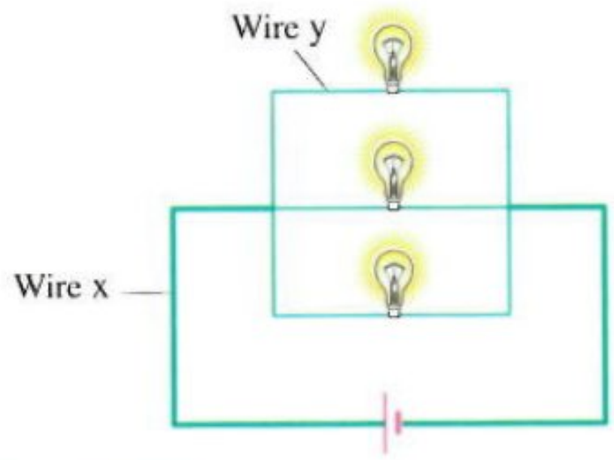
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Second Essay questions

1 The opposite figure shows an electric heater which is equipped with two heating components. **Are** the two components connected in parallel or in series? **And why**?



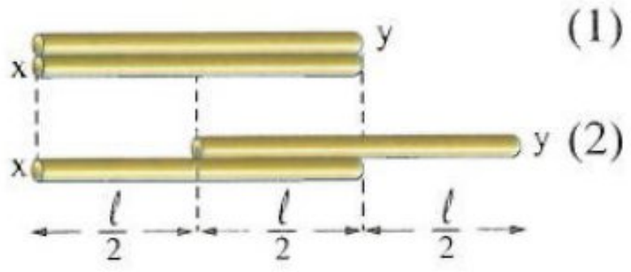
2 The opposite figure represents an electric circuit. **Explain why** the thickness of wire x is preferable to be greater than that of y.



3 A wire of uniform cross-section and resistance R is reshaped in the form of a rectangle abcd that its length ab is double its width bc as shown in the opposite figure, **which** two points are connected to an electrical source to obtain the greatest equivalent resistance?

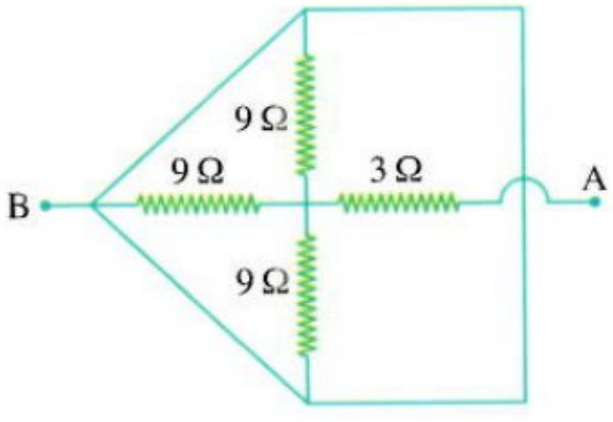


4 Two non-isolated identical wires, each of uniform cross-sectional areas, length ℓ and electric resistance R, touch each other side by side in ways (1) and (2) as shown in the opposite figure. **What** is the ratio of the equivalent resistance between point x and y in the two cases $\left(\frac{R_1}{R_2}\right)$?

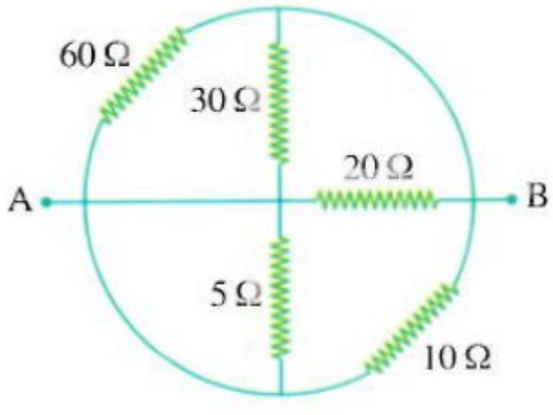


5 **Calculate the equivalent resistance between points A and B in each of the following cases:**

(a)



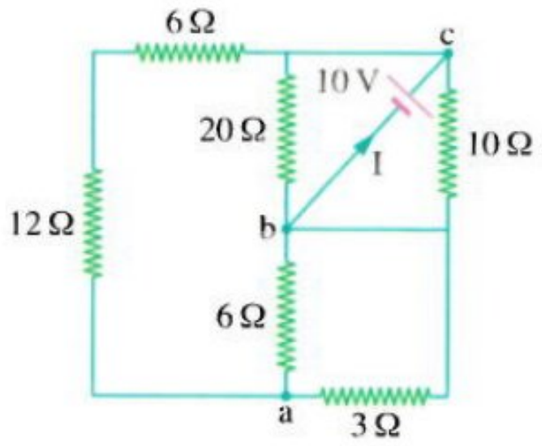
(b)



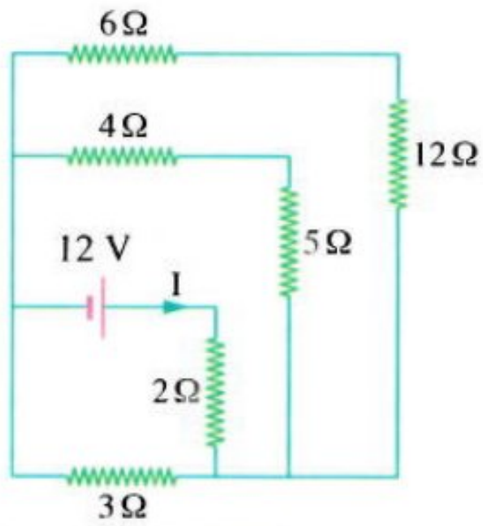
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6 **Calculate the intensity of current (I) in each of the following circuits:**

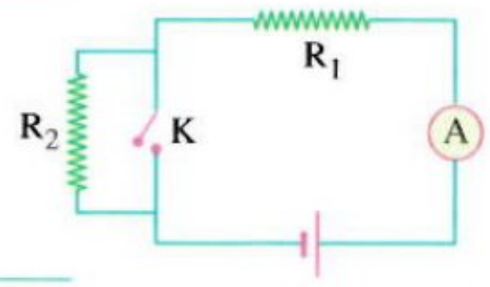
(a)



(b)



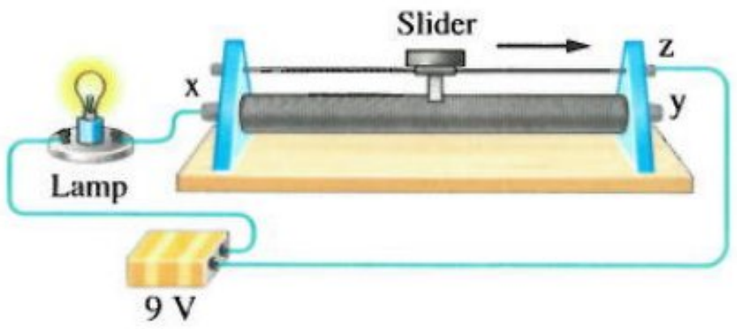
7 In the opposite electric circuit, **explain what happens to** the reading of the ammeter when switch K is closed.



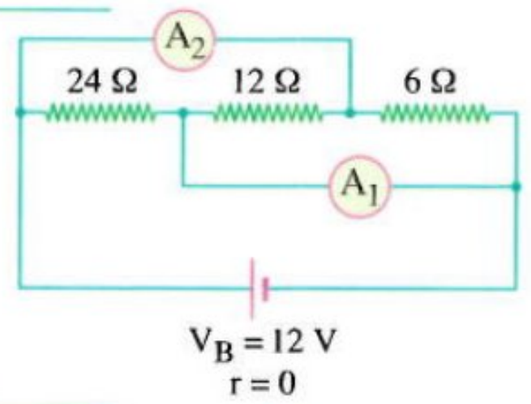
8 An electric circuit consists of a source whose electromotive force is 130 V and its internal resistance is negligible. It is connected to two resistors 300 Ω, 400 Ω in series, **calculate** the reading of the voltmeter whose resistance is 200 Ω if it is connected between the terminals of:

(a) Resistor 300 Ω. **(b)** Resistor 400 Ω.

9 The opposite figure shows a sliding rheostat connected in series to an electric lamp and a battery 9 V of negligible internal resistance, **what happens to** the potential difference between the terminals x and z when the slider moves towards terminal z? **Why**?



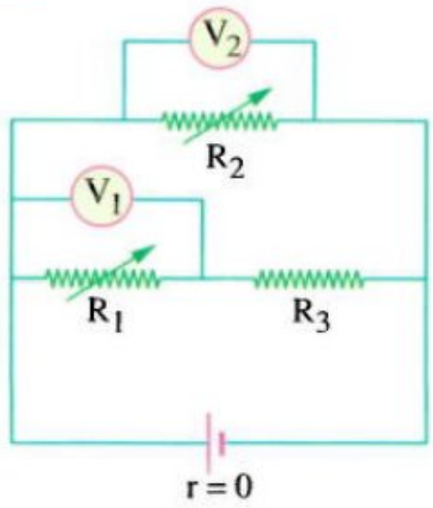
10 In the opposite electric circuit, **calculate** the ratio between the readings of the ammeters $\left(\frac{A_1}{A_2}\right)$.



11 **In the opposite electric circuit, explain what happens when increasing:**

(a) The variable resistance R_1

(b) The variable resistance R_2

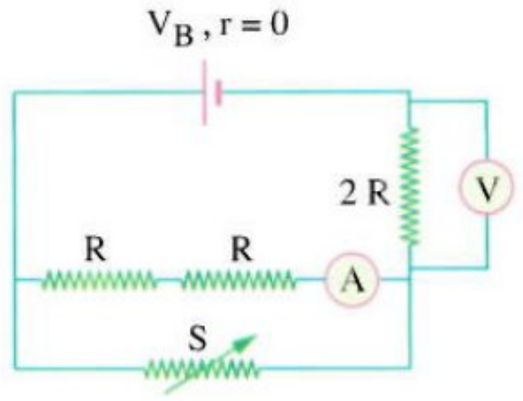


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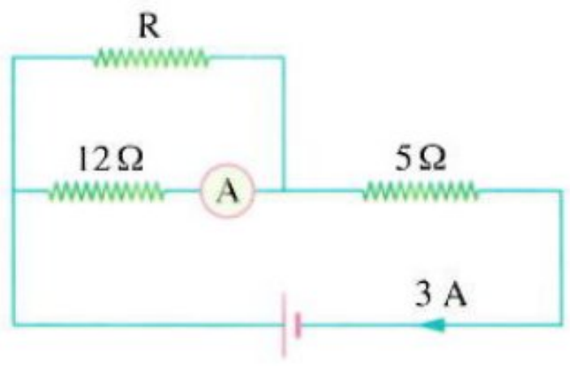
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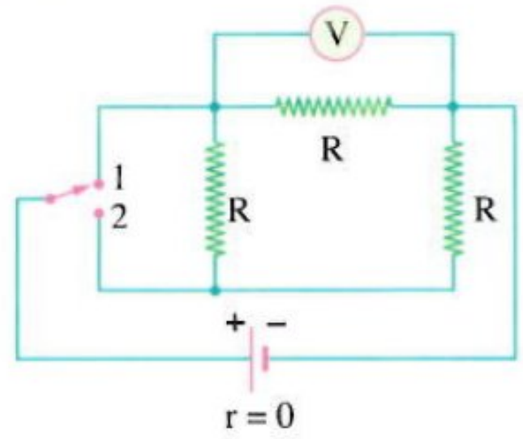
12 The opposite figure represents a closed electric circuit, **explain why** a change in the readings of the ammeter and voltmeter occurs when the resistance of the variable resistor (S) increases.



13 In the opposite circuit, if the ammeter reading is 1 A. **Calculate** the equivalent resistance in the circuit.



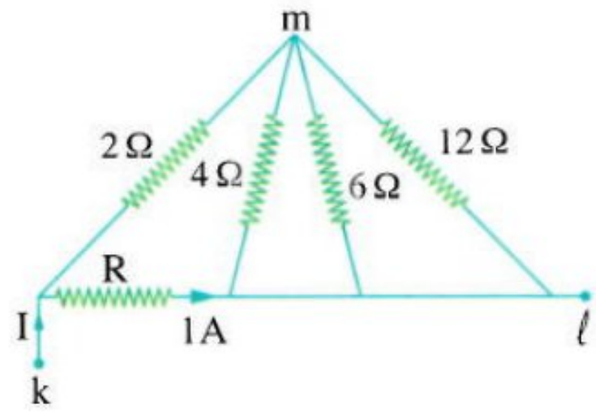
14 In the opposite electric circuit, **explain what happens to** the reading of the voltmeter when changing the switch position from position (1) to position (2).



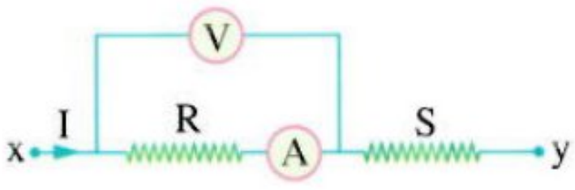
15 The opposite figure represents a part of an electric circuit where the potential difference between the two points k and ℓ equals 50 V and the current intensity passing in the resistor R equals 1 A, **calculate:**

(a) The potential difference between the points k and m.

(b) The resistance of resistor R.



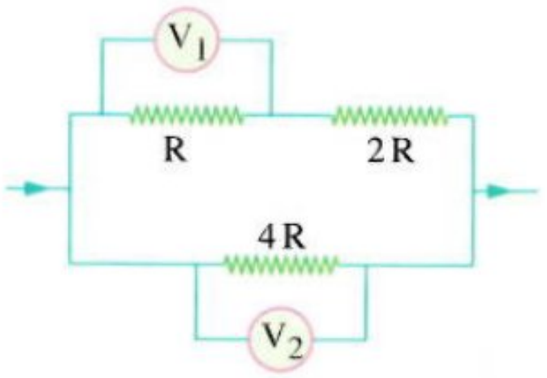
16 In the opposite figure, if the potential difference between the points x and y equals 20 V and the ammeter reading is 1 A while the voltmeter reading is 5 V. **Calculate** the value of each of resistances R and S.



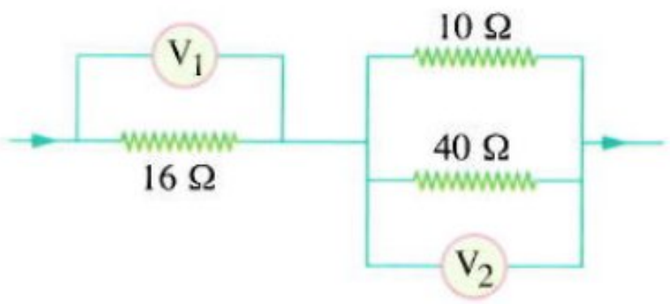
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17 **In each figure from the following, if the voltmeter reading V_1 equals 12 V. Calculate the reading of voltmeter V_2:**

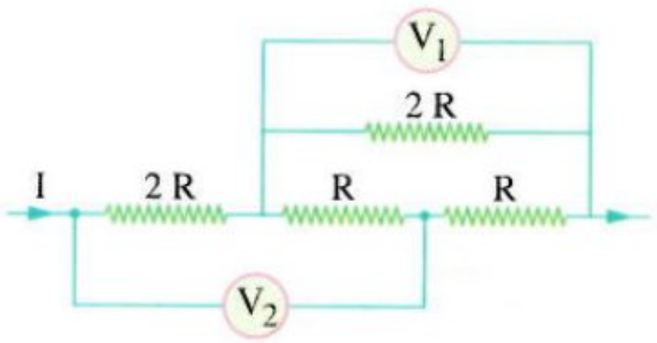
(a)



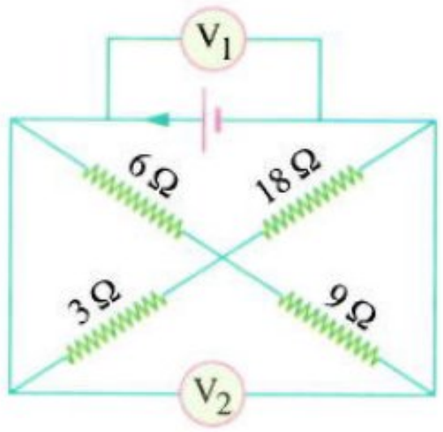
(b)



(c)

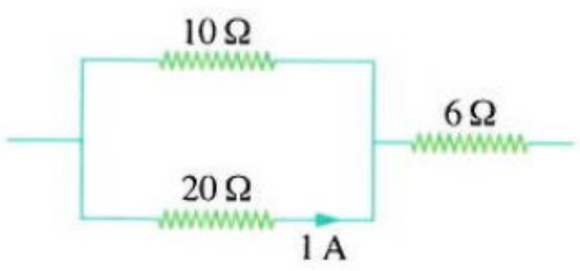


(d)

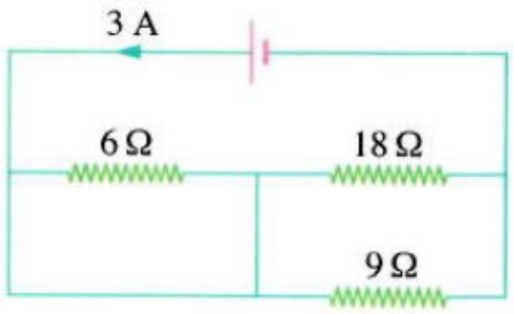


18 **Calculate the electric current intensity (I) passing in the 6 Ω resistance in each of the following:**

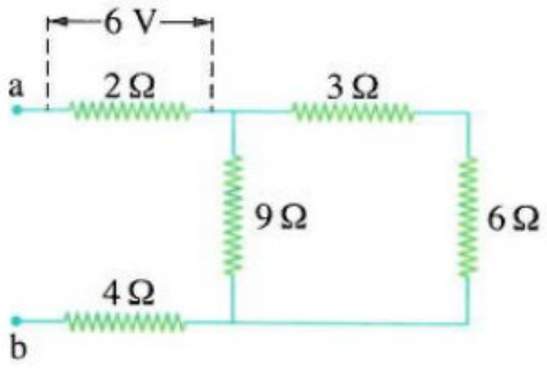
(a)



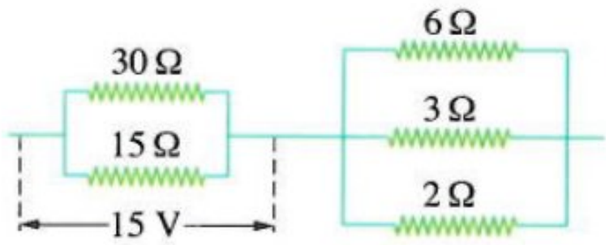
(b)



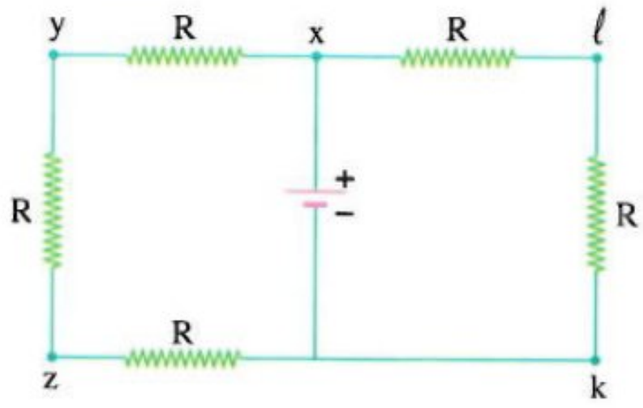
(c)



(d)



19 In the opposite electric circuit, **arrange** the points x, y, z, k, ℓ descendingly according to the electric potential of each one.



20 Six identical bulbs, each one works on a maximum potential difference of 100 V. The bulbs were required to be operated on a potential difference of 200 V, **explain how** these bulbs can be connected together such that each of them carries its maximum current?

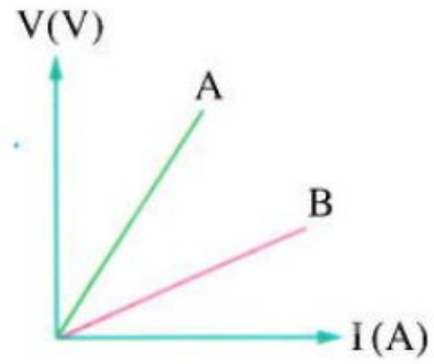
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21 Three resistances 10 Ω, 20 Ω and 40 Ω are connected together in a certain method to an electric source, so electric currents of intensities 0.4 A, 0.5 A and 0.1 A pass in these resistances respectively. **What** is its equivalent resistance?

22 The opposite graph illustrates the relation between the potential differences (V) across two wires A and B and the electric current (I) which is passing in each of them.



(a) **Which** of the two wires has larger electric resistance? **And why**?

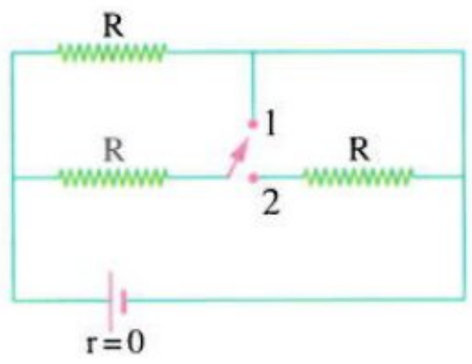
(b) If the two wires are connected together in parallel with an electric source, **which of them** consumes higher electric power? **And why**?

23 Four lamps each of resistance 6 Ω are connected together in parallel. Then the combination is connected to a battery of an emf 12 V and negligible internal resistance, **calculate:**

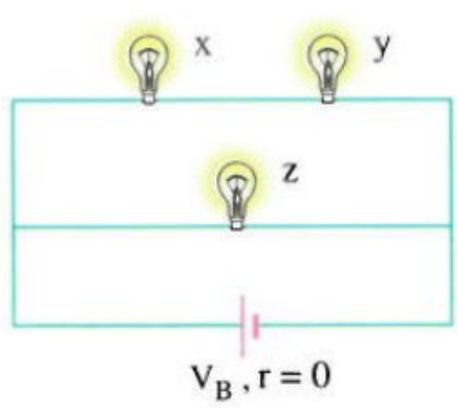
(a) The total charge which is leaving the battery in 10 s.

(b) The consumed power in each lamp in 10 s.

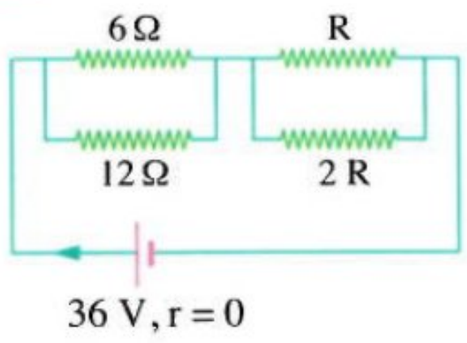
24 In the opposite circuit, **deduce** the ratio between the consumed power in the circuit in the case of closing the switch in position (1) and that in case of closing the switch in position (2).



25 Three lamps x, y and z of resistances R, 2 R and 4 R respectively are connected with an electric source V_B of negligible internal resistance as shown in the opposite figure. **Determine with explanation** which of the three lamps consumes larger electric power.



26 In the opposite circuit, if the consumed power in the 6 Ω resistance is 24 W, **calculate** the consumed power in resistance R.



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27 Two students (N) and (M) constructed an electric circuit using three identical lamps, a switch and an electrical source as shown in the following two figures:

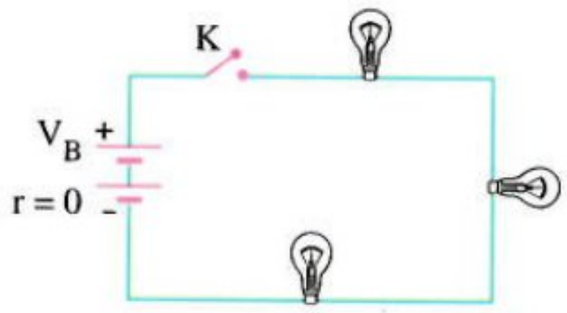


Figure (1)

The circuit of student (M)

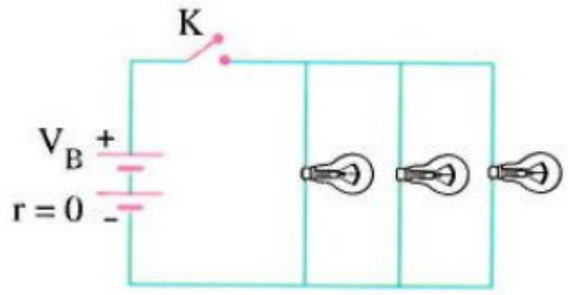


Figure (2)

The circuit of student (N)

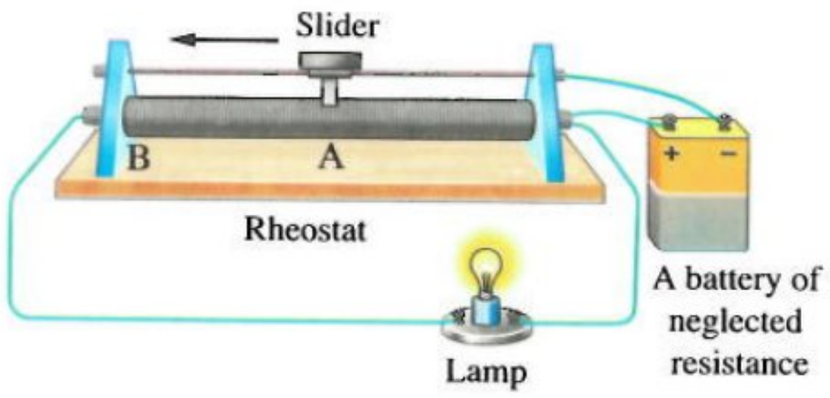
(a) **Compare between** the brightness of lamps in each of the two circuits when closing the switch K.

(b) **What happens for** the brightness of lamps in each of the two circuits when one of them is damaged?

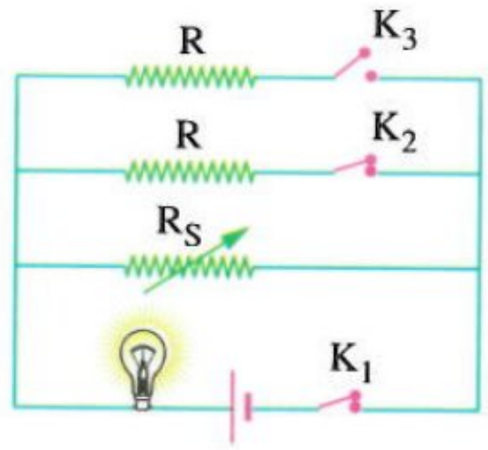
28 In the opposite electric circuit, **explain what happens to** the brightness of the lamp when:

(a) Moving the slider of rheostat from position A to position B.

(b) Replacing the positions of lamp and battery and moving the slider from A to B.



29 In the opposite electric circuit, **how** to increase the brightness of the lamp by two different methods?

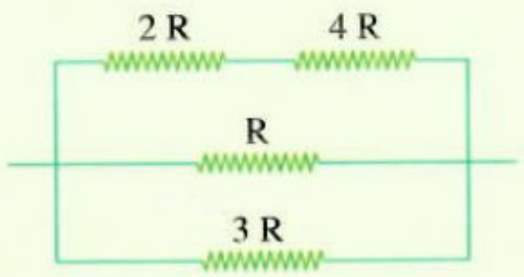


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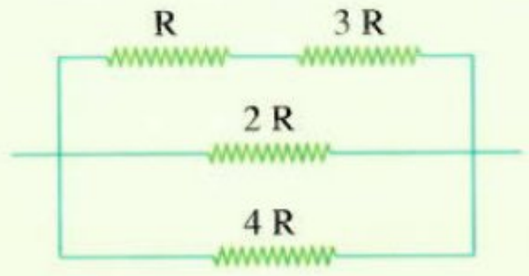
Questions of the Previous Years Exams on Lesson 2

1 Which combination of the following resistors gives an equivalent resistance of value R?

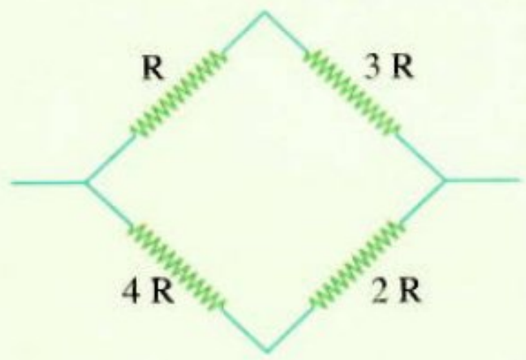
(Exp. May 21)



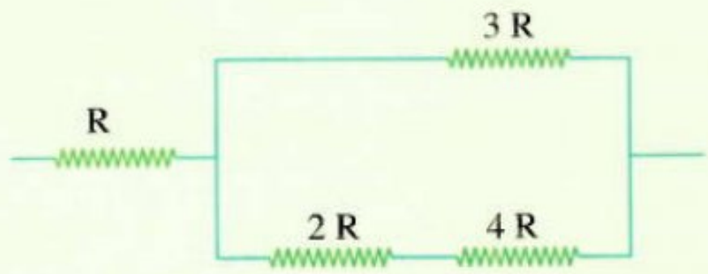
(a)



(b)

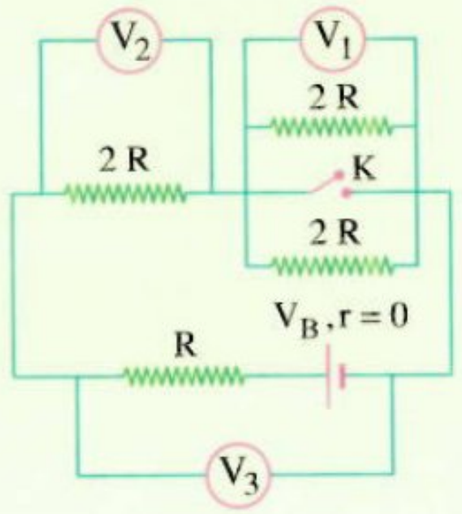


(c)



(d)

2 In the opposite electric circuit, when closing the switch (K), which of the following rows shows what happens to the readings of the voltmeters V_1, V_2 and V_3 correctly? *(Exp. May 21)*

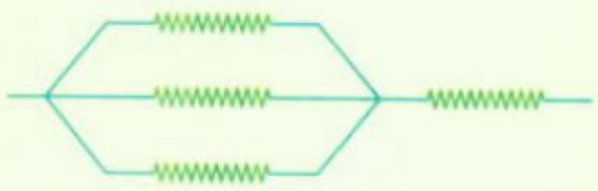


	V_1	V_2	V_3
(a)	Becomes zero	Increases	Decreases
(b)	Increases	Increases	Decreases
(c)	Becomes zero	Decreases	Increases
(d)	Increases	Increases	Increases

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3 Four equal resistors are connected together as in the following figures, so the correct arrangement of the figures from the highest equivalent resistance to the least one is ..... .

(Exp. June 21)



(1)



(2)



(3)



(4)

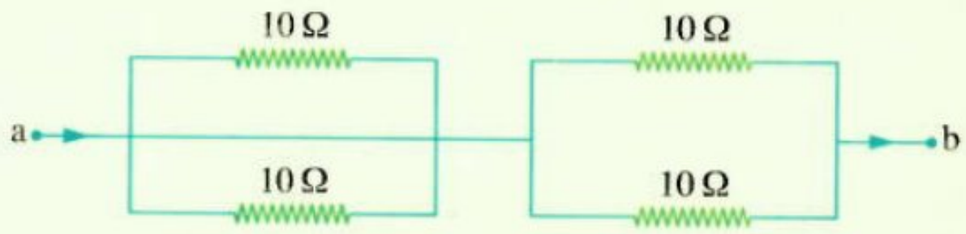
(a) $4 < 3 < 2 < 1$

(b) $1 < 2 < 3 < 4$

(c) $4 < 1 < 3 < 2$

(d) $1 < 4 < 2 < 3$

4 In the opposite electric circuit, the equivalent resistance between points a and b equals ..... . *(Exp. June 21)*



(a) 5 Ω (b) 10 Ω (c) 20 Ω (d) 40 Ω

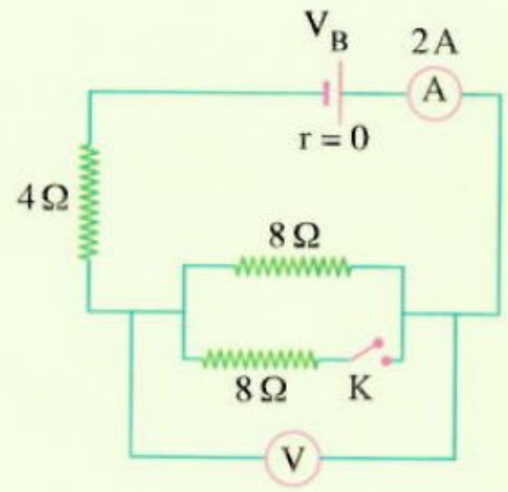
5 In the opposite electric circuit, when the key (K) is closed, the voltmeter reading equals ..... . *(Exp. June 21)*

(a) 12 V

(b) 8 V

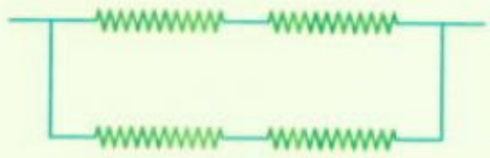
(c) 6 V

(d) 4 V

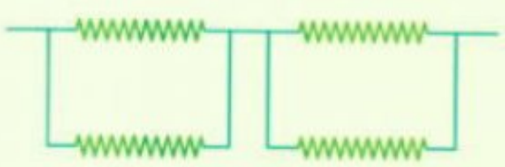


6 Four equal resistors are connected together as in the following figures, which figure gives the least equivalent resistance?

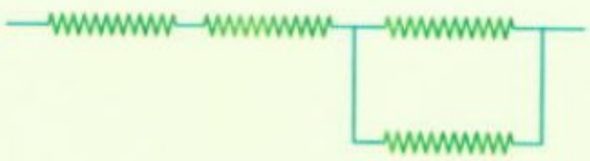
(1st session 21)



(a)



(b)



(c)



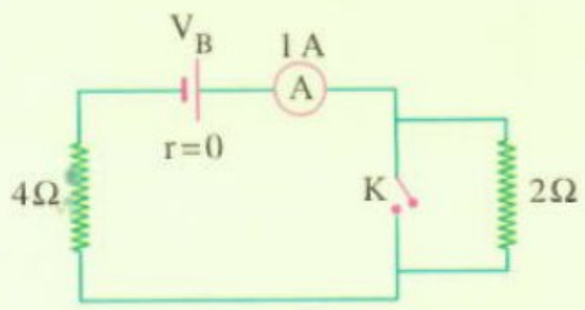
(d)

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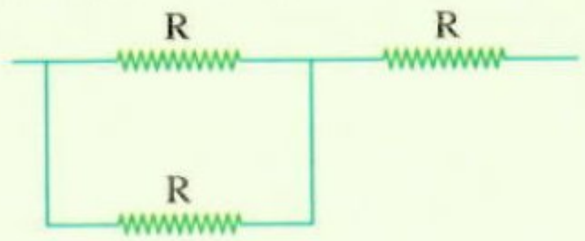
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7 In the opposite electric circuit, when the key (K) is closed, the ammeter reading becomes ..... . *(1st session 21)*

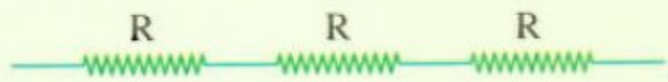


(a) 0.5 A
(b) 1.5 A
(c) 2 A
(d) 0.75 A

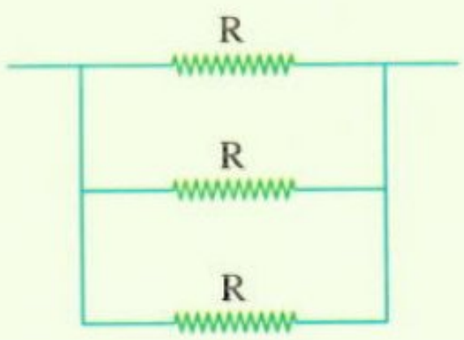
8 Arrange the figures according to their equivalent resistance ascendingly given that the resistors are identical. *(2nd session 21)*



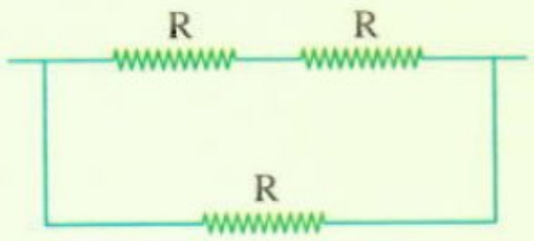
(1)



(2)



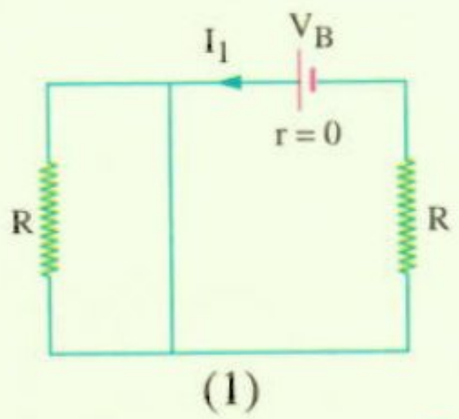
(3)



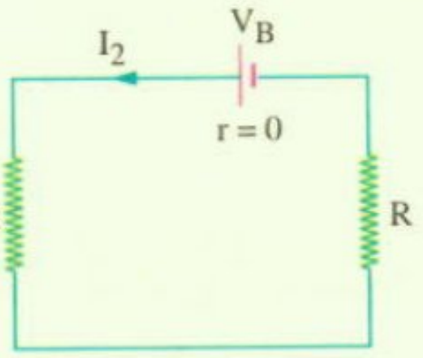
(4)

(a) 3 < 4 < 1 < 2
(b) 2 < 4 < 3 < 1
(c) 1 < 3 < 4 < 2
(d) 4 < 3 < 2 < 1

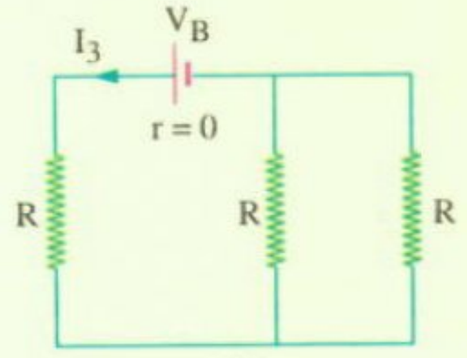
9 In the shown three electric circuits (1), (2) and (3). Which one from the following relations is correct? *(2nd session 21)*



(1)



(2)



(3)

(a) $I_1 = I_2$
(b) $I_1 > I_3$
(c) $I_2 > I_3$
(d) $I_3 > I_1$

10 A uniform cross-sectional copper wire was shaped as a rectangle {kyxm} as in the figure, its length (ky) is double its width (mk), which two terminals should be connected to the electric source to obtain the greatest electric resistance? *(1st session 22)*



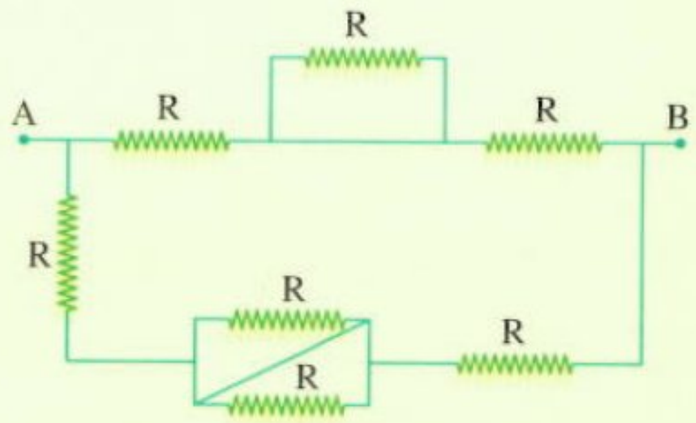
(a) m, k
(b) k, y
(c) x, y
(d) k, x

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11 Two electric resistors, the resistance of the first is three times as that of the second. When they are connected in parallel, they give an equivalent resistance of 3 Ω, then when they are connected in series gives ..... . *(1st session 22)*

(a) 12 Ω (b) 16 Ω (c) 8 Ω (d) 4 Ω

12 The figure shows a part of a closed electric circuit that contains a group of identical resistors. The equivalent resistance between the two points (A) and (B) equals ..... . *(2nd session 22)*



(a) $\frac{6\,R}{5}$ (b) $\frac{5\,R}{4}$

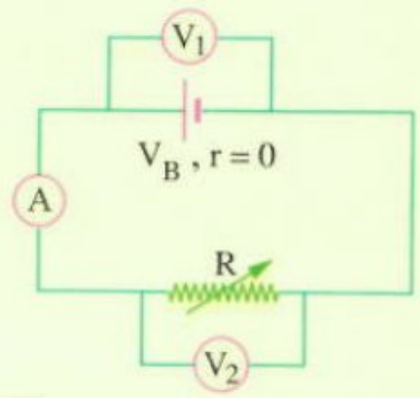
(c) $\frac{3\,R}{2}$ (d) R

13 You have three resistors R_1, R_2 and R_3 as shown in the figure. When they are connected in parallel, they give an equivalent resistance 4 Ω, then when they are connected in series, they give an equivalent resistance ..... . *(2nd session 22)*

$R_1 = 3\,R$ $R_2 = 4\,R$ $R_3 = 6\,R$

(a) 9 Ω (b) 27 Ω (c) 13 Ω (d) 39 Ω

14 In the electrical circuit in front of you, when the value of resistance (R) increases. Then, the reading of V_1 ..... and the reading of V_2 ..... . *(1st session 22)*



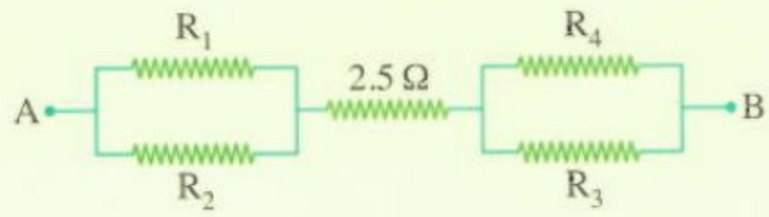
	V_1	V_2
(a)	does not change	does not change
(b)	increases	increases
(c)	increases	does not change
(d)	does not change	increases

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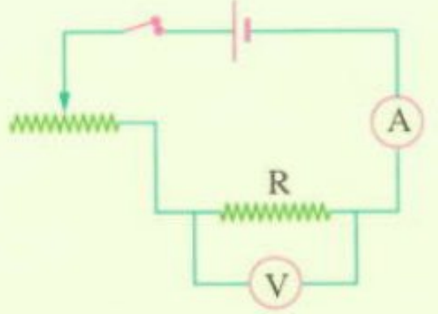
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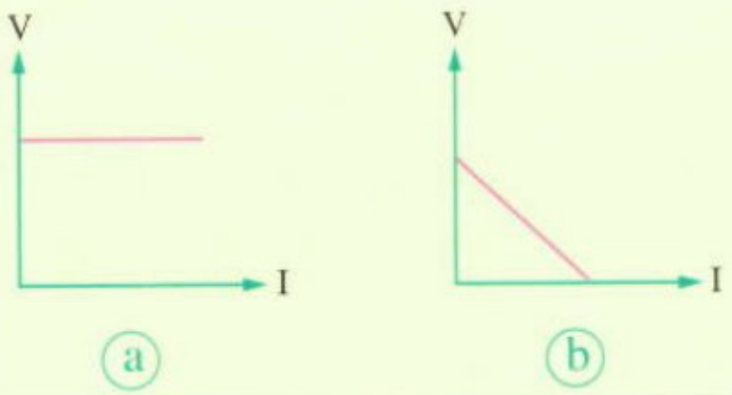
15 In the opposite figure, which of the following choices makes the resistance between A and B equals 5 Ω? *(Exp. 23)*



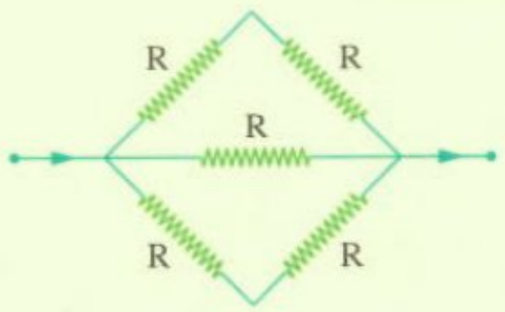
	R_1 (Ω)	R_2 (Ω)	R_3 (Ω)	R_4 (Ω)
a	2	9	8	2.5
b	1	9	2	8
c	1	2	8	9
d	8	1	9	5

16 Which graph represents the correct relation between the electric potential difference (V) across the two terminals of the fixed resistance (R) and the ammeter reading (I) at the same temperature? *(1st session 23)*





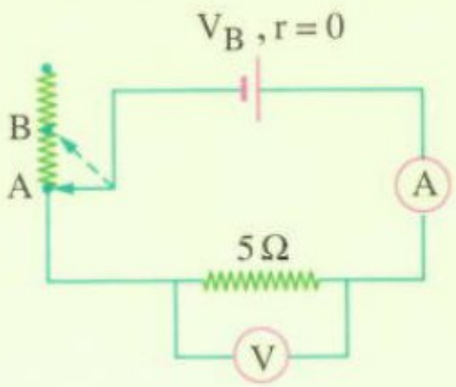
17 The figure represents a part from a closed electric circuit. The equivalent resistance of the shown combination equals ..... . *(1st session 23)*



(a) R (b) 2 R

(c) $\frac{R}{2}$ (d) $\frac{3R}{5}$

18 In the electric circuit, if the voltmeter reading when the slider of the rheostat is at (A) was 12 V and when the slider is moved to point (B), the voltmeter reading was 3 V. The value of the resistance taken from the rheostat at (B) is ..... . *(1st session 23)*



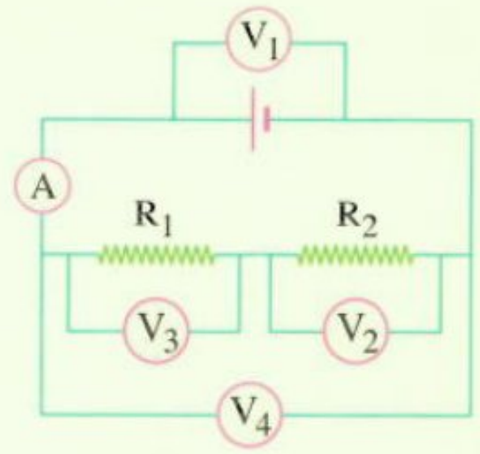
(a) 25 Ω (b) 30 Ω

(c) 15 Ω (d) 20 Ω

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19 In the shown electric circuit:

Which readings of those voltmeters are equal? *(2nd session 23)*



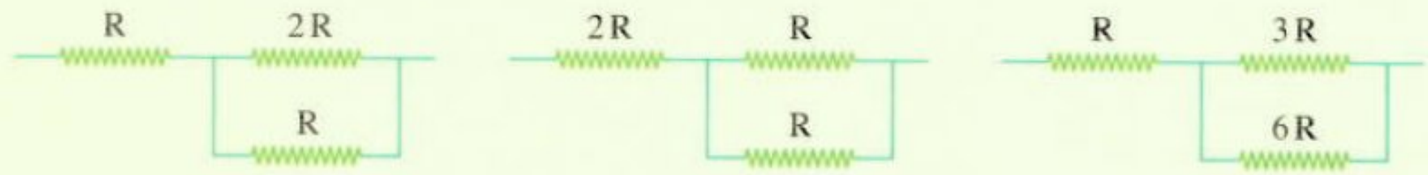
(a) V_2, V_3

(b) V_2, V_4

(c) V_2, V_1

(d) V_1, V_4

20 The figures show a number of resistors connected to each other.



Which of the following expressions is correct concerning the equivalent resistance of each combination? *(1st session 24)*

(a) The equivalent resistance of combination (X) equals that of combination (Y).

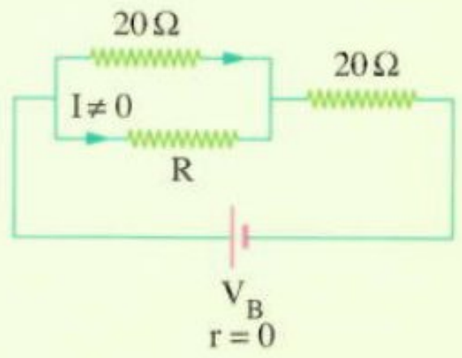
(b) The equivalent resistance of combination (X) is less than that of combination (Y).

(c) The equivalent resistance of combination (Z) is less than that of combination (X).

(d) The equivalent resistance of combination (Z) is more than that of combination (Y).

21 From the opposite electric circuit:

Which choice is possible to express the total resistance of the circuit? *(2nd session 24)*



(a) 19 Ω (b) 25 Ω

(c) 15 Ω (d) 40 Ω

22 When two different resistors are connected together in parallel, their equivalent resistance will ..... . *(Practice 25)*

(a) be equal to the sum of the resistors

(b) have an average value between the values of the two resistors

(c) be less than the smaller resistance

(d) be greater than the larger resistance

23 A uniform wire of resistance 120 Ω is cut into a number of equal segments and when these wire segments are connected together in parallel, their total resistance becomes 1.2 Ω, so the number of these segments of the wire equals ..... . *(Practice 25)*

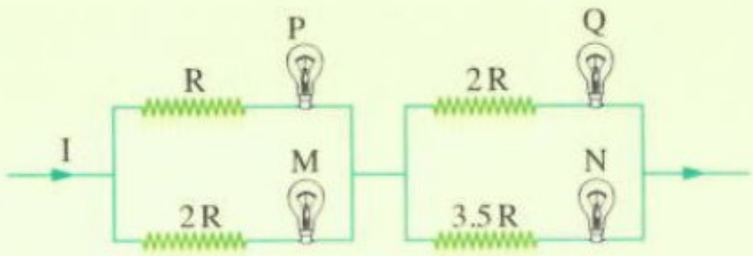
(a) 6 (b) 10 (c) 12 (d) 24

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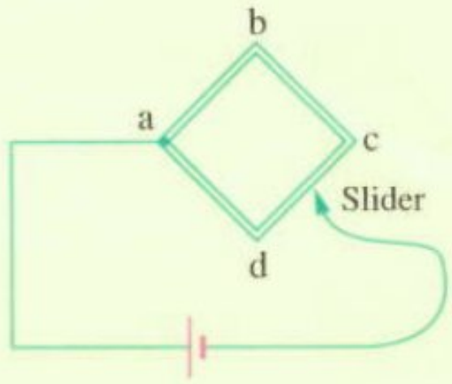
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24 Four identical bulbs N, M, P and Q, the resistance of the filament of each is R, are connected to different resistors as shown in the opposite figure, then the brightness is the same for ..... .



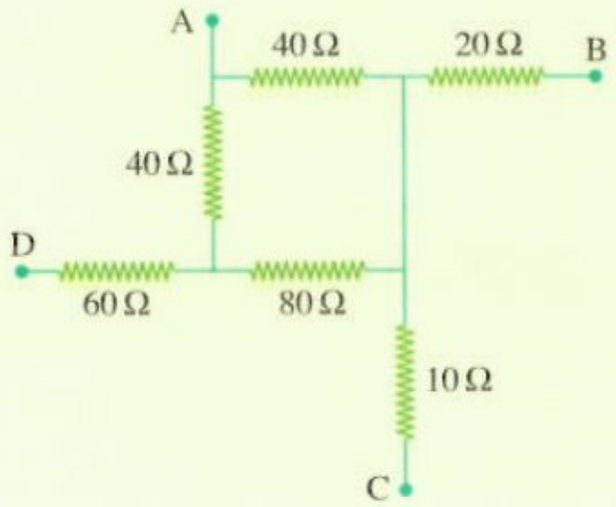
(a) both bulbs Q and M
(b) both bulbs M and N
(c) both bulbs P and N
(d) all the bulbs *(Practice 25)*

25 A wire of uniform cross-section is bent to form a square abcd, of which each side has a resistance R, one of the poles of a battery is connected to point a on the frame. Then the point at which the other pole of the battery is connected so that the lowest current intensity passes in the circuit is ..... . *(Practice 25)*



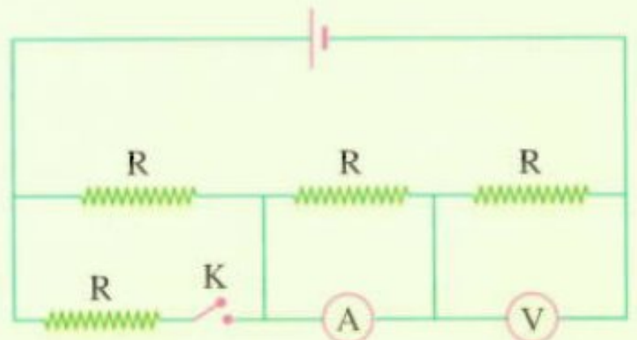
(a) a (b) b (c) c (d) d

26 Resistors are connected as shown in the opposite figure. To which two points of the following should a battery terminals be connected in order to get the lowest equivalent resistance? *(Practice 25)*



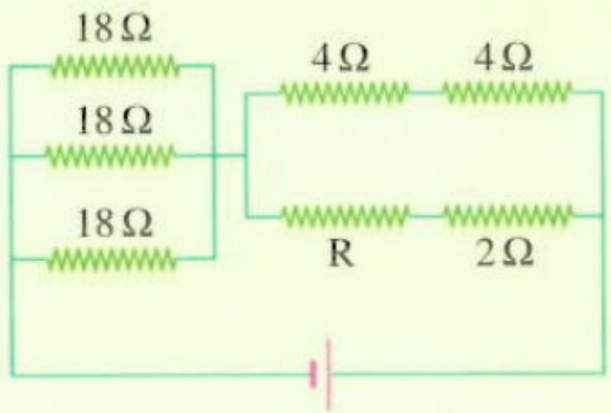
(a) B, C (b) A, B
(c) C, D (d) A, D

27 In the opposite electrical circuit, if the voltmeter resistance is infinite and the ammeter resistance is negligible, what happens to the reading of each of the ammeter and the voltmeter respectively when switch (K) is closed? *(Practice 25)*



(a) Increases, increases. (b) Decreases, decreases.
(c) Decreases, increases. (d) Increases, decreases.

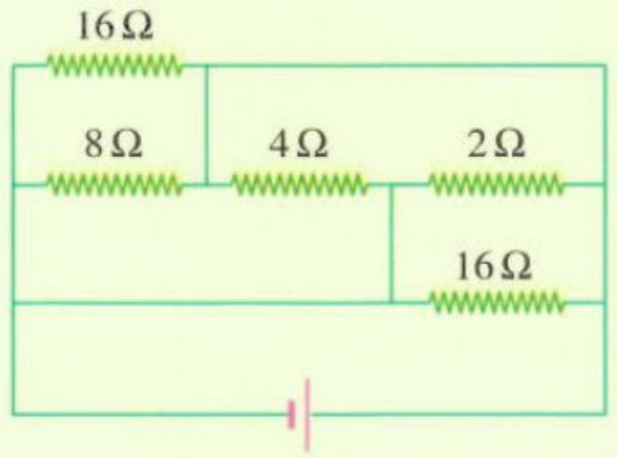
28 The figure shows a number of resistors connected together. If their equivalent resistance is 10 Ω, the resistance of resistor R is equal to ..... . *(1st session 25)*



(a) 4 Ω (b) 2 Ω
(c) 6 Ω (d) 8 Ω

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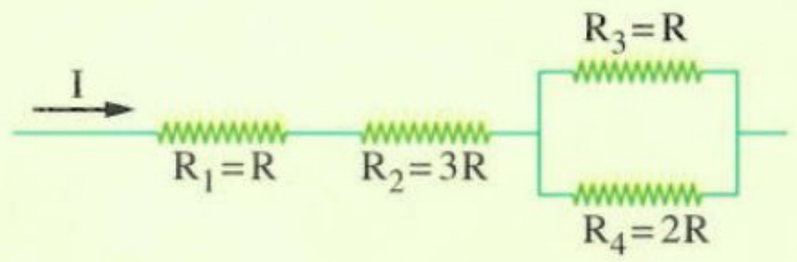
29 A number of resistors are connected in an electrical circuit as shown in the figure. By using the given values of resistance, the equivalent resistance of this combination is equal to ..... . *(2nd session 25)*



(a) 6 Ω (b) 1 Ω

(c) 8 Ω (d) 16 Ω

30 The figure represents a part of an electric circuit with four resistances R_1, R_2, R_3, R_4.
Which of the following statements is correct? *(Practice 26)*



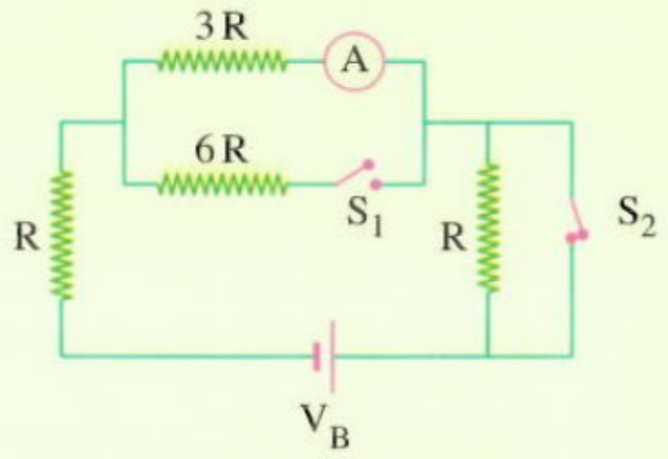
(a) The current intensity passing through resistance R_1 is equal to that passing through resistance R_3

(b) The current intensity passing through resistance R_2 is three times that passing through resistance R_4

(c) The potential difference across resistance R_2 is three times that across resistance R_3

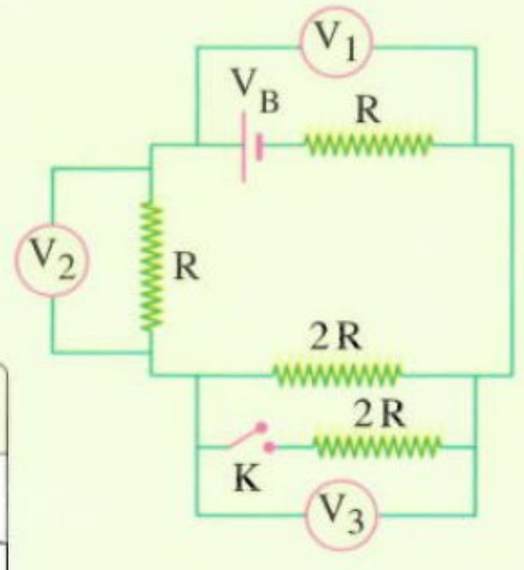
(d) The potential difference across resistance R_1 is equal to that across resistance R_4

31 The electric circuit shown in the figure contains several resistors, an ammeter, a battery of electromotive force V_B and negligible internal resistance and two switches, S_1 and S_2. When S_1 was open and S_2 was closed, the ammeter reading was I, so when closing switch S_1 and opening switch S_2, the ammeter reading becomes ..... .



(a) I (b) $\frac{2}{3}$ I (c) $\frac{1}{2}$ I (d) $\frac{1}{3}$ I *(Practice 26)*

32 The diagram shows a closed electric circuit containing several resistors, a battery of negligible internal resistance, and an opened switch K. What happens to the readings of voltmeters V_1, V_2, and V_3 when switch K is closed? *(Practice 26)*



	Reading of V_1	Reading of V_2	Reading of V_3
(a)	Remains constant	Increases	Decreases
(b)	Increases	Decreases	Increases
(c)	Decreases	Decreases	Increases
(d)	Decreases	Increases	Decreases

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